Factors Affecting Distance Running Performance

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The sport of distance running has a long history that has been closely tied with the rise of sports science. The founder of modern physiology Nobel Peace Prize winner A.V. Hill chose running as his platform to develop the concept of oxygen consumption (Bassett & Howley, 2000). Even with this long history of investigating the mechanisms behind performance and ways to enhance it in sports science, many questions remain unanswered and the exact factors that govern performance are still debated. In addition, unlike other activities such as weight lifting, the optimal way to train distance runners, including both the effects of training at different intensities and how to periodize that training, remains unknown. The current function of science in training is not to be used as a way to prescribe training, but instead as a way to explain why training used by coaches or athletes works.

The purpose of this review is two fold. First, it is to establish the variety of mechanisms that control and limit performance in the sport of distance running. Secondly, it is to look at the current training methods used by trained athletes and evaluate their impact on the physiological factors that govern performance. By analyzing the factors that affect performance and the current training trends, limitations in the training of competitive distance runners will become apparent. I'll take you through how we run, from the nervous system down to what happens on the muscular level.

Additionally, a sub goal is to bridge the gap between the Sports Scientists/Physiologists of the world and the coaches of the world. Both groups do outstanding work, but it is as if a gap exists with each group going about their business with an air of superiority while completely ignoring or dismissing the other group. This has resulted in two completely different ways of training endurance athletes, with the coaches prevailing up to this point in superiority of performance, in my opinion. This strange battle should not be against each other, but rather a cooperative one aimed at finding out how to optimize performance. Once egos are put aside and everyone acknowledges that we do not have all the answers, then I feel that new performance levels will be met.

Chapter 1: How Running Happens

When broken down in its simplest form, running is nothing but a series of connected spring like hops or bounds. A certain amount of energy needs to be imparted into the ground to propel the runner forward, continuing the running movement. The amount of energy needed depends on the pace that the athlete is running; as the pace increases a greater amount of energy is needed. This energy comes about through two primary mechanisms called active and passive mechanics. Active mechanics refers to what we all think about when it comes to what drives running, actively recruiting muscles that generate force via muscle contraction. On the other hand, the body, namely the muscles, tendons, and ligaments, act in a spring like manner temporarily storing the energy that comes about from the collision of the foot with the ground. During the subsequent push off, or propulsion, phase of running, this energy is utilized and released, contributing to forward propulsion. These two mechanisms combine to provide the necessary force and energy to power the running movement.

In simplistic terms a runner can keep going at the same pace as long as they can produce the necessary kinetic energy, whether this comes from force production or passive mechanical energy. Once they cannot impart enough energy, the pace has to slow as they are not able to cover the same amount of ground with the same stride rate. Simply stated, they fatigue. Before looking at performance and fatigue, it's important to grasp the intricacies of the running movement. While the previous description provides the framework, lets delve into each step of how running actually occurs from the Central Nervous System all the way down to a single muscle contraction. Once this has been done, the potential limiters along the route can be identified.

Motor Programming

The brain, or to be more exact the Central Nervous System (CNS), is where movement starts. The simple act of moving one finger is an incredibly complex task involving multiple systems, so completely understanding a complex dynamic movement like running is a daunting task. What we do know is that movement originates in the nervous system. The nervous system consists of higher levels of organization, such as the brain itself, and lower levels of organization, such as the spinal cord and brainstem. These two levels of the nervous system combine to decide how movement takes place.

Movement occurs with a combination of pre-conceived motor programs and slight tweaks or alterations based on sensory information. Essentially, the body has a gross general plan of how to go about doing a certain movement, and then tweaks that plan based on the sensory information it is constantly receiving, such as the ground surface, the position and length of your limbs, and a

whole load of other sensory sources. The movement pattern serves as the rough basis for how that particular movement should take place. The sensory information provides for the on the go adjustment like that seen when running on a road versus the sand, or if an unexpected root pops up in front of you. If the body simply worked of pre-determined motor programming without the ability to use feed forward or feedback information to adjust, we'd all be in trouble.

On the higher level, the brain works in an integrated way in that several areas of the brain combine to work dynamically to create the running pattern, using feedback and feed forward information to provide the details on what the muscles should do. At the lower spinal cord level, movements that occur reflexively or without the need for sensory feedback or feed forward information are developed. At the spinal cord level, Central Pattern Generator's (CPG) guide movement patterns and muscle activity without the need for sensory input (Molinari, 2009). These two processes integrate using both active and passive mechanics to decide how movement ultimately takes place.

In establishing the movement pattern, what muscles to activate, how often to activate them, and in what order activation takes place is all determined. While running, the CNS uses a complex amount of sensory information including; external stimuli such as the ground surface, limb movement and position, such as how the foot is striking, and internal stimuli such as the length of various muscles throughout the movement or even the buildup of fatiguing products in the muscles themselves. All of this information, combined with the basic motor programming, results in an on the fly adjustment of how you are moving. A variety of adjustments are made including: what type of motor unites (groups of muscle fibers) are recruited, the recruitment pattern, how long a rest to work cycle a motor unit has, the relaxation of opposing muscles, and the manipulation of non-propulsive fibers to minimize the effect the impact with the ground has. The CNS is constantly using all of the sensory information, comparing the intended movement with the actual movement and making slight tweaks or adjustments. Not only does it make adjustments based on the movement, but also on what to alter when fatigue is building up. While this will be covered in depth in the next chapter, how the body deals with fatigue is ultimately a motor control issue. Have you ever wondered why you might start leaning back, swing your arms wildly, or reach out with your foot during the end of an exhausting race? This happens because of a combination of conscious and subconscious control in which you are trying to compensate for fatigue by a variety of biomechanical adjustments. Part of training is teaching the body how to accurately adjust the movement pattern to fatigue. From a motor programming standpoint, doing all out workouts where form is broken down completely might lead to negative motor programming, or in laymen terms bad habits.

At first the movement pattern is rough, uncoordinated and inefficient, but as a person becomes better trained, this process is refined and improved. Initially, the exact recruitment pattern or how to relax the opposing muscle is not known or refined. Slowly, the body becomes more

efficient at determining exactly what muscles need to be working and for how long. This refinement results in a smoothing out of the movement and is an improvement in neuromuscular control which creates an efficient movement pattern that enhances performance via improving efficiency. This process is called motor learning, and contrary to popular belief, running is a skill that needs to be learned and refined.

While previously it had been thought that improvement in motor learning only occurred at the higher levels such as in the motor cortex in the brain, recent evidence has demonstrated that even at a spinal cord level, the movement pattern can be refined (Molinari, 2009). The movement pattern is generally improved by a better coordination of activating just the right amount of motor units to do the work, improving the cycling of motor unit activation, and decreasing the level of co-activation (when the opposing muscle is active at the same time as the main muscle). Additionally, as a movement becomes well refined, it is believed that the CNS becomes better at using all of the sensory information that is receiving, essentially weeding out the pertinent versus inconsequential information better than when first learning how to move.

Running Around with your head cut off:

The phenomenon of chickens running around after their heads cut off shows that a general movement pattern for running is available at the spinal cord level. Studies with other animals and even historical reports with humans when the guillotine was in use, have confirmed this phenomenon. This points to the conclusion that activities like running and walking might have an ingrained motor program that has developed through evolutionary process.

Sending and Receiving the Signal:

The actual process of activating muscles occurs via the nervous system sending neural signals called action potential. Action potentials work as the communication system between the nervous system and the muscles. Both Neurons, which are nerve cells, and muscle cells create action potentials. In neurons, the action potentials serve as a communication device, and in the case of movement send the signal to contract all the way to the muscles. An action potential works via differences between the electrical charge inside the cell and outside the cell. A cell has a resting voltage called the resting membrane potential. In its resting state, a cell is polarized in that inside the cell has a negative charge compared to the outside of the cell. This happens because of a greater concentration of negative ions inside the cell and/or a greater concentration of positive ions outside the cell.

When this charge is reversed, or depolarized, to a significant enough degree so that now the inside of the cell has a positive charge, an action potential arises (Brooks & Fahey, 2004). Full depolarization has to occur for an action potential to be generated. This means that a significant

enough stimulus needs to be received to change from a negative to a positive electrical balance inside the cell. Once the action potential is generated, repolarization occurs, returning the membrane potential to resting levels. The action potential flows down the cell and communicates to the next cell in line via neurotransmitters. This process occurs over and over as the signal makes its way from its origin through the nervous system and down to the muscles.

The manipulation of the electrical charge inside and outside of the cell occurs mainly from differences in sodium and potassium inside and outside of the cell. The Potassium maintains the negative internal charge, while the sodium keeps the positive external charge. The manipulation of the sodium and potassium levels inside and outside of the cell creates the differences in charges. The primary mechanisms for controlling these two substances are the Sodium-Potassium pump, and a greater ease of movement through the cell membrane of potassium (Brooks & Fahey, 2004). For depolarization to occur, sodium gates open which causes the shift in electrical charge to occur. The opening of the sodium gates in motor units is caused by the neurotransmitter Acetylcholine (ACh). Neurotransmitters are chemicals that allow for communication between cells. Following the action potential creation, the potassium gates open causing potassium to exit the cell, which causes repolarization, returning the cell to normal. As we become better trained, the signaling via the neurons can improve. A change in the excitability of a neuron is one possibility, as is a change in the speed of the signal being sent.

Muscle Contraction

The nerve signal eventually reaches an α motoneuron, which is a neuron that innervates a group of muscle fibers. The entire α motoneuron and connected muscle fibers make up what is called a motor unit. The action potential travels down the neuron until it reaches the gap between the neuron and the actual muscle fiber and is called the neuromuscular junction. This junction is where communication between the neuron and the muscle takes place. As previously mentioned, this occurs via the releasing of neurotransmitters (in this case ACh) that travel across a small gap between the neuron and the muscle and bind to special receptors on the muscle. The binding of the neurotransmitter causes an action potential to occur in the muscle cell, which can lead to the depolarization process described previously.

In the muscle cell, the depolarization causes Calcium that is stored in a structure called the sarcoplasmic reticulum to be released. The Calcium quickly spreads throughout the muscle fiber with the goal of eventually reaching the actual contractile parts of the fiber. Deep within the cell is its basic unit the myofibril. The myofibril consists of two main filaments, actin and myosin. Actin is referred to as a thin filament while myosin is the thick filament because it has myosin heads on it which can attach to the actin. At rest, the myosin heads cannot attach to the actin because the attachment site is blocked. However, the calcium released frees up the attachment site and allows the myosin head to attach to the actin. When it does this, it essentially pulls on the actin, causing

contraction. The repeated pulling and releasing that goes on is what causes muscle contraction. Without calcium release this interaction cannot occur. To help conceptualize this process, think of a stationary person (the myosin) pulling on a rope (the actin) to try and drag a heavy object towards them. The person's hand represent the myosin head as they grab the rope, pull it some, let go, and then grab it again to pull the object closer.

But that's not the entire story. This whole contraction process requires energy. Energy in the form of ATP is required so that the myosin head can pull on the actin. This movement requires the release of energy. However, in terms of supplying energy, ATP needs to be supplied once the myosin head has completed its pull to allow for it to release and be ready for the next pulling cycle. Thus, the process of supplying energy is one of replenishment. Without the resupplying of ATP after the myosin head's pulling has occurred, the continual process of attaching, dragging, and releasing can not occur. In our conceptualization, without energy, the actual pulling of the rope takes energy, but if we did not supply energy at the end of a single pull, then our person would not be able to move his hands further up the rope and pull again. This is the process of a single contraction of a muscle fiber. Once the contraction occurs, relaxation has to occur before a subsequent contraction occurs. Relaxation is dependent on the calcium being transported back into its holding site, the sarcoplasmic reticulum. Until the calcium returns, another contraction can not occur.

Energy Needed

As you can see from the process of contraction, chemical energy needs to be generated in the form of ATP to resupply the myosin heads during the contraction process. We only have a limited amount of stored ATP in the muscle, so instead we have several processes to recycle ATP. These processes are a series of chemical reactions that take the various products left over from the energy release that occurs when contraction takes place and recycles them into ATP. When energy is used by the myosin head, the ATP, which consists of an Adenosine molecule and three Phosphates (Pi), is broken down to ADP (Adenosine + 2 Pi) and a Pi. The releasing of one of the Pi is what causes energy release. In the end we are left with ADP and Pi floating around, or in some cases AMP (Adenosine + 1 Pi) and Pi. The energy systems work to use these and other building blocks to recreate ATP.

The energy systems all use a series of chemical reactions to produce ATP. With each chemical reaction, enzymes are required to convert the initial products into the final products. Enzymes accomplish this by speeding up the rate of the reaction. Therefore the quantity of certain enzymes is one trainable factor that can enhance performance, as the ability to perform certain essential chemical reactions is improved.

Each energy system differs in complexity in terms of how many reactions are needed to finally get to ATP, and on what the initial fuel source is. Obviously, with a greater number of reactions, it takes longer to go through the entire process. Additionally, there are more steps

involved, which means more chances of slow down, and more substances needed for each reaction. On the other hand, the supply of the products used during the energy systems matters. With our simple, one or two step reaction systems they can produce ATP very quickly, but the fuel supply is limited and thus used quickly. On the other hand, the complex multi reaction systems have fuel supplies that are much larger which means while they cannot produce ATP as quickly, they can do so for a much longer time. Lastly, one other difference is in the by-products that are produced with each system. Each system results in additional products besides ATP. Some of these products can interfere with energy production or muscle contraction, and function in signaling to the CNS that the body is out of homeostasis. Thus it is a balancing act between by-product build up, the speed and power of the system, and the endurance of the system. Let's look at some specific details for each system.

The first system is actual several different small systems that are termed the immediate energy systems. The quickest and easiest immediate energy source is stored ATP. The muscle simply uses stored ATP as a quick and easy energy source. The problem is that the amount of stored ATP in a muscle is extremely low, enough to power contraction for only a second or two (Brooks & Fahey, 2004). The next immediate system is what is referred to as the Phosphagen system. It consists of the simple one step reaction of Creatine Phosphate (CP) and ADP which yields ATP and Creatine. Once again, the supply of CP in the muscle limits the use of this highly powerful system to only 5-6 seconds of work (Brooks & Fahey, 2004). Lastly, the myokinase system takes two ADPs and creates one ATP and one AMP. Even when all of these systems are combined, they can only provide energy for muscle contraction for between 5-15 seconds, far too short for any endurance event (Brooks & Fahey, 2004). Besides the short capacity of these systems, they also take a long time to recover. These immediate systems essentially act as borrowing systems and cannot be repaid unless intensity is low enough. For example, after total exhaustion of its resources the Phosphagen system requires the aerobic system to turn Creatine back into CP so that it can be used again. At rest this takes up to several minutes.

To supplement the immediate energy systems, Glycolysis, which is sometimes referred to as the anaerobic system, takes part of the work load. Glycolysis is a system that requires no oxygen and has intermediate speed, power, and capacity. Essentially, it is the middle distance runner of the energy systems. Glycolysis works by breakdown of glucose or stored carbohydrate, glycogen. The breakdown of glycogen requires an extra step and is called Glycogenolysis. Unlike the immediate energy systems, Glycolysis involves 12 sequential chemical reactions that take us from Glucose to Pyruvate. From here there is a fork in the road where pyruvate can either be converted to lactate or to acetyl-CoA. The conversion to acetyl-CoA allows for that substance to enter the mitochondria and be used by the aerobic energy system. Contrary to popular belief, the decision on which way the system goes at this point is not based on whether oxygen is present or not (Brooks & Fahey, 2004). Instead the quantity of enzymes that convert it to lactate or acetyl-CoA and whether or not there is

sufficient mitochondrial activity to handle pyruvate and other products produced by glycololysis (namely NADH) are the major determining factors on which way Glycolysis goes. Thus mitochondria, which will be discussed shortly, are important in determining the end route of Glycolysis. More on this process and the conversion to lactate will be covered in subsequent chapters.

One major drawback to Glycolysis, especially when it goes the lactate route, is that by-products are produced which can interfere with the energy systems, contraction itself, or even serve as a signaling mechanism to the brain that fatigue is immanent. While lactate itself does not cause fatigue, certain accompanying products, namely Hydrogen ions (H+), have been shown to contribute to fatigue. Lactate is in fact a fuel source that can be used aerobically. In the next chapter, the build up of these products will be discussed. While the build up of fatiguing products is one down side to Glycolysis, the amount of energy produced is another downside. With each cycle through Glycolysis, 2 total ATP are produced, which is far less than the amount produced aerobically. Glycolysis is thus an intermediate system that delivers a moderate amount of ATP fairly quickly, but with some negative consequences.

The last energy system is commonly referred to as the Aerobic system. You may notice that I am not terming it aerobic Glycolysis. This is because Glycolysis is a process by itself that does not require oxygen. You have to go through Glycolysis to get the necessary products to proceed with aerobic energy breakdown, but it is confusing to think of aerobic and anaerobic Glycolysis, because in reality there is one Glycolysis, the last step just differs. As mentioned previously, pyruvate is converted to acetyl-CoA, which then has to be transported into the mitochondria for use. The mitochondria is a different organelle and is commonly referred to as the power house of the muscle cell. Once inside the mitochondria, the acetyl-CoA enters what is called the Krebs cycle. The Krebs cycle is the first step of the aerobic system. It consists of a series of 10 chemical reactions that function to produce an ATP source and a series of products that can be used in the second step of the aerobic system. The important products are NADH and FADH. From here, these products enter the second step of the aerobic system, the electron transport chain. The Electron Transport Chain consists of a series of reactions that basically take the NADH (or FADH) and another H+ ion and react it with Oxygen, creating ATP, NAD, and water. While the process is more complex than this, the important thing to remember is that it is only this last step in which Oxygen is required. As can be seen, the aerobic system requires a large number of steps and transport of products to and from a different part of the muscle cell then where contraction is actively taking place. Due to these factors, the aerobic system produces a large amount of ATP (~30 in total) but is more time consuming. An advantage is that the subsequent products are kept to a minimum. Thus the aerobic system has a large fuel supply, causes little by-product fatigue, but takes longer to produce energy

While the above steps have dealt with using Glycolysis and thus carbohydrate to produce acetyl-CoA, fat and protein can also be broken down to acetyl-CoA which can enter the Krebs cycle

and produce aerobic energy. The problem with Fat is that it is an even more complex chemical process than carbohydrate breakdown, thus while the energy production is very high, it can only work at low to moderate intensities. Once a runner gets much beyond around half marathon pace, the fat usage is very minimal. With protein, there is no natural storage of protein like fat or carbohydrate. Thus when using protein as a fuel source, proteins that are meant for another use are being consumed. An example would be the breakdown and use of muscle protein.

Lastly, let's look at how the energy systems work. As already mentioned, the energy systems are not mutually exclusive but interact. To conceptualize how they interact, let's go through starting an exercise or a race. It's best to think of all the energy systems being started up at the very start of exercise, it just takes some longer to rev up and reach full capacity. The amount of contribution is dependent on the intensity, or energy required. The systems dynamically combine to provide the total amount of energy that is required. The exact contribution is not static but changes over time.

The immediate energy systems take up the slack during the initial few seconds providing the majority of the energy and tapers off to near zero contribution by 10-15sec, depending on the overall intensity. At lower or moderate intensities, the full immediate energy store is not used up as the energy requirement is low and Glycolysis quickly steps in. Glycolysis takes around 20-30sec to reach maximum capacity, while the Aerobic system takes around 90sec- 2min to reach maximum capacity (Duffield et al., 2005). In looking at a race, in terms of the dominant energy system, the cross-over point where aerobic energy is the majority supplied occurs at around 45sec (Hill, 1999). For this reason, if we look at relative energy system contribution for different races, anything over 400m uses the aerobic system to supply the majority of its energy. For competitive males a variety of studies have found that the energy contribution between aerobic and anaerobic systems is 60% aerobic and 40% anerobic for the 800m, 77%/23% for the 1,500m, and 86%/14% for the 3,000m races (Hill, 1999, Duffield et al., 2005).

The timing of the energy system's reaching capacity is not the only issue. The amount of energy required also plays a role. The aerobic system is limited in its total energy supply rate capacity. Therefore if the exercise is at an intensity that is higher than the maximum rate of energy production for the aerobic system, Glycolysis has to step in and cover the energy requirement gap that is present. This results in the ever accumulating by-products that can eventually lead to fatigue. For this reason, increasing the capacity of the aerobic system to produce energy is a beneficial training adaptation to delay fatigue. If the gap between aerobic energy production and needed supply can be shrunk, that means less Glycolytic energy is needed and less by-product accumulation.

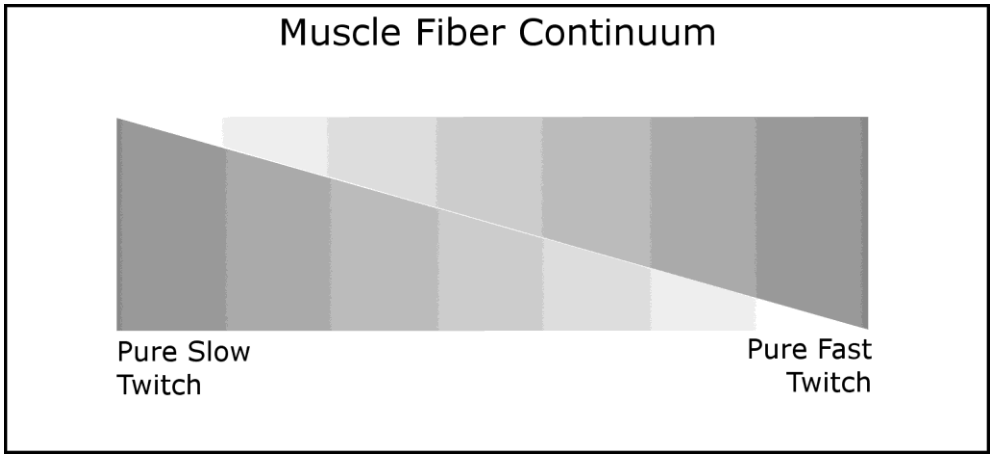
As you can see, contrary to popular belief, you do not get more "anaerobic" during a race, but actually rely increasingly on aerobic means. This is a confusing issue for some, as they've been taught that lactate accumulation from going anaerobic causes fatigue. While lactate does not cause fatigue, the other by-products that can cause fatigue increase in accumulation at shorter high

intensity races toward the end of the race. It isn't that they are being produced at a higher rate, it is just that the ability to use or clear these items is increasingly diminished. Additionally, some of these products cause a reduction in the ability to produce energy through Glycolysis, which as that system begins to "falter" and the aerobic system is maxed out, something has to give because the energy can not be supplied. The thing that gives is the pace, as a slowing of the pace decreases the energy demand. More on the actual causes of fatigue are discussed in the following chapter.

Muscle Fiber Types

Having gone from the brain down to the microscopic organelles in the muscles themselves, let's look at the different kinds of muscle fibers we have. Fiber types are generally classified as Slow Twitch (ST) or Fast Twitch (FT) with several subdivisions of FT fiber types depending on the classification method (FT-a,x,c are common classifications). The main method in classifying fibers is based on what type of a protein called myosin the fiber predominately has. The problem is that each muscle fiber type does not contain only one kind of myosin form, but instead most have a mixing of a variety fast and slow forms. Therefore it is not a distinct division in fiber types like most believe. As was stated in Brooks, Fahey, and Baldwin's seminal text on Physiology, "considering the number of possible combinations of MHCs (myosin heavy chain) and myosin light chains, there are innumerable fiber types (pg. 412)."

While scientists like to break things down into a nice distinct fiber type classification system, the reality is that fiber types are more like a spectrum. On one side of the spectrum we have what we'd call a pure ST fiber and on the opposite is the pure FT fiber. In between these two extremes is a range of fibers with different ratio's of FT/ST characteristics, and is where the majority of fibers fall. Where exactly a fiber falls depends on its individual characteristics which include mitochondria density, capillary density, oxidative and Glycolytic enzyme activity, creatine phosphate stores, and contraction velocity.



Along this continuum, with training fiber type can shift to either the aerobic (ST) or anaerobic (FT) side. With acute training, the shifts are very small, but with long term training a larger change

can occur. How large a change is up for debate. For a long time Scientists thought that we could not change fiber types at all. Once this was disproved it became that distinct conversion from FT to ST can't occur. The reality is that we most likely can change fiber types, but it can't be proven in humans because it is a long process. Several pieces of evidence lead to this conclusion.

First, animal studies on both rats and rabbits have demonstrated complete fiber type changes with chronic muscle stimulation. What they've found is that a large amount of damage is required for a complete transformation (Pette & Vrbova, 1999). For example, in rabbits a complete transformation from FT to ST occurs with chronic muscle stimulation only if the researcher goes in and creates lesions in the muscle fiber. In rats, the FT to ST conversion occurs, but takes a large amount of chronic stimulation causing considerable muscle damage. What these animal studies demonstrate is that a complete conversion probably can occur, it just takes a lot of damage, or in other words a long period of time of training.

Secondly, fiber type percentages have been altered in elite skiers when tested during their beginning stages of training and then 8 years later. A longitudinal study by Rusko et al. found that after 8 years of training and a doubling of training volume, the percentage of ST fibers in a group of Cross-Country Skiers increased by 11% (1992). Put these two together and it makes sense that a large amount of volume (or in other words, damage) is needed to change fiber types.

This could partially explain why runners tend to reach peak performance levels later than sprint or power type athletes. It could be because distance runners need large amounts of training, or damage, to complete this muscle fiber type conversion. Additionally, this fiber conversion could also partially explain the success of African distance runners. Perhaps their large volumes of unofficial training early on, such as running to school or even the active lifestyle that they live, allows for this conversion to take place at a younger age. Africans tend to reach peak performance at younger ages then their Western counterparts and this could partially explain that phenomenon.

A Recruitment Issue:

How muscle recruitment takes place is another important piece of the puzzle. As discussed earlier, each muscle has a number of motor units which contain a large amount of individual muscle fibers. These motor units are what gets signaled to contract and when this occurs all of the muscle fibers in the unit contract. An entire muscle will never activate all the motor units it contains. If this occurred, catastrophic damage to the bone and surrounding tissue could occur. Thus the brain always keeps some motor units in reserve, even during maximal contractions. This reserve can be thought of as the body's safety system. We have all heard stories of people lifting boulders or mom's lifting cars to save a child. It is believed that in these rare occasions full activation of the muscles occurs. It's likely that the brain does a kind of cost benefit analysis and decides that using the full reserve and risking muscle or tissue damage is better than the alternative, death. The interesting thing is that trained athletes can use a larger percentage of their maximum recruitable

fibers then the untrained. Essentially, the well trained person slowly convinces the body that it's okay if it activates a little more muscle. But how does the body decide how many and what type of motor units are used?

First, muscle recruitment is dependent not on speed or intensity, although they are related, but on force output required. The amount of force needed is what predominately determines muscle activation. If more force is required, the brain activates a larger amount of the muscle fiber pool. This explains why running slightly slower up a hill activates more muscle than running on the flat ground at a slightly faster pace. But it's just not about how much muscle is recruited but what type and in what way.

What type of fibers recruited has always been thought to work based on Henneman's size principle, which states that the easier to recruit fibers (ST) take up the work first, while the harder to recruit fibers are activated as force or duration increases. But this isn't always the case. There are instances when FT fibers are recruited before ST fibers, particularly in situations when a high amount of force is needed in a short amount of time, such as ballistics exercises or during sprinting. Similarly, the force rule of muscle fiber recruitment can be violated during prolonged activity when ST fibers fatigue (glycogen is depleted for example), and then the FT fibers are recruited to take up the slack, despite the relatively low intensity and force recruitment.

Finally, the way recruitment occurs is important. Muscle fiber, or more properly termed Motor Unit, recruitment for endurance events happens in a cyclical manner. That means within a whole muscle, some units are recruited to do the work and then they can cycle off and recover, while other units cycle on and take up the workload. At low intensities, the number of units active at the same time is very low, but as intensity increases more units need to be active at the same time to result in a great enough force output, thus less units are recovering or inactive. Additionally, as recruitment increases the "resting" time between both when a fiber is recruited and between individual contractions of that fiber decreases. Understanding the muscle fiber type spectrum and the ways in which muscles are recruited will be important in designing training to combat fatigue and individualize training.

Passive Mechanics:

Up until now, the focus has been on active mechanics, in that active muscle contraction initiated by the brain is supplying the force for movement. Traditionally this has been considered the main contributor to forward movement. The problem is that due to the short ground contact time and then the even shorter amount of time in which the body is in position to actual push and contribute to propulsion another energy source is needed. This is where passive mechanics come in. Passive mechanics can be separated into elastic energy storage, reflexes, and passive mechanical principles.

When a runner's foot collides with the ground, energy is absorbed from the impact. That energy can either be re-cycled and utilized or dissipate. Instead of wasting the energy, the body has a remarkable system that allows for us to reuse the energy in a similar way to a spring. In this case, the muscles and tendons function as the spring, compressing and storing energy when ground contact is made and subsequently rebounding and releasing it upon push off. This spring like mechanism provides a large amount of the energy required for forward propulsion. The key is putting the body in proper position to get the most elastic energy return with the least amount of energy dissipation. This is one reason why running mechanics should be optimally developed. This concept will be developed further in the Chapter on Efficiency.

Two similar concepts that rely on energy storage and return called the stretch reflex and the stretch shortening cycle (SSC) also help contribute to forward propulsion. The SSC basically refers to energy storage and release in the muscle them selves. It occurs when a muscle is actively stretched and then is immediately followed by a muscle contraction. During the pre-stretch portion, energy is stored in the elastic components of the muscle and then the energy is released during the contraction part. Essentially it is a spring like mechanism with storage and release of energy greater than if just contraction took place. The amount of elastic energy return is dependent on several factors including the length and speed of the stretch, the stiffness of the muscle, and the time between the stretch and the subsequent contraction (Saunders et al., 2004). The most obvious example is of the calf during hopping or running. Upon landing the calf is stretched and the quickly contracts upon push off.

The stretch reflex is a similar phenomenon that occurs because of sensory receptors in the muscle called Muscle Spindles. The spindles detect changes in length of a muscle. When a muscle is rapidly stretched, the spindle basically sends a signal for the muscle to contract. This is done to maintain a relatively constant muscle length. It's common to experience this phenomenon during a High School or College class, or when tired on an airplane. You may start to nod off slightly and your head dips forward, only for you to reflexively jerk it back up. In this case, the muscles on the back of the neck were rapidly stretched and thus the signal was sent to contract them, jerking the head back upwards and startling yourself and those around you.

One of the most overlooked contributions to human movement is how the lower legs function from a purely mechanical viewpoint. The leg can be thought of as multiple segments connected via simple joints. Taking the muscles out of it, these limb segments will function based on pure mechanical principles. This means that Physics and in particular Newton's laws of motions help dictate how the legs will move.

On this front, interesting research by Tad McGeer has demonstrated the passive dynamic properties of the lower legs. He built robots that were essentially just the lower leg segments of the body and without an engine can walk in a human like fashion as long as a single initial force is applied. Once the single initial force is applied, or if the robot is on a slight downhill, the machine

will walk in human like fashion without any assistance. These robots are a great demonstration of the passive dynamic properties of the lower limbs. Just like in human walking, due to the physics of the two joint system, the lower leg will fold up slightly on the recovery phase allowing for it to clear the ground and land in front of the body.

The translation to running should be obvious. We tend to think of the entire running cycle happening because of active mechanics, but the reality is simple mechanical principles dictate much of our movement. When walking for example, there is no need to contract the hamstring to a large degree to bring the lower leg up slightly towards the butt during the recovery phase. Similarly during running, if the push off and hip extension is large, the lower leg will fold up automatically and does not need to be pulled up with the hamstring. If the runner chooses to pull with his hamstring, then he's just wasting energy.

The combination of both passive and active dynamics helps creates movement. In looking at running performance or the biomechanics of running, it is easy to focus on only one particular part of the picture. When discussing how to improve performance the conversation inevitably focuses on active muscle contraction, without giving any regards to the contribution of passive dynamics to performance. Similarly, when looking at how a person should run, often the stretch reflex or passive dynamics are not considered. Keep this in mind as we delve further into what limits running performance. Now that we know how running happens, let's look at why we fatigue or tire during running.

Chapter 2: Fatigue: Friend or Foe?

What limits performance? That is the central question that a coach, athlete, and sports scientist all want to answer, and the answer is fatigue. Fatigue is a nebulous catch all term that basically means we slow down. While we have a good idea of what causes fatigue under a variety of isolated situations, the exact mechanisms behind fatigue in a dynamic activity such as running are unknown. The truth is that each individual likely has a different limiter based on their training and individual physiology.

Understanding where fatigue comes from and how it develops is crucial for the coach. The name of the game is limiting fatigue so that we can race longer or faster. By knowing how it occurs, a coach can plan training to adapt the body to resist fatigue.

How Fatigue manifests itself:

The obvious consequence of fatigue is that you slow down. Why do we slow? The simple reason is that it is a way for our body to protect us. We slow because we are well beyond homeostasis, that nice natural equilibrium the body likes, and if we go much more beyond that homeostasis then dangerous consequences could occur. Fatigue is thus a protective mechanism.

We don't need science to tell us that we slow down, so what actually happens? When broken down, running is nothing but a series of spring like hops. A certain amount of force needs to be imparted into the ground to propel the runner forward. The amount of force needed depends on the pace that the athlete is running. At a faster pace, more force is needed. This force comes about mainly through the muscles themselves generating force via muscle contraction. In simplistic terms a runner can keep going at the same pace as long as they can produce the force necessary on the ground. Once they cannot impart enough force, the pace has to slow as they are not able to cover the same amount of ground with the same stride rate. But what causes a drop in force? The drop in force occurs when muscles fatigue. What exactly causes this fatigue and subsequent drop in force?

First, understanding that total force output is a result of not only of individual muscles, but also of how many motor units (groups of muscle fibers) are recruited, and the way that the movement takes place. First, the most basic contributor to total force production is in the individual motor unit. A motor unit generates force via contraction. This occurs when a strong enough nerve signal is sent to the motor unit which tells all muscle fibers in that unit to contract. The process is complex and in depth but to quickly summarize a motor unit is first stimulated via a neural signal, that signal causes changes in Sodium and Potassium levels which trigger an action potential. As action potentials increase, they signal Calcium release which clears the way for contraction to occur. During the process of contraction itself, energy (ATP) is required (Brooks, 2004). Fatigue is believed

to occur along many steps of the process, although much attention is focused on supplying energy for the actual contraction to occur.

The last portion of muscle contraction is important when discussing fatigue. Many theorize that since the last step of muscle contraction is dependent on ATP supply, energy supply is what limits performance. Or in other words, in order to delay fatigue, the recycling of ATP must keep up with the demand of ATP by the muscles. If supply can not keep up with demand, then fatigue occurs. As you all know, we have several different energy systems to recycle ATP. This is where the energetic theory of fatigue comes into play. If we cannot regenerate ATP at a sufficient rate, fatigue occurs. The importance of the energy systems derived from these ideas.

While single muscle fiber contraction is how force is developed in isolation, the reality is that total force develops through the integration of many different motor units being active. Thus, the second factor that determines force is how many motor units are recruited to do the work. As previously mentioned, the Central Nervous System (CNS) sends the signal to the muscle to initiate recruitment. Thus, the CNS is in charge of deciding how many motor units need to be recruited. The CNS can regulate exact force production in several ways. First, the type of motor units recruited and their individual characteristics plays a role. Muscle fiber types are typically broken down into several distinct types, but it is best to think of them as a spectrum ranging from pure Fast Twitch (FT) to pure Slow Twitch (ST). The more FT a fiber is the higher force production from that fiber, but the lower fatigue resistance. For this reason, ST fibers are generally initially recruited while FT fibers are reserved for later recruitment or very high force requirement activities like sprinting.

Secondly, the total amount of motor units recruited influences force production. A greater amount of motor units recruited means more muscle fibers able to do the work. Obviously whether they are FT or ST fibers also plays a role, but in general the more motor units recruited the more force that can be developed. Lastly, the way in which muscles are recruited plays a role. Muscle recruitment can either happen synchronistically or asynchronistically. For endurance events, recruitment general happens in an asynchronous fashion in which we rotate the work among motor units as some contract while others rest. Once the working units become fatigued, the resting units take over the workload and let the fatigued ones rest (Maglischo, 2003). In this way, force output is kept constant. While these aren't the only ways the body regulates force output, for fatigue these are the main ones to consider.

One other factor plays a role in how much force output occurs and it has to do with the biomechanics of the movement. There are certain biomechanical phenomenons which aid in force output. The use of elastic energy is chief among them. The most well known is that of the Stretch Shortening Cycle (SSC). The SSC occurs when a muscle is initially stretched and then immediately followed by a subsequent contraction. When this occurs, the force output is much higher than if the muscle had simply contracted from a resting position. It essentially acts in the same way that a

rubber band would. Not only do the muscles use elastic energy, but the tendons do too. Another form of elastic energy storage and return occurs when ground contact is made. Upon ground contact, a large amount of energy is transmitted up the body that can be stored and used subsequently to push off from the ground. In particular, the tendons are useful in this respect. The two main contributors are the Achilles tendon and the arch of the foot (Ker et al., 1987). While the details of these will be discussed later, it's important to understand that elastic energy storage and return plays a large role in total force output. Additionally, this system can experience fatigue. Muscle or tendon damage are one way that the storage and use of elastic energy can decrease. Now that we understand the basics of force output, it's time to move onto fatigue.

How Fatigue occurs:

There are three basic ways that fatigue is believed to occur. The first is what I call the by-product buildup, the second is depletion, and the third is regulation, which includes the Central governor theory.

The By-Product Buildup:

The By-Product buildup idea states that it is the build up of certain by products or substances that cause fatigue at several different levels. The most common, although wrong, example of this theory is the build up of lactic acid. In essence, the build up of certain products creates fatigue by impairing force output at any number of the different steps to get from muscle recruitment to contraction. Whether these products directly or indirectly cause fatigue is another subject that will be covered shortly.

There are many different products that can potentially cause fatigue. Many of the earlier ideas centered around the energetic model, in which supplying ATP to the muscle for contraction is the main component. As is well known, the three basic energy systems are the immediate (Phosphagen and Myokinase), Glycolysis, and Aerobic system. Each system requires a series of chemical reactions that ultimately result in ATP reformation. The belief is that the energy systems, mainly the Glycolytic one, create by-products which inhibit subsequent energy production. As the use of these systems increased, by-products that cause fatigue increased to a degree that they interfered with total energy production from all the systems. In the energetic model, the enzymes that catalyze the numerous chemical reactions become less active, thus directly slowing energy production.

Looking at it from a contraction standpoint, these by product increases can interfere with contraction via methods outside of energy production. Any number of the steps to get from sending the signal to actual muscle contraction can be impaired. The accumulation of such products as ammonia and potassium are prime examples, as these alter muscle fiber excitability. The impairment of muscle contraction can occur at any number of different sites, ranging from

impairing the action potential to changes in Calcium release or uptake which can delay contraction and relaxation (Hargreaves & Spriett, 2006).

While lactate was initially seen as the culprit, hydrogen ions (H+) and the corresponding drop in pH are more compelling examples. As pH drops, the rate of ATP replenishment drops due to a reduction of PFK and ATPase activity, as well as an increase in the amount of Calcium needed during muscle contraction (Maglischo, 2003). This is just one example, and while it is beyond the scope of this review to give every product and site of fatigue, some include increased levels of: H+, ammonia, Potassium, Phosphate, Calcium and ADP (Hargreaves & Spriett, 2006).

Depletion

The idea of depletion is basically the exact opposite of the by-product buildup idea. Instead of the accumulation of products that cause fatigue, it is the depletion of products that lead to fatigue. It's best to think of this theory in terms of fuel sources. Whenever a vital fuel source is running low, then fatigue is going to occur because the runner will need to slow down and switch fuel sources to make sure that total depletion does not occur.

Two prime examples of this are on the opposite ends of the energetic spectrum. The first is with the immediate energy systems. As we run out of stored ATP, we have to switch to get another energy system revved up and going. If that didn't occur, then we'd hit some sort of catastrophic failure. The other example is in that of glycogen use. There is only a limited amount of glycogen stored in the body. When that fuel source starts to run low, the body has to switch to less productive fuel sources, such as fat, to keep going. This occurs during the marathon or longer races. Thus the depletion of glycogen causes fatigue. Other examples include a decrease in blood glucose or Branch Chain Amino Acids (Hargreaves & Spriett, 2006).

The reality is that fatigue is a dynamic process and does not occur in isolation. Thus it depends on the individual and the race which they are running. It is likely that it is a combination of products that build up and other products that deplete which cause fatigue.

Regulation:

The last idea on fatigue is an integrated model. The idea is espoused by Tim Noakes in his Central Governor model and in other dynamic models such as Samuel Marcora's. The central idea of such models is that fatigue is not directly caused by any such build up or depletion of certain products. Instead those products serve as feedback for either a conscious or subconscious controlling mechanism.

Whatever the controlling mechanism is, the point is that exercise is not limited but rather regulated. The body uses the changes in homeostasis of the various products to regulate fatigue. Depending on the model it does this through regulation of muscle force output. Finally tying it back to the force output mechanisms, remember back to the various methods of regulating force

output. Noakes' CGM states that the body uses the feedback it receives to regulate performance via controlling force output through the various methods already mentioned, chiefly muscle recruitment. If the by-products build up too quickly, muscle recruitment is decreased, and thus the runner slows. In Noakes' model, this occurs via a subconscious regulatory system. More on Noakes's model will be discussed in the chapter on VO2max.

For the conscious controlling mechanism, Marcora refers to pain perception and the level of motivation. The increase or decrease in products causes changes in pain perception. As the pain increases, runners consciously control their speed, and thus force output. The body in essence is creating ever increasing levels of pain to protect itself. At some point, pain levels will increase to such a high level that the athlete will be forced to give in and slow. The degree and when this occurs depends on the level of each runner. This model will be covered more in depth later.

While regulation theories are intriguing and gaining more ground in the scientific community, in terms of practical application to training, it changes little. It does not matter if the build up of a certain by-product or pain directly causes fatigue or indirectly causes it through regulation, the training goal is still to delay that build up.

Oxygen's role:

This basic understanding on how and why fatigue develops is critical in understanding the development of the scientific models of endurance performance, and in figuring out what limits performance. Using the concepts expressed above, to prevent fatigue we must reduce the rate of by product build up or fuel source depletion, or alternatively increase the level of build up or depletion that we can withstand before we consciously or subconsciously start reducing performance.

Exercise science has been based on the energetic model which states that it is energy supply that is important in delaying fatigue, due to the fact that contraction is dependent on having energy. Having explained the energy systems in Chapter 1, energy is recycled via three main energy systems, the immediate systems, Glycolysis, and the aerobic system. The immediate systems consist of the phosphagen and the myokinase systems which are simple reactions. They provide quick energy, but are limited by their fuel supply and by the build up of by-products which slow recycling of energy. Glycolysis refers to the breakdown of glucose to pyruvate, which either turns into Acetyl-CoA or lactate. Glycolysis involves many more reactions but is still relatively quick and powerful. It's limited by both supply and by by-product build up. Lastly, we have the aerobic or oxidative system which can be an extension of Glycolysis. This is because it can take the Acetyl-CoA that Glycolysis can produce and runs it through many reactions in the Krebs cycle, which then allows for ATP production during the Electron Transport Chain when oxygen is combined. Additionally, the aerobic system can use other fuel sources such as various fats to produce energy. The aerobic system takes longer to rev up and has a lower rate of energy output. Lastly, it is limited by supply, and does not produce by-products that cause fatigue at any significant level.

One key concept to understand with energy systems is that they don't work independently. As soon as hard exercise starts they are all on, it just takes longer for some to rev up to full capacity and take their workload. Thus the immediate systems and then Glycolysis carry the early road during the start of the race until the aerobic system gets revved up in distance races. Research has demonstrated that this crossover point where the aerobic system becomes predominant is usually around 90 seconds (Spencer & Gastin, 2001). Thus, contrary to what popular literature states, in middle and distance events, we get more energy aerobically as the race progresses. Due to the predominance of the aerobic system in middle and long distance events and the fact that it does not build up by-products that can lead to fatigue at the same rate as the other systems, oxygen has been given central importance in running performance. Therefore the supply and utilization of oxygen so that the aerobic system can recycle energy is crucial. This leads to the next chapter that focuses on oxygen transport and utilization.

From here on out, all of the processes that can inhibit any of the steps involved in preventing fatigue by minimizing the rate of depletion or build up of products and modifying neural regulation will be looked at. We'll start with the traditional factors focused on the energy supply model, and branch out to neural factors. As mentioned, we'll start with an analysis of oxygen delivery and use in the VO2max chapter. From there, we'll look at how the body deals with lactate, not because lactate causes fatigue itself, but rather because lactate can be used as a fuel source and correlates with the buildup of fatigue causing by-products. To conclude the traditional trio, Running Economy and the idea of it not only being the total supply of energy that is paramount but the rate of energy production that matters will be discussed. Finally, we'll venture away from the traditional ideas and explore some of the other aspects of fatigue and what might cause them, including the nervous system, fuel use, and psychological factors. The goal will be to look at every step along the way to see what the weak links are that may limit performance.

Chapter 3: An Oxygen problem?

Ask almost anyone what limits running performance and the inevitable answer is oxygen. That may suffice for the general public, but does that vague answer mean breathing in oxygen, transporting it, or utilizing it in the muscles? In the first two chapters, oxygen's role in energy production and fatigue were discussed. Understanding oxygen's role, lets first look at how we measure oxygen's impact and then at each step from intake in the lungs to utilization in the muscles.

As a coach, oxygen's importance is probably obvious, but it's important to delve deeper and realize that oxygen goes through numerous steps from intake down to utilization. Understanding this process and where the potential road blocks lie will allow you to better train your athletes. Workouts can be designed to address the specific limitations for each individual athlete, as it is likely that some will have problems with oxygen utilization while others might have an intake limitation.

The Measurement: VO2Max

"What's your VO2max?" For any endurance athlete this common measurement seems to define your talent or potential. VO2max refers to maximum oxygen consumption and is one of the most commonly measured components when tracking changes in endurance performance. We've all heard of the stories of Lance Armstrong's or Bjørn Dæhlie's incredibly high VO2max measurements, but should it really garner that much attention? In short, no. Contrary to popular belief, VO2max is simply a measurment and does not define fitness or potential. In fact, among well trained runners, it is impossible to discern who is the fastest by VO2max. That does not mean that oxygen transport and utilization is not important, it simply means that the measurement of VO2max does not accurately reflect these processes. Before dispelling some of the VO2max myths, lets look at the contributing factors of VO2max.

As discussed in the previous chapter, oxygen is used to create energy aerobically. Aerobic energy has fewer negative side effects, thus for endurance performance one of the goals is to increase our ability to work aerobically. VO2max refers to the maximum amount of oxygen used, and is calculated by taking the amount of oxygen taken in and subtracting the oxygen exhaled out (Bassett & Howley, 2000). The measurement of VO2max is used to quantify the capacity of the aerobic system. It is potentially influenced by a variety of factors as Oxygen makes its way from the environment all the way to the mitochondria in the muscles. Vo2max is calculated by using the Fick equation, where Q equals Cardiac Output, CaO2 equals arterial oxygen content, and CvO2 means venous oxygen content:

$$VO2max = Q\ (CaO2 - CvO2)$$

Cardiac Output is a measure of Stroke Volume (the volume of blood pumped with each beat) and heart rate. Arterial oxygen content is the amount of oxygen in the blood that is being delivered to the muscles, while venous oxygen content refers to the amount of oxygen in the blood that is returning to the heart and lungs. When the arterial and venous oxygen content are subtracted, the amount of oxygen being taken up by the muscles is known. As demonstrated by the Fick equation, Vo2max is influenced by a variety of factors.

While the measurement of VO2max is of little practical interest for reasons to be discussed in the subsequent chapter, maximizing the body's ability to get and then use oxygen in the muscles is of the utmost importance in running performance. The components affecting and limiting this process can be divided into oxygen intake, transport, and utilization. Oxygen's route from the air all the way to the muscles is largely driven by pressure gradient differentials, meaning that oxygen likes to travel from an area of high concentration to one of low concentration. The major steps include:

- Oxygen intake
  - Air intake to lungs
  - To bronchioli and aveoli where it diffuses to capillaries (blood)
- Oxygen Transport
  - Cardiac output-pumps blood throughout
  - Hemoglobin concentration
  - Blood volume/shunting
  - Capillaries to diffuse the Oxygen into muscles
- Oxygen Utilization
  - Transport to mitochondria
  - Use in Aerobic respiration and Electron Transport Chain

Oxygen intake

The first step along oxygen's journey is obviously getting it into the body and ultimately the blood stream, which occurs mostly due to the operation of the respiratory system. Air is taken in through the mouth or nose due to subconscious control that relies on the pressure differential of the outside air and that in the lungs. From here, the air proceeds down the pharynx and into the lungs. In the lungs, the air travels first through the bronchi and then into smaller tubes called the bronchioli.

At the end of the bronchioli tubes in the lungs are Alveoli. This is the site where oxygen transport from the lungs to the blood occurs. Oxygen transport from the alveoli to the blood occurs with diffusion via pulmonary capillaries. The capillaries are very small blood vessels that allow for exchange to larger blood vessels via diffusion. The amount of Oxygen diffused depends on both the pressure difference between the alveoli and the pulmonary capillaries and the total amount of pulmonary capillaries. The amount of capillaries plays a role, especially in well trained athletes,

because it allows for a longer period of time in which inflowing blood is in contact, meaning that there is a longer time for oxygen to diffuse into the blood.

Even at high intensities, the oxygen saturation in the blood is normally above 95% (Powers et al., 1989). This has been used as evidence that oxygen intake and transport from the lungs to the blood is not a limiting factor since saturation is near full. However, in some well trained athletes, a phenomenon known as Exercise Induced Arterial Hypoxemia (EIAH) occurs. EIAH causes oxygen saturation levels to drop by as much as 15% below resting levels during heavy exercise. This greatly affects VO2max, as each 1% drop in oxygen saturation below 95%, decreases VO2max by 1-2% (Dempsey & Wagner, 1999). EIAH occurs because the large Cardiac Output of well trained individuals causes the blood to move through the pulmonary capillaries so quickly that there is not enough time for full diffusion of oxygen and thus saturation to occur. Therefore, in some highly trained athletes, oxygen intake and diffusion can be a limiter to VO2max.

Do it yourself Physiology:

Is your limiter in your lungs?

A simple way to find out if you have Exercise Induced Arterial Hypoxemia is to buy a pulse oximiter. These devices simply clip on your finger and take the oxygen saturation in the blood in a noninvasive way. Most models give a pulse reading and an oxygen saturation level. These devices can be found for under $50. To find if you have EIAH use one of these devices during a set of progressively faster repeats. A good test would be to do 1,000m repeats with short rest at progressively faster speeds. In between each repeat clip the device on your finger and record an oxygen saturation level. If the level drops by 5 or more percent from the resting value of 98%, you likely have EIAH and may be limited by the diffusion of oxygen from the lungs to the blood.

EIAH is an overlooked phenomenon and could have implications in elite endurance performance. As researchers, we often look for one main limiter, however the limiter of a certain system or pathway changes based on the individuals physiological makeup and training. Due to the enormous training loads of elite endurance athletes which greatly increases oxygen transport, it is likely that there is a shift in the limiter from more of a transport issue to an intake and diffusion issue.

Diffusion is an issue that is dependent on both oxygen intake and on cardiac output (Wagner, 1996). The degree to which oxygen is saturated is often thought of only in terms of the pressure gradient between the air and the lungs and then the lungs and the blood. However, while the pressure gradient and the attraction, or binding affinity, of Oxygen to Hemoglobin play roles, the Cardiac Output of the heart also contributes. As already mentioned, with a larger Cardiac Output blood moves through the area where oxygen exchange occurs quicker, resulting in less time available for loading and unloading. This can be partially offset due to an increase in pulmonary capillaries. The larger cross section of the capillaries allows for a longer amount of time for oxygen saturation to occur. If you find that an athlete has reduced oxygen saturation levels, or EIAH, it

might be beneficial to try and enhance pulmonary capillarization through various exercises designed to stress the pulmonary system.

Additionally, respiration could interfere with oxygen saturation. High rates of ventilation can result in a mixing of the gases in the lungs resulting in a ventilation-perfusion inequality (Wagner, 1996). In his review on VO2max, Wagner notes that in the majority of normal subjects there is evidence of some sort of pulmonary dysfunction (1996). The results of which can be seen in the increase in alveolar-arterial PO2 difference with increasing exercise intensity.

The muscles that aid in respiration also play a role during exercise. The oxygen cost of respiration contributes significantly to VO2max. At moderate intensities, respiration accounts for 3-5 % of VO2, while at heavy intensities it accounts for ~10% of VO2max in untrained individuals (Aaron et al., 1992). However, in well trained athletes, it was found that respiration accounted for 15-16% of VO2max during intense exercise (Harms et al., 1998). These findings that respiration has a higher oxygen cost in well trained athletes lends credence to the idea that demands and limits in trained versus untrained athletes are different.

Another potential reason that respiration could limit performance is that the respiratory muscles compete for blood flow with skeletal muscle. Due to this competition, diaphragm fatigue can occur at intensities greater than 80% VO2max (Johnson et al., 1993). In a study by Harms et al. they investigated this possibility by artificially increasing and decreasing the respiratory muscle load. They found that endurance was significantly increased by 15% in the decreased respiratory muscle load trial, while endurance was decreased by 14% when increased respiratory muscle resistance was applied (2000). While such studies demonstrate in theory that respiratory fatigue occurs, what happens in the real world? A study by Romer et al. provides an answer as it found that global inspiratory muscle fatigue occurred during 20 and 40km cycling time trials (2002).

The question then becomes if the respiratory muscles can be trained and if such training can lead to performance improvements. The study by Romer set out to evaluate this as well, finding that using inspiratory muscle training improved performance over the 20km and 40km trials by 3.8% and 4.6% when compared to a control group, and resulted in a reduction of respiratory muscle fatigue post trial. Other studies investigating respiratory training have been mixed. Several studies have seen increases in performance in cyclists and rowers (Volianitis et al., 2001; McConnell & Sharpe, 2005). On the other hand, several authors have found no change in performance with training (Williams et al., 2002; Inbar et al., 2000). In a review on the subject, Sheel found that out of 8 studies done, six showed improvements in performance following respiratory training (2002). The contrasting results are likely due to whether oxygen intake and respiration were the participant's main limiter. As we have seen, the degree to which the respiratory muscles contribute to VO2max varies based on training level. For higher level runners, it is likely that respiratory fatigue or EIAH occurs due to changes discussed previously. For this reason, higher level runners should consider respiratory training, while lower level runners probably will not see the same degree of benefit.

Applying it your training:

Several new devices can be used to train the respiratory muscles. The cheapest way is to use pursed lip breathing focusing on breathing using the entire diaphragm. Start with long inhalations and exhalations, gradually increasing the speed of the pursed lip breathing until it is very rapid. Throughout be sure to use the entire diaphragm as most people neglect the lower half.

Another option is the powerlung which is a device that you breathe through that adds resistance on inhalation and exhalation. This device costs around $100. A third, more expensive device, costing around $900, is the SpiroTiger respiratory trainer. It is backed by several research studies and trains the respiratory muscles in a slightly different way. Essentially it uses more of an endurance training model.

Oxygen transport

Since A.V. Hill's first experiments measuring VO2max, oxygen transport has always been considered the major limiter of VO2max (Bassett & Howley, 2000). Oxygen transport refers to transporting the oxygen from when it enters the blood stream all the way to the muscles that will take it up and use it. Di Prampero calculated that oxygen transport accounted for 70-75% of the limitation of VO2max (2003). The rate of oxygen delivery is dependent on several factors.

Heart adaptations

The heart's Cardiac Output (CO) refers to the amount of blood that is pumped out of the heart each minute and is usually regarded as the major limiter of VO2max. CO is dependent on two factors, as it is calculated by multiplying Heart Rate (HR) and Stroke Volume (SV). Thus to increase maximum CO, one of these factors would have to be modified. Maximal HR is a factor that does not change due to training or even lowers slightly, while sub maximal HR is lowered with training (Brooks et al., 2004; Levine, 2008). However, with endurance training, SV increases at rest and all intensities.

The increase in SV is primarily due to an increase in heart size and contractility (Brooks et al., 2004). These changes to the heart cause an improvement its ability to rapidly fill and an increase the end-diastolic Volume (EDV), which is the amount of blood present at the end of filling. According to the Frank Starling mechanism, the greater the stretch on the heart (or EDV), the greater the subsequent contraction is. Think of it as a rubber band like effect. This means that an increase in EDV, which would create a greater pre-stretch, would increase the subsequent ejection, or SV. Increasing EDV thus plays a central role in increasing SV. In addition, endurance athletes have an increased ability to rapidly fill the heart at high intensities, which is important as at higher intensities there is less time between heart beats for the heart to fill (Levine, 2008). Further supporting this idea, work by Levine et al. showed that in endurance athletes, their increased SV was almost entirely a result of EDV increases due to enhanced compliance of the heart (1991).

Traditionally, the increase of CO during progressive maximum exercise has been seen to occur with an initial rise in SV which then plateaus, while HR continuously increases to maximum. However, recent evidence points to some elite endurance athletes having a continual rise in SV and not exhibiting a plateau (Rowland, 2009). The data is conflicting on whether this phenomenon occurs and if it occurs the reasons for it. Rowland suggests that if it does occur, the likely mechanism is an increase in End Diastolic filling, which when accompanied by an enhanced Frank Starling mechanism, would result in an increasing SV (2009). This increasing CO throughout exercise is an interesting phenomenon and could point, once again, to the individual nature of VO2max limitation.

One other mechanism that increases end diastolic filling is an increase in blood volume. This was first demonstrated in animal models (Wagner, 1996). A study done by Krip et al. manipulated blood volume in endurance trained and untrained individuals and studied its effect on cardiac function (1997). They found that blood volume increases and decreases caused significant alterations in maximal diastolic filling rate, SV, and CO. These results demonstrate that increases in blood volume, which is an effect of endurance training, improve SV and CO via enhanced diastolic filling.

As mentioned previously, a high CO can have negative consequences in regards to oxygen delivery. With a high output, the diffusion time for oxygen in the lungs and muscles is greatly reduced because of how fast the blood moves through the oxygen exchange zone. The body must do a balancing act to elicit optimal oxygen delivery. This balancing points to a limit of effective Cardiac Output increase if a subsequent increase in diffusion capacity does not occur. This can be seen at extreme altitudes where in theory, despite increases in Cardiac Output, oxygen uptake would not increase due to diffusion limitations (Wagner, 1996).

Future Advances in measuring Cardiac Output:

Traditionally, measuring CO is hard to and not practical during intense exercise. However, a new device called the PhysioFlow has shown promise. It's a device that has EKG leads coming from a small battery pack that can be strapped around the waist. This allows for Cardiac Output, Stroke Volume, and other hemodynamic parameters to be calculated while running.

Hemoglobin

Another major factor in oxygen transport is the oxygen carrying capacity of the blood itself. This is dependent on the Red Blood Cell (RBC) mass and the Hemoglobin (Hb) concentration, which serve as the major carrier of oxygen in the blood. Red blood cells serve as the blood cells that carry oxygen, while hemoglobin is an iron containing protein within red blood cells that binds with oxygen to transport it to the muscles. The degree to which oxygen binds to hemoglobin is dependent on the partial pressure of Oxygen in the blood and the affinity between oxygen and Hb.

The affinity of Hb and Oxygen is how strongly attracted to each other each component is and is affected by temperature, pH, hydrogen ion concentration, and carbon dioxide concentration. Hb affinity is once again a situation of balance. While having a high affinity would seem like a positive adaptation as it would allow more Oxygen to bind to Hb, it would also make it more difficult for the Oxygen to detach from the Hb when going from the blood to the muscle.

An increase in Hb would improve performance via allowing for an increased transport of oxygen to the muscles. Research has demonstrated this relationship via seeing how reductions in Hb affect performance (Calbet et al., 2006). For example, a reduction in Hb because of anemia has been shown to decrease VO2max (Lamanca & Haymes, 1993). Interestingly, in a series of studies by Ekbolm et al., after reducing Hemoglobin levels via withdrawal of blood and seeing the initial drop in Hb, VO2max, and endurance, after 2 weeks VO2max returned to normal despite Hb and endurance performance being still reduced, (1972). This may point to an adjustment made when blood withdrawal occurs that acutely compensates for the drop in Hb that restores VO2max. The fact that VO2max could be restored while still having low Hb raises some interesting questions and shows the remarkable adaptation potential of the body and reminds us that there are many different ways to optimize oxygen delivery to reach VO2max. In addition, the return of VO2max but not performance, points to the idea that VO2max and endurance performance can not be considered synonymous.

On the other end of the spectrum, in studies artificially increasing Hb levels, VO2max and performance have been shown to increase (Calbet et al., 2006). In one particular study by Buick et al., eleven elite distance runners showed significant improvements in time to exhaustion and VO2max following a blood transfusion that increased Hb levels from 15.7 to 16.7 g . 100 ml-1 (1980). In a study on blood doping, which artificially increases Hb levels, improvements in VO2max have been between 4% and 9% (Gledhill 1982). When combined, these results show the impact that Hb levels have on VO2max.

Common Misconception with Blood tests:

When you get a regular blood test, Hemoglobin and Hematocrit (the ratio of RBC to the rest of the blood) are the ones endurance athletes obsess over. The problem is that these tests give Hemoglobin per 100 milliliters and do not represent total Hemoglobin levels.

With endurance training, Hemoglobin increases, but plasma volume usually increases to a greater extent. Thus, trained runners will usually see Hemoglobin and hematocrit levels drop. In fact, elite Kenyan runners have had levels far below normal of both of these parameters. What this means, is that you should not obsess over these two parameters as they mean little without knowing changes in plasma volume.

Blood volume and blood shunting

With an increase in the hemoglobin, the blood itself becomes more viscous, or thicker, as more of the blood contains red blood cells as opposed to plasma. The hematocrit is a measure of the

ratio between red blood cells and plasma. With this increase in viscosity, the rate of blood flow is slowed, thus decreasing oxygen and nutrient delivery to the muscles. Therefore, too high a hematocrit could potentially decrease performance via a slowing of the delivery rate. As can be seen, if is once again a balancing act between the rate of blood flow and the carrying capacity of the blood on what best delivers the most oxygen and nutrients to the muscles.

With endurance training, blood volume is normally increased along with hematocrit and Hb. Blood volume can be increased by as much as 10% with training (Costill & Trappe, 2002). The body seems to self regulate in creating an optimal hematocrit to allow for increased carrying capacity of the blood while also having adequate blood viscosity. However, it is not clear whether a high hematocrit with higher viscosity or a lower hematocrit with lower viscosity is better for endurance performance. Athletes using illegal drugs, such as Erythropoietin (EPO), to artificially boost red blood cell production have performed well with dangerously high hematocrits (Calbet et al., 2006). On the other hand, several world class African runners perform well with very low hematocrit and Hb levels that would normally suggest anemia (personal communication, R. Canova, Oct. 2007). It is likely that this is because of their large blood volumes from endurance training and it is possible that this reflects an alternative way of adaptation to altitude which will be discussed below. The question on which delivers oxygen the best and improves performance the most is unanswered.

It is possible that the optimal way to deliver oxygen in terms of blood Hb and Hematocrit varies based on the individual. In looking at the way particular groups of people adapt to altitude, it is not a stretch to see a similar variation in whether groups respond best to higher Hb/Hematocrit or lower Hematocrit/higher plasma volume. There is an Ethiopian specific pattern of adaptation to altitude to go along with the previously found Andean and Tibetan patterns (Beall et al., 2002). The Andean response consists of erythrocytosis (RBC increase) with arterial hypoxemia (reduced oxygen saturation). The Tibetan pattern shows normal Hb concentration with arterial hypoxemia and the Ethiopian pattern consists of maintenance of Hb concentrations and oxygen saturation levels. The maintenance of oxygen saturation levels at high altitudes points to some improvement in oxygen diffusion ability from the lungs to the blood or increased Oxygen-Hb affinity. The exact mechanisms are unknown but the fact that several different patterns of adaptation have been found at altitude points to the idea that different adaptation mechanisms could occur in optimizing the blood parameters mentioned above.

One other consideration is that when exercising the body reroutes a larger portion of the blood volume to working muscles and essential organs. This is called blood shunting and allows for more blood and oxygen to be available for the working muscles. At rest only 15-20% of the total blood volume goes to the muscles, while 85-90% goes to the muscles during exercise (Maglischo, 2003). This process occurs because of dilation and constriction of the arteries, with dilation opening up the arteries thereby reducing restriction of blood flow to the working muscles.

Capillary density

Lastly, oxygen in the blood stream has to get into the muscles. This process is aided by capillaries, which are the body's smallest blood vessels. In this case, they surround muscle fibers providing a place where substances can diffuse from the capillaries into the muscles or vice versa. This diffusion is how oxygen gets into the muscle fibers, and lactate and carbon dioxide are transported from the fibers to the capillaries. As mentioned already in regards to oxygen saturation in the blood, a larger density of capillaries allows for a longer amount of time for oxygen diffusion to occur. Capillary density has been shown to increase with endurance training, particularly intensive endurance training (Billat, 2003). In addition, capillary density has previously correlated with VO2max (Basset & Howley, 2000).

Oxygen utilization

Once oxygen is delivered to the muscles it then must be utilized. Oxygen utilization takes place in the powerhouse of the cell, the mitochondria. It is here that oxygen gets used to produce energy. The arterial-venous oxygen (a-vO2) difference tells the difference between the oxygen in the arterioles and the content in the veins. This allows for quantifying the oxygen utilization by the muscles. Oxygen utilization by the muscles is generally not considered to be a major limiter of VO2max. This conclusion has been reached from two main lines of reasoning. First, peak a-vO2 differences in elites and non elites are not very large (Hagberg et al., 1985). Secondly, in looking at the a-v O2 difference, it can be seen that there is not much oxygen leftover in the veins. Arterial oxygen content is around 200 mL of O2L-1, while venous oxygen content is approximately 20-30 mL of O2/L (Bassett, 2000).

In simply looking at the numbers listed above, common sense would have us ask why there is even 20-30mL of O2 leftover. While it may not be the main governor of Vo2max, a-vO2 is a factor that can be improved with training. While contributing much less to VO2max, it is important to remember that even a fraction of a percent improvement is crucial in the world of elite athletics. In several studies, a-vO2 difference has been shown to increase with long term endurance training by as much as 11% (Bassett & Howley, 2000; Saltin, 1973; Wilmore et al., 2001). Recognizing this fact, it appears that while not a major limiter to VO2max, it does play a role and therefore should be considered a relevant training adaptation.

In addition, the question still remains as to why there is not full extraction of oxygen by the muscle. Venous Hb concentration has been reported as ~15% in normal subjects (Wagner, 1996). There are several possibilities for limitations in oxygen transport from the blood to the site that uses it in the muscle, the mitochondria. These include the capillary density, which would allow for greater contact time and diffusion, the diffusion distance, and myoglobin concentration, which serves to transport oxygen from the cell into the mitochondria. In his review on the subject, Wagner concludes that distance to the mitochondria is not a major factor because of the high intracellular

Oxygen conductance (1996). Animal studies, for example one by Bebout et al. using dogs, suggest that muscle capillary density is the main structural factor affecting muscle Oxygen conductance (1993).

Future advances in measuring muscle tissue saturation:

Recently a way to measure muscle tissue saturation that is both portable and non-invasive has been developed. The Portamon Near Infrared Spectroscopy (NIRS) is a small device which you strap next to the muscle you want data from. It uses NIRS to measure tissue saturation, oxy-hemoglobin, deoxy-hemoglobin and total hemoglobin. The system transmits data wirelessly so it allows for data on what is going on in that muscle while exercising with zero invasiveness. The applications for such technology are almost endless, both from a sports science and coaching perspective. It's one step closer to knowing exactly what is going on in the muscle during exercise.

Oxygen's final destination in the muscle cells are the mitochondria. The Mitochondria is the site in the muscle cell in which aerobic energy generation takes place. To be transported across the muscle cell to the mitochondria, myoglobin is required. Myoglobin transports oxygen from the cell membrane of a muscle fiber to the mitochondria. Greater myoglobin concentrations would allow for more oxygen transport to the mitochondria, potentially enhancing oxygen delivery and thus performance.

Oxygen is used in the mitochondria during the Electron Transport Chain. Therefore the amount of mitochondria plays a large role in aerobic energy generation. In theory, the more mitochondria, the more oxygen utilization and extraction that can occur in the muscle.

However, many studies have shown that while mitochondria enzyme's increase significantly with training, the corresponding change in VO2max is much less. Mitochondrial enzymes function to aid the chemical reactions needed to eventually generate energy. In one study monitoring changes with training and detraining, mitochondrial capacity increased by 30% with training, while VO2max increased by only 19%. However, VO2max improvements lasted much longer during the detraining phase than mitochondrial capacity (Henriksson & Reitman, 1977). Mitochondria enzyme concentrations are much more likely to affect other factors in performance to a greater degree than VO2max, like Lactate Threshold and substrate utilization (Bassett & Howley, 2000; Klausen et al., 1981).

The VO2max limiter

Having gone through the possible limiters of VO2max, it is important to remember that there is not a single limiting factor and that VO2max can be impeded at several points as oxygen travels from the ambient air to the mitochondria. In his review on the subject Wagner eloquently stated that "The concept of 'the' limiting factor to VO2max is no longer an appropriate concept. All parts of the O_2 transport pathway are involved in determining VO_2max. Thus a change in O_2

conductance of any one component will change VO_2max in the same direction (1996, pg. 29)." This idea should be expanded out to include that the impact of each step will vary based on the individual, training status, exercise, and environment.

Several examples of this can be seen in comparing different groups of subjects, some of which have been briefly mentioned. Oxygen diffusion from the lungs to the blood is rarely considered a limiter of exercise in healthy individuals at sea level. However, the phenomenon of EIAH occurs in many well trained endurance athletes. This drop in oxygen saturation in well trained individuals significantly impacts VO2max. Thus the relative importance of oxygen saturation's impact on VO2max changes with training status (Dempsey & Wagner, 1999).

In a similar demonstration, a study by Powers et al. showed the variation in the main limiter of VO2max between highly trained and normal subjects (1989). They tested VO2max while inhaling normal air and oxygen enriched air in both groups. In the normal group, VO2max was not significantly different while inhaling either air. In contrast, the highly trained group saw a significant increase in VO2max (from 70.1 to 74.7 ml/kg/min) when inhaling oxygen enriched air. This leads to the conclusion of oxygen intake playing a role in limiting VO2max in highly trained people, but not in normal subjects, thus showing the impact of training status and endurance capacities. Lastly, the different adaptation responses to altitude in Ethiopian, Tibetan, and Andean groups shows that several different mechanisms can occur to reach the same goal of maintenance of Oxygen delivery.

While the focus for almost a century has been on oxygen transport being the limiter, oxygen utilization could play a role to some degree, as increases in peripheral factors, such as oxygen extraction measured by a-vO2 difference, have been shown to cause a small, but significant, increase in VO2max (Wilmore et al., 2001). In conclusion, the take away message is that the individual physiology, the training done by a person, and potentially other factors can shift the limitation of VO2max to a number of different sites, or alter the degree to which the various sites limit oxygen transport and even oxygen utilization. The bottom line is that VO2max represents a summary of many different processes that effect fatigue and thus performance. The absolute total number is not as important as how each path along the route to oxygen utilization in the muscles functions.

Instead of worrying about the absolute number, for performance worry about ways to increase the amount of energy that is produced aerobically, and that means improving each step of the process. By understanding the different spots of potential slow down along oxygen's route, a coach can design training around those athletes specific needs. As can be seen, the weak point is likely to change as a runner develops, so the training will have to shift slightly too. Without understanding the entire process of oxygen delivery and utilization, we end up relying on generic one size fits all training recommendations for improving the aerobic system, which as you can hopefully see is incorrect.

What does this all mean?

It means the limiter to VO2max changes based on the individual's physiological makeup and training status. The takeaway is that if each person has a different "weakness" in their VO2max, that different types of workouts should be used. Yet with most training programs suggested by coaches the same workouts or workout types are target as VO2max workouts. Similarly, in the research there has been a large focus on finding the key workouts to maximize VO2max. There is no such magic workout. It is going to vary quiet vastly based on the individual.

The takeaway message should be don't focus your training on improving some parameter like VO2max, instead focus on providing the right training stimulus to improve race performance.

Chapter 4: The Fallacy of VO2max

In a comprehensive review on training, Midgley and McNaughton's first sentence state's "The maximal oxygen uptake (VO2max) has been suggested to be the single most important physiological capacity in determining endurance running performance" (2006). Based on this notion, training for distance runners has become fixated on the concept of VO2max. Training to enhance VO2max is the subject of numerous review articles and popular coaching material. A whole theory of training has evolved based on the idea of training at the speed that corresponds with VO2max, and at certain percentages of VO2max (Daniels, 2005). Given the emphasis on this particular parameter one would assume that it must be very closely tied with performance and fatigue. It's not.

In the following chapter the limitations of VO2max will be discussed. Including the legitimacy of the variable itself, why it arose to such prominence, the efficacy of basing training paces off of it, should we even train to improve it, and how closely it ties to performance. The discussion focuses on Vo2max as a measurement, not necessarily the importance of oxygen transport and use. This is of the utmost importance for coaches because VO2max training has become a staple in the training lexicon.

How the VO2max concept developed

The ability to measure oxygen consumption first arose in the early 1920's. It was in 1923, when A.V. Hill and his partner H. Lupton came up with the idea of there being an upper limit on oxygen consumption. In an experiment which consisted of Hill running at various speeds around a grass track while measuring VO2, it was found that Hill reached a VO2max of 4.080 L/min at a speed 243m/min (Bassett, 2000). Despite increases in speed, his VO2 did not increase, leading Hill to conclude that there is a maximum limit to oxygen consumption, or in his words:

"In running the oxygen requirement increases continuously as the speed increases attaining enormous values at the highest speeds: the actual oxygen intake, however, reaches a maximum beyond which no effort can drive it… The oxygen intake may attain its maximum and remain constant merely because it cannot go any higher owing to the limitations of the circulatory and respiratory system" (Noakes, 2008, pg. 575).

These findings led to two lasting conclusions. First, that VO2max is limited by the circulatory and respiratory system. The second conclusion was the result of trying to device a laboratory test for determining VO2max, in which thirty years later, Taylor et al. decided that during a graded exercise test, a VO2max was obtained when a plateau occurred in VO2 (Noakes, 2008). However, in Taylor's original definition, a plateau was not a true plateau but it rather consisted of a VO2

increase of less than 150ml/min from one workload to the next. These findings led to the idea that in order for a true VO2max to be reached, a plateau of the VO2 should occur.

Understanding how the VO2max test came about is important as it impacts the way we currently view and use the parameter. The fact that VO2max was first measured during exercise by one of the pioneers of Exercise Science in the 1920's goes a long way in explaining the level of importance ascribed to it. Whenever a new parameter is discovered or introduced, a large degree of emphasis is put on that parameter in the research. The initial reaction by many scientists is to ascribe a great deal of significance to the newly discovered parameter, as if it will answer all of the questions that we have. It is almost as if it is human nature to go through this process of discovery and then exaggeration of the importance of the new finding. This can be seen in many instances in a wide degree of scientific fields. In Exercise Science, this may best be demonstrated by the rise of the anaerobic or lactate threshold during the 1970's, 80's, and 90's. With the ability to portably test lactate, research was centered on ways to improve lactate threshold and the various methods to test for it. Coaches also devised ways of using lactate testing as a way of manipulating the training of their athletes. Over time, the use of lactate testing and the lactate threshold have slowly decreased and settled into a level of importance that it most likely deserves. Whenever something is new, it is overemphasized, before it usually settles into its rightful place of importance over time.

Due to the very early development of the VO2max concept, a large amount of early research and study was focused on it, escalating the importance given to the parameter. In addition, theories were developed utilizing the VO2max concept very early on as it was one of the only things that was measurable. The problem is that early development of the VO2max concept created a situation where there was enormous amount of data and research surrounding it, in essence creating a concept that is too large to break down. It is almost as if the field of Exercise Science was built upon the VO2max concept.

Recently, the legitimacy of VO2max as a measurement and the acceptance of VO2max as a practical measurement of cardio-respiratory endurance have been called into question. Their contention is that VO2max is not actually a representative measure of the maximum ability to transport oxygen, but is rather controlled by a central governor. In Tim Noakes' Central Governor Model (CGM), the CGM predicts that the body regulates exercise to prevent myocardial ischemia during exercise. This is accomplished by limiting the blood flow to the periphery which the brain accomplishes by regulating muscle recruitment (Noakes & Marino, 2009). Therefore, VO2max reflects this regulation of muscles recruitment. In essence, a central governor acts as a regulator for exercise instead of exercise being limited by some parameter.

The basis of the CGM can be explained with the quote from Noakes that states that "The CGM predicts that a wide range of biological signals are monitored to ensure that exercise always terminates before the loss of homeostasis in any bodily system. (2009, pg. 339)." In essence, the key to understanding the CGM is that exercise is regulated, not limited, and that this regulation is done

to insure protection of vital organs. Instead of the production of so called by products such as H+ directly causing fatigue, these substances act as feedback to a central governor that then adjusts the workload to insure that homeostasis is maintained.

There are several theoretical arguments for this model or another similar integrated model. Noakes and other CGM proponents point to the fact that fatigue is seldom catastrophic like would be predicted in traditional models. Instead, the body uses various feedback information and past experiences to modulate power output or in the case of running, pace. The idea of pacing being prevalent in endurance events and the fact that a finishing kick, or end spurt, occurs are given as further evidence to support this model (Noakes, 2003). Interestingly, evidence of alterations in pacing strategy and EMG, which measures muscle activation can be seen from the very beginning of performance, such as that seen when racing in warm versus cool weather, which leads credence to the anticipation model of fatigue (Noakes, 2008).

An increase in muscle activation is also seen during the last segments of races, which should not occur if the muscle is "failing" due to fatigue. Noakes' hypothesis is that at the end of a race, the body's feedback says that it is near completion so that is can push slightly deeper into its capacity (Noakes, 2008). Evidence for this hypothesis can be seen in a study that found that when completing a 20-km cycling trial in normoxia versus hyperoxia, the improvement in power output in hyperoxia was proportional to the increase in iEMG that also occurred, which the authors cited as evidence that control of muscle activation was one way in which performance was regulated (Tucker et al., 2007).

Another interesting point raised in the CGM debate is the effect hypoxia has on Cardiac Output. Exercise in hypoxic conditions show a reduction in peak Cardiac Output, due to both a decrease in HR and SV (Calbet et al., 2003). According to the conventional model Cardiac Output, since it is regulated by muscle oxygen demands, should not be reduced. However in the CGM, Cardiac Output is reduced as a regulatory mechanism and is determined by the work done by the muscles (Noakes, 2004). Thus, a reduction in Cardiac Output in hypoxia is due to a decrease in muscle activation. While when supplementary oxygen is taken, Cardiac Output immediately increases to normal levels (Noakes, 2004). This immediate increase in Cardiac Output demonstrates that there is a regulatory mechanism in control and one has to question why Cardiac Output is reduced at altitude when oxygen demand by the muscles should be higher.

In regards to VO2max and how it is tested, Noakes has pointed out that in most cases the original requirement of seeing a plateau in VO2max during an incremental exercise test does not occur (Noakes, 2008). Demonstrating this lack of plateau, in a study on world class cyclists only 47% reached a plateau, prompting the authors of the study to state that their limitations might not be oxygen dependent (Noakes, 2008). It is amusing that some authors have commented that motivation may be the reason some athletes do not reach a plateau (Shephard, 2009). This could be a valid statement if the subjects were sedentary, however since the above study was with world

class cyclists, it seems a bit ludicrous to suggest that motivation during a maximum test would be a problem in such athletes. In addition, in other studies, one by Hawkins et al., there have been individual variations in VO2max levels between the traditional incremental test and a supramaximal test (Noakes, 2008). While in the average of the whole group there were no differences between the tests, the fact that certain individuals showed different VO2max is interesting and shows that the traditional test does not always give the highest VO2.

Combining the fact that a plateau does not occur in many subjects and the fact that some individuals reached higher VO2max values during a supramaximal test than the standard incremental one, the use of the standard incremental VO2max should be called into question. Other studies show that knowing or not knowing when a test or trial will end significantly effects physiological parameters, which lends credence to the aforementioned idea. In a study by Baden et al. they demonstrated that Running Economy significantly changed, along with perceived exertion, during a sub maximal run based on whether the group knew they were running 20 minutes or whether they did not know, even if they ended up running 20 minutes (2005). The VO2max test is one in which participants do not have an exact finish distance or time, so it is likely that this degree of uncertainty could affect the physiological parameters measured. The study also points to the importance of feedback and anticipation and that it can affect physiological variables.

One final theoretical problem is why variation exists based on exercise testing mode (Basset & Boulay, 2000). A runner tested running versus another modality such as cycling will have different VO2max values. There is great individual variation too, between 0 and 13% in the aforementioned study. If we recognize that regardless of exercise the oxygen cascade from the air through delivery via Cardiac Output are central adaptations and should not be different between the exercise modes, then the change in VO2max must either happen on the muscular level or it is regulated via muscle recruitment. This would explain why elite cyclists reach higher percentages of treadmill VO2max when testing cycling VO2max compared to lower level cyclists (Basset & Boulay, 2000). Lastly, the fact that muscle mass activation seems to be the major reason for variations in VO2max among a whole variety of testing methods, shows that muscle activation may play a significant role in determining VO2max, at least to a certain point (Dalleck et al., 2004).

Considering this new theory of fatigue, and the fact that the requirement used for reaching VO2max does not occur in many subjects, the use of VO2max as a testing parameter should be called into question. In addition, if VO2max is regulated, then the question arises if it even accurately reflects cardio-respiratory endurance. If we accept this to be true, then using VO2max and percentages of VO2max for training might not give the training response that we think it does.

Efficacy of basing training paces off of VO2max

With the rise of VO2max research, training systems developed based on the parameter in two ways. First, training at the speeds that elicit VO2max has become the magic training intensity which supposedly elicits the most improvements. Secondly, training at percentages of VO2max has become en vogue as a way to quantify training intensity.

In regards to training at VO2max, this arose because of a review of research that showed that the largest improvements in VO2max occurred when training at an intensity that corresponded with the parameter, irregardless of duration of the exercise (Wenger & Bell, 1986). This finding was subsequently used to demonstrate that training at VO2max was the best intensity for improving endurance in all groups of people. There are two problems with this conclusion. First, the studies findings are generalized to all groups, even though, as we will talk about later, VO2max does not improve in well trained individuals. Secondly, VO2max and endurance performance are used almost synonymously, which is not true, as discussed earlier VO2max may not even measure cardio-respiratory endurance and is certainly not the only factor in endurance performance.

Despite these concerns, training at VO2max has risen to prominence. In looking at the research, there are countless studies and reviews that focus on training at this intensity (Midgley et al., 2006). It has gone so far, that maximizing the time spent at VO2max during a training session has garnered much attention (Midgley et al., 2006). Researchers have studied the various interval training programs with the sole goal in seeing how much time at VO2max each subject spent during the training, which in itself is interesting because it shows the emphasis on the parameter instead of performance. The thought is that time spent at VO2max is the stimulus needed to improve VO2max. However, this theory has not been substantiated by research. For instance, in a study by Billat et al. after 4 weeks of training using an interval program designed to elicit time at VO2max, VO2max and, more importantly, performance did not improve (1999). In addition, even in untrained people, the original review by Wenger and Bell stated that improvements in VO2max at high intensities were not dependent on the volume of training (1986). Despite these facts, researchers continue to press on with the idea that time spent at VO2max is the key ingredient for improved endurance, even though no research backs up this theory.

Using %Vo2max and %vVO2max to quantify intensity is an accepted practice in research and is used in many training programs, such as those prescribed by Jack Daniels and Joe Vigil (Vigil, Daniels, 2005). The problem with this approach is that each individual will have a wide range of adaptation, even if training at the same percentage of VO2max. This occurs due to differences in the individual's physiology. For instance, lactate threshold can occur at wide range of %VO2max, even in trained individuals (Brooks and Fahey, 2004). As an example, if two trained runners both performed at a fixed intensity at 80% VO2max, one can be below lactate threshold and one above. This would substantially impact the energetics of the workout, as can be seen in a study that showed there was a 40-fold range for increases in lactate levels at 70% VO2max among individuals (Vollaard et al., 2009). In a recent study by Scharhag-Rosenberger et al. they tested

whether exercising at the same %VO2max resulted in similar metabolic strain. They found large individual variance in the lactate response at the fixed intensity, even if groups were matched for similar VO2max values. This led them to conclude that the use of percent VO2max values for training or research should not be used if the goal is to have similar metabolic strain by the exercisers.

In addition to lactate differences, factors such as the individual's substrate use, fiber type, and other physiological variables will all vary considerably at a fixed percent of VO2max. This was demonstrated in a study by Vollaard et al. (2009). The study showed that while on average improvements were seen in a variety of endurance parameters after six weeks of endurance training, the individuality of the response was widespread with some showing even negative responses to the training, even though the training was at the same 70%VO2max intensity for all subjects (Vollaard et al., 2009). The study showed that there was a wide range of adaptation in maximal and sub maximal tests including VO2 parameters, muscle enzyme activity, and metabolite levels. An interesting finding in the study is that low responders for an increased VO2max were not low responders in other parameters. The change in VO2max did not correlate with the change in performance on a time trial, which is a significant finding demonstrating that perhaps more attention should be paid to changing in performance instead of manipulating physiological parameters such as VO2max. One has to question the training recommendations based on training designed at improving parameters such as VO2max, with the assumption being that performance will improve because of it, when studies show that change in VO2max are often not linked with a change in performance. This phenomenon of varied response is not new and can be seen in a wide array of training situations, such as altitude training for example (Chapman et al., 1998).

Knowing the wide variance in adaptation that can occur when training at a fixed percent of VO2max, its use has to be called into question. In fact, Vollard et al. questioned the use of %VO2max as a way to standardize intensity and suggested standardization on parameters that more directly effect power output (2009). These findings combined with those by Scharhag-Rosenberger et al. suggest that the use of %VO2max should be eliminated if the goal is to standardize intensity. One has to really wonder about training programs that use %VO2 to prescribe training as what adaptations will take place are almost a crapshoot. This does not seem like a scientific way to train, as it is portrayed. In practical terms for trained distance runners, it probably makes more sense to standardize paces in relation to their recent race performances or based on percentages of goal race pace in well trained runners.

Should we train to improve VO2max?

As mentioned previously, studies have shown that training at VO2max elicits the most improvement in VO2max. This has been used as reasoning for training at VO2max because, as

previously discussed, VO2max is the traditional measurement for endurance. The logic is that if VO2max is increased, endurance performance increases. This is not the case. In addition, the question arises if VO2max actually improves in well trained runners? It doesn't.

Showing the separation of VO2max and performance, the Vollaard et al. study found that the change in VO2max was not related to the change in time trial performance (2009). Studies demonstrate improved performances without changes in VO2max (Daniels et al. 1978). Also, studies show that VO2max can improve without changes in performance, which is seen in a study by Smith et al. that showed improvements in VO2max by 5.0% without an improvement in performance over either 3,000m or 5,000m (2003). In addition, in looking at long term changes in performance in elite athletes, changes in performance occur without subsequent changes in VO2max.

In highly trained athletes, many studies have shown that VO2max does not change, even with performance improvements. In one of the only studies done on a large group (33) of elite runners, Arrese et al. tracked changes in Vo2max across three years. Performance improved by an average of 1.77% in men, and .69% in women, with VO2max remaining essentially unchanged (~76.56 vs. ~76.42 in men, and ~70.31 vs. ~70.05 in women) (Legaz Arrese et al., 2005). This points to improved performance in elite runners without changes in VO2max. Furthermore, it has been shown that among homogenous groups, such as well trained runners, VO2max does not correlate well with performance and can not be used to distinguish what runners are faster (Legaz-Arrese et al., 2007).

Further evidence can be seen in two case studies on elite runners. In a study on a female Olympic level runner, Jones showed that while the athlete's 3,000m time improved by 46 seconds, their VO2max decreased from 72 ml/kg/min down to 66 ml/kg/min (Jones, 1998). Another study by Jones, this one on the current women's marathon world record holder, found that while VO2max varied some based on the time of testing, it was essentially stable at 70 mL · kg–1 · min–1 from 1992 to 2003 (Jones, 2006). The fact that Radcliffe's Vo2max was essentially stable despite her training volume and intensity increasing substantially is intriguing. Her training increased from a modest 25-30 miles per week (and her VO2max was already 72 at the time) to 120-160 miles per week. The fact that VO2max did not change despite this massive increase in volume and intensity shows the short time course of changes in VO2max. Most importantly, Radcliffe's performance improved dramatically during that time period.

The rapid change in VO2max can even be seen in untrained individuals. In a study by Smith and Donnell, they evaluated the changes in VO2max over a 36 week training period (1984). VO2max substantially increased by 13.6%, but all of those gains were seen in the first 24 weeks of the study with no further increases during the final 12 weeks. Similarly in a study by Daniels et al. in untrained subjects VO2max increased during the first 4 weeks of training, but did not increase after that even with a further increase of training, despite continued improvements in performance

(1978). Given the evidence that VO2max does not change in elite runners and does not correlate with performance, training focused on improving VO2max does not seem like a logical idea for well trained runners.

Vollaard et al. may have put it best when they came to the conclusion that "Moreover, we demonstrate that VO2max and aerobic performance associate with distinct and separate physiological and biochemical endpoints, suggesting that proposed models for the determinants of endurance performance may need to be revisited (2009, pg. 1483)". There recognition that aerobic performance and VO2max are not direct equals or even well linked is a step in the right direction and needs to be acknowledged to a much greater degree. Combining these findings with Noakes' CGM creates a situation where VO2max may not be measuring what we think it is. Adding the facts that using %VO2 to classify training results in a wide range of adaptations and changes in VO2max do not occur in trained athletes, one has to question basing entire training programs on VO2max. The point isn't to question the importance of oxygen or the aerobic system, but instead to show that VO2max does a poor job of representing the aerobic contribution

The bottom line question that needs to be asked is why is so much of training focused on a variable that does not change in well trained athletes, barely changes in moderately trained, levels off after a short period of time, and does not even correlate well with performance? Does this sound like a variable that we should be basing all of our training off of?

Chapter 5: Lactate, Acid and other by products

For decades the main culprit of fatigue and the nemesis of all athletes was a foe called lactic acid. Unfortunately, for decades we were dead wrong. Instead, lactic acid, or more correctly termed lactate, has given way to the idea that the build up of other products could potentially cause fatigue. If you recall the chapter on fatigue, this is the build up theory. The revelation that lactate is not the enemy doesn't mean that our training needs to be thrown out or all of the research based on lactate should be banished. While Lactate does not directly cause fatigue, it highly correlates with it, so that as fatigue increases lactate increases. This is partially due to the linear relationship between by products such as H+ increases and lactate increases. One of the keys to running performance is to delay the build up these accompanying products. If the rate of those products accumulation can be decreased, fatigue can be delayed. So while lactate is not the culprit, it corresponds with the build up of products that can cause fatigue and given the vast amount of research done on lactate, we'll delve into what causes lactate build up and what can be improved to delay it's build up. First let's look at how the build up of such products affects performance.

Coaches commonly use workouts to enhance lactate tolerance or the lactate threshold. While the traditional method of hard intervals for the former and tempo runs for the later has merit, knowing the underlying mechanisms of this process will help coach better design workouts to enhance those variables.

Buffering Capacity and Dealing with Acidosis

In events that are run at speeds that require more energy than the aerobic system can provide anaerobic energy sources must take up the slack. As previously mentioned, middle distance events have a significant anaerobic component. Due to this anaerobic component, certain products will accumulate in the body, potentially causing fatigue. Previous studies have demonstrated that an increase in H+, which is a proton that dissociates from lactate and would decrease the pH, may impair muscle contractility (Mainwood & Renaud, 1985).

While the previously accepted notion that lactate played a direct role in fatigue, essentially acting as a "poison" to the muscles, has been disproved, it does not discount the entire acidosis concept of fatigue (Noakes, 2007). While lactate itself may not cause fatigue, it corresponds with other products in the body which may have contribute to fatigue, thus lactate can still be used in studies as a marker of fatigue. For instance, as mentioned already, an increase in circulating blood lactate corresponds well with a decrease in pH and increase in H+. An increase in H+ has been

shown to reduce the shortening speed of a muscle fiber, while a reduction in pH impairs Ca2++ reuptake in the Sarcoplasmic Reticulum and may inhibit phosphofructokinase (PFK) (Hargreaves & Spriett, 2006; Brooks et al. 2004). In addition, a decrease in pH could stimulate pain receptors (Brooks et al., 2004). All of these actions could potentially cause fatigue.

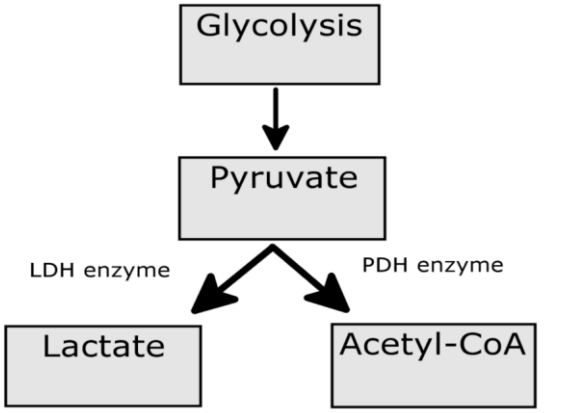
The ability of the muscle to buffer the H+ could potentially delay fatigue. As mentioned above, a rise in H+ concentration causes many processes that could potentially lead to fatigue. In particular, its inhibitory effect on PFK would impact ATP production through Glycolysis. Training has been shown to increase buffering capacity in both recreational and well trained athletes (Laursen & Jenkins, 2002). Furthermore, in one study, the buffering capacity of 6 elite cyclists was found to be significantly related to their performance in a 40km time trial (Weston et al., 1996). This demonstrates the importance of dealing with by-products and buffering capacity when it comes to performance. Due to the impact of acidosis on energy production and performance, much of the coaching and scientific literature focuses on delaying this process. Of particular interest is the concept of the lactate threshold.

The Lactate Threshold

In this review, Lactate Threshold (LT) is defined as the fastest running speed at which blood lactate levels remain in a relative steady state. Or stated in another way, it is the fastest speed in which lactate production and clearance are in equilibrium. LT has been given many different names, from Maximum Lactate Steady State (MaxLass) to Anaerobic Threshold, and has been given many different definitions based on fixed lactate readings and extrapolation from a lactate curve. Originally, LT was defined as a fixed lactate reading at 4.0 mmol/L, but more recent research has shown that lactate levels at LT can vary as much as 6 mmol/L, between 2 and 8 mmol/L (Beneke, 2000). In addition, LT can be expressed as a percentage of VO2max.

The lactate threshold is dependent on many factors relating to the production and clearance of lactate. This interaction is what governs the amount of lactate in the blood and ultimately the LT (Billat et al., 2003). Lactate production within the individual fibers occurs via Glycolysis. At the end of Glycolysis, pyruvate is formed, which can go one of two routes: it can be converted into lactate via the enzyme Lactate Dehydrogenize (LDH), or it can be converted into acetyl-CoA via PDH and enter the mitochondria to be used in the Krebs cycle and produce energy aerobically. Whether Pyruvate is converted to acetyl-CoA and enters the mitochondria or is converted to Lactate depends on several factors, including LDH and PDH enzyme concentration and activity (Hargereaves 2006). Contrary to popular belief, conversion to lactate occurs even when in the presence of adequate oxygen.

Path of Lactate



It is important to note that we are measuring blood lactate when defining LT, not muscle lactate. That means that LT is dependent not only on how much lactate a muscle produces but how much actually makes it into the blood stream. When lactate is produced it can either stay in the muscle, travel to adjacent muscle fibers, move into the interstitial space between muscles, or travel to the blood stream. How much travels to the blood is partially dependent on both the difference between lactate levels in the blood and muscle and on the lactate transporter activity, which will be covered later. Lactate appearance in the blood also depends on exercise intensity and the amount and type of muscle mass activated. The greater the intensity means a greater reliance on Glycolysis without as much aerobic respiration taking place. Also, the more intense an effort, the greater amount of Fast Twitch fibers are recruited, which because of their characteristics are more likely to produce lactate.

On the other side of the coin is lactate removal which occurs via several mechanisms. Within the muscle cell itself lactate can be used as fuel by being taken up and oxidized by the mitochondria. Therefore lactate can be consumed by the muscle fiber, or it can be transported to adjacent fibers to be used. Additionally, it can be transported to interstitial spaces (the space surrounding/between muscle cells). In these instances, lactate produced in the muscle would not increase the blood lactate levels as it would either be consumed by the producing muscle or adjacent fibers, or it would be sent to interstitial spaces.

Lactate that makes it to the blood stream can be removed via several ways. Muscle fibers that are on the Slow Twitch side of the muscle fiber spectrum can act as consuming fibers that take the lactate from the blood and use it as an energy source. Muscle fibers that are not taxed to a high degree also are used to take up lactate from the blood stream. In addition the Heart, Brain and liver all play an active role in clearing lactate from the blood. The Heart and Brain use lactate as a fuel source, while the liver, through the Cori cycle, acts to convert lactate to pyruvate and then ultimately the fuel source glucose. This process has been termed the lactate shuttle, in which lactate

is produced in a muscle then sent through the blood stream where it can be taken up and used as a fuel source by muscle fibers or other tissues and organs. It has been theorized that this is one of the body's way of transporting energy stores to different muscles, as glycogen in its stored form can not be transported.

Common Misconceptions and Lactate:

Contrary to popular belief, lactate does not cause fatigue. In fact, as explained in the text, lactate is a fuel source and likely delays fatigue. While acidosis may play some role in fatigue, it is other products such as Hydrogen ions which contribute to decreasing muscle pH. Lactate actually helps by consuming those Hydrogen ions.

Lactate also does not cause muscle soreness. Lactate removal after exercise occurs fairly quickly and does not linger for days like many people think. Following an intense exercise bout lactate can be back to near normal levels after an hour or less and much quicker if easy aerobic exercise is done as that speeds removal.

Why measure lactate if it doesn't cause fatigue? The answer is because the appearance of lactate corresponds almost exactly with by-products that do cause fatigue, which are harder to measure. A goal of training can still be to decrease lactate, although what you are really trying to do is decrease products like Hydrogen ions.

Muscle fiber Characteristics

Muscle fiber type will play a role in lactate production and ultimately LT. Fast Twitch muscle fibers are more likely to produce lactate due to the individual characteristics of the fiber that make it prefer Glycolysis, such as having a larger amount of Glycolytic enzymes (Bishop et al., 2000). In addition, having less mitochondria, myoglobin, and greater LDH activity makes it where the pyruvate is more likely to go the route of lactate. Similarly for Slow Twitch fibers, mitochondria concentration plays a role in enhancing the LT (Billat et al., 2003).

An increase in mitochondria allows for more pyruvate to be converted in acetyl-CoA and enter the mitochondria. Because of these factors, the fiber type of the athlete and mitochondria concentration will help determine the amount of lactate produced and the LT. Not only does an increase in mitochondria size cause a decrease in lactate, but an increase in mitochondrial enzymes decreases lactate too (Bassett & Howley, 2000). This is likely due to an increase of the pyruvate that is converted to Acetyl-CoA instead of lactate, or an increased ability for lactate oxidation. Going beyond just changes in lactate concentration, several studies have established a relationship between mitochondrial enzyme activity and the LT (Coyle, 1995). Lastly, a study on detraining found that the drop in LT that occurred closely mirrored the drop in mitochondrial enzyme activity (Coyle, 1985). These results show the close relationship between lactate levels and mitochondrial enzyme levels.

Monocarboxylate transporters

On the other hand, lactate clearance occurs via the lactate shuttle. Lactate produced in a muscle fiber can be used as a fuel source by several different organs; including being shuttled to adjacent muscle fibers, the liver, cardiac muscle, and other muscles via the blood stream (Stallknecht et al., 1998). This shuttle system allows for the transport of an energy source, which is important because stored glycogen in a muscle fiber cannot be converted back into glucose to be transported. This transport is mediated by lactate transport proteins.

So far, two different primary lactate transporters have been found that help facilitate transport into and out of the muscle cell. Monocarboxylate 1 (MCT 1) has a high affinity for lactate and is primarily found in ST muscle fibers. It is believed that MCT1 has a role in the oxidation of lactate and therefore the transport of lactate into the cell (Billat et al., 2003; Hargereaves, 2006). For this reason, MCT1 quantity has been found to correlate with endurance performance (Bentley et al., 2009). MCT4, another transporter protein, is found primarily in FT fibers. It has a lower affinity for lactate and is believed to have a role in lactate transport out of the muscle cell (Billat et al., 2003; Hargereaves 2006). Both of the MCT types increase with training, thus showing that lactate dynamics can be improved.

Maximum Lactate Steady State

Lactate steady states are generally only considered to happen during longer duration running. In particular, it is often cited that the fastest speed that one can run while keeping lactate readings steady is a speed that a runner can hold for about one hour (Billat et al. 2003; Daniels, 2005). However, field data provided by Italian coach Renato Canova during a simulated 5k and 10k race in elite Kenyan athletes, one of which is a world record holder, suggests that elite runners may be able to run at a lactate steady state in the middle kilometers of a 5k or 10k, which take around ~13 and 27~ minutes for these athletes to complete. This goes against conventional wisdom, as it is generally accepted that blood lactate levels increase steadily throughout the race. This may be true for most athletes, but if the elite African runners have this ability to run at a steady state in the middle of the 5k or 10k race this could partially explain their dominance.

In addition, there was a relationship between the ability to increase speed in the final kilometer and the ability to increase lactate by a large degree in the final kilometer. Essentially the athletes were able to run at a moderate amount of lactate during the middle of the race and then during the final stretch of the race, increase their speed and lactate levels to a large degree. Perhaps this gap between lactate during the middle of the race, and the ending lactate explains the ability to kick at the end of a race. The African runners are notorious for being able to finish very fast last laps in both championship and fast races. Many times their last lap times are very close to their self

reported fastest times they can run one lap fresh. This phenomenon is one that needs to be explored in a controlled environment.

Lactate Testing

The best way to measure LT is up for debate (Billat et al., 2003). Traditionally, the lactate test has consisted of a series of ~5min steps that gradually increase in pace. However, research is unclear on what the best testing method is, and if using a step test, how long each step should be and how much velocity should be increased with each step. It all comes down to how much time is required to reach a reasonable degree of steady state. This has led some researchers to suggest long stages of up to 20min (Billat et al., 2003). However, the problem with this suggestion is that it is not be practical, and if one does not have a reasonable approximation of the velocity of the individuals LT, then it could take many long stages to find a LT. .

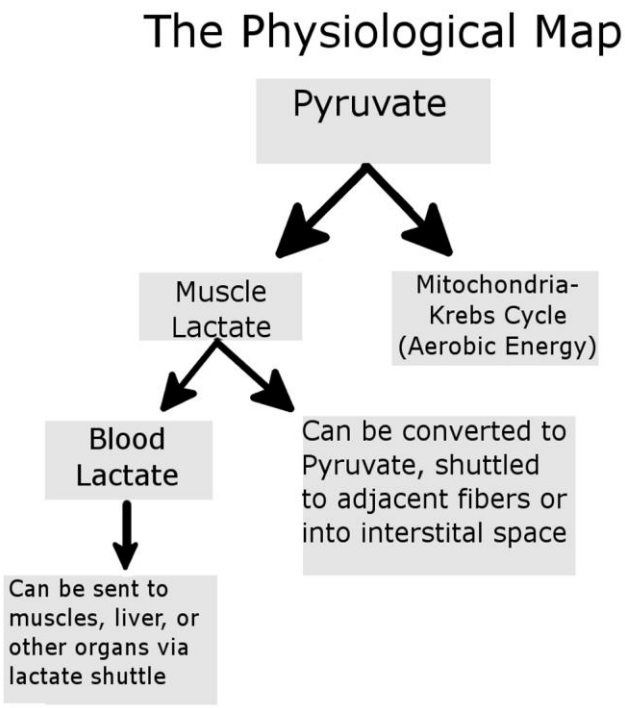
At best the tests used to find a true lactate threshold are essentially educated guesses. The only true way to find a lactate threshold (or Max Lass) is to run at a constant speed for around 20-30min. and take blood lactate samples and make sure there is very little increase in lactate from start to finish. Then, do this again at a faster pace. You have to keep doing this until you find the fastest pace that you can maintain without an increase in blood lactate. Practically, it's pretty much worthless and too time consuming. So we throw that idea out the window. Instead scientists have developed all sorts of ways to guess what the LT is based on a lactate curve. Some ways are more accurate than others, but most involve a step test. For example a common test is to run 5x mile with 1-2min rest starting at an easy/moderate speed and increasing speed by 15sec per mile or so. For example running 5:35, 5:20, 5:05, 4:50, and 4:35 for someone who has a threshold of around 4:55-5:00. This gives you a lactate level for each speed which is then plotted on a speed vs. lactate level graph. A variety of methods are used to determine where lactate threshold is, but they all essentially look at what speed lactate readings start to increase significantly over baseline. Some involve simply looking and guessing while others involve mathematical formulas, but the bottom line is all are guesses.

Since we've established that there is a problem with finding the lactate threshold, let's look at the next problem with lactate threshold research. That would be the problem of knowing what that nice little lactate curve actually means. The traditional way to read a curve is that a shift to the right in the curve means a better LT, or better endurance, and a shift to the left means deteriorating endurance. The problem with that idea is it is too simplistic. According to Jan Olbrecht, who outlines lactate testing in his excellent book *Science of Winning*, there are two opposing forces that mainly act upon the lactate curve. These opposing forces are the aerobic and anaerobic capacity. Aerobic capacity is essentially how well the aerobic system functions to minimize lactate levels. If

you recall the previous section on lactate dynamics, lactate can be minimized via several mechanisms. First, when pyruvate is formed, it can be converted to acetyl-CoA and be used to produce ATP aerobically, thus minimizing the amount of pyruvate that becomes lactate. Also, lactate that is produced in the muscle can be taken up and used as aerobically via adjacent muscles or other tissues, and lactate that is in the blood stream can be taken up by Slow Twitch muscle fibers to be used as fuel aerobically. Thus aerobic capacity can be changed based on any improvements that enhance these processes, such as increased aerobic enzymes, mitochondria, or oxygen delivery.

In essence Anaerobic Capacity refers to the maximum amount of pyruvate that can be produced. An increase in Glycolytic capacity by improvements such as increases in Glycolytic enzymes is one example. The theory is that maximum pyruvate production impacts the LT because if more pyruvate is produced without a change in the amount that can be turned into Acetyl-CoA (or aerobic ability), then more pyruvate automatically gets converted to lactate. Therefore a change in LT can occur without any change in aerobic abilities.

These two factors interact to produce the lactate curve. An increase in anaerobic capacity would shift the curve to the left (meaning more lactate produced at each speed), while an increase in aerobic capacity would shift the curve to the right (meaning less lactate produced at each speed). The strength of each of these opposing forces determines where the curve ends up. While this view is still a simplistic actualization, it acknowledges the anaerobic contribution to the lactate curve and provides a better model for looking at the lactate curve. That is an important point, because traditionally all changes in the lactate curve have been thought to be because of changes in aerobic abilities. This model by Olbrecht shows that changes in the curve depend on the push and pull of two factors, not just one. You can clearly see that not only do aerobic abilities, such as how much pyruvate gets converted to lactate or how much lactate can be used by other muscles or organs, but also anaerobic abilities like how much pyruvate is produced play a role. Shown below is a simple diagram of this model:



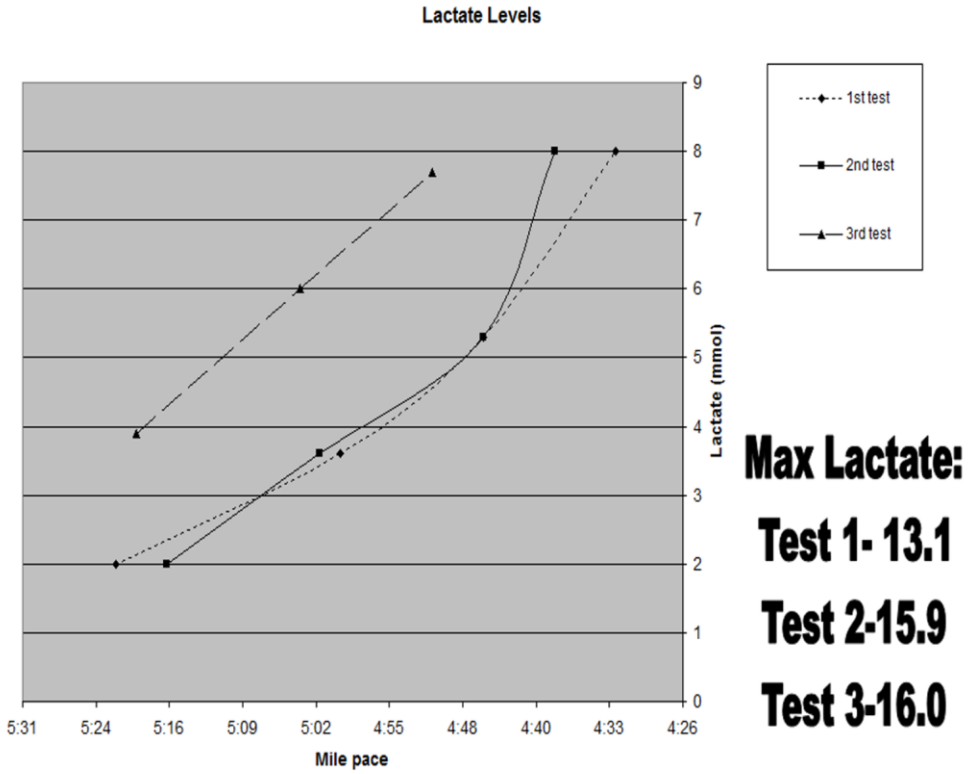
Now that we have this model, what does it mean and how do we use it? Starting at the top of the diagram, the first thing that can be affected is pyruvate production. From this point forward, I will refer to maximum pyruvate production as Anaerobic Capacity. Aerobic Capacity can be seen in the amount of pyruvate that is taken up by the mitochondria and lead to aerobic energy.

Looking at these two capacities, it can be seen how the model works in diagnosing training. In order to increase energy available we can increase the Anaerobic Capacity. That will allow the athlete to produce more pyruvate. However if the aerobic capacity stays the same, then this means more lactate will be produced at that effort level. If we increase the aerobic capacity, then we can see that more of that pyruvate is shuttled into the mitochondria resulting in aerobic energy increase and less lactate at the given effort. Using this model, then lactate is an interaction of both the anaerobic capacity and aerobic capacity, not just the aerobic capabilities which is widely recognized as the only source in traditional training models.

The question then becomes how do we analyze the LT curve to see if it was a change in aerobic or anaerobic capacity that caused a change in the LT? While there is no simple flaw proof way, the use of an anaerobic capacity test makes everything a bit clearer. An anaerobic capacity test consists of running an all out 400-600m and taking lactate readings every 2-3min after until a maximum reading occurs. This usually occurs within 5-9 minutes following a max test. Combining the maximum lactate number and the speed of your time trial, you get a good baseline idea of your

anaerobic capacity. The higher the lactate and faster the speed, the higher the anaerobic capacity is. A decrease in maximum lactate levels and speed shows a decrease in anaerobic capacity.

Once you have this information, do a traditional LT test to get a lactate curve. My suggestion would be 5x5minutes with 60-90 seconds between repeats, starting at a moderate speed and increasing the speed by 10-15sec per mile until the last two repeats are faster than LT. Plot the results in a speed versus lactate graph to get a curve. The first test serves as a baseline. The real information comes when you repeat this test in a month or two. To analyze it, first look to see if there was a change in the lactate curve to see if it shifted left or right. Next, look at the anaerobic capacity to see if it changed. Remember that an increase in anaerobic capacity shifts the curve to the left, while a decrease shifts the curve to the right. The degree to which each capacity changes also matters; in that a moderate increase in anaerobic capacity with a stronger increase in aerobic capacity would result in the lactate curve moving slightly to the right. The interaction is what matters. Below is an example interpretation:



In the above graph, we have three separate lactate tests performed. The first test provides a baseline for the subsequent tests. In test two, it can be seen that lactate levels are essentially the same throughout, with a very slight difference only at the fastest speed. In the traditional way of evaluating the test, these two tests would give the same LT and the conclusion would be that LT did not improve so the training was ineffective. However, note that the max lactate increased dramatically from 13.1 to 15.9mmol. We can conclude that since anaerobic capacity improved by so much thereby pulling the curve leftwards, the aerobic capacity must have improved by a similar

amount since the lactate levels did not change. Thus the conclusion is that anaerobic and aerobic capacities were both improved and the training was effective if that was the goal.

Next, test 3 shows the lactate levels shifted upwards and left, as more lactate was produced at every intensity. If we look at the maximum lactate, it shows a very similar number to the previous test so the anaerobic capacity did not change. Therefore, since the lactate levels increased without a change in anaerobic capacity, it must be concluded that the aerobic capacity deteriorated. This is just one example of how lactate curve analysis can be done using this new model. The takeaway message is that lactate levels reflect a balance of opposing aerobic and anaerobic forces.

Chapter 6: Efficiency

While most of the time the focus is on capacities, or how big the engine is, the real key is often how efficient a runner is. Coaches and scientists often talk about the importance of efficiency, but what it actually is and how to improve it is not well understood. Instead of one catch all term, there are actually several types of efficiency including metabolic, neural, and biomechanical. These three types of efficiencies combine to create total efficiency. Total efficiency is useless unless the components that make it up are understood. By understanding the three different types of efficiency, a runner can find and improve their particular weakness. Before delving into the different types of efficiency and how they can be improved, let's look at the measurement commonly used for total efficiency.

The Measurement: Running Economy

The last of the traditional big three physiological parameters for running performance is Running Economy (RE). Running Economy is the measurement used to classify total efficiency in research. The measurement uses oxygen intake to represent energy use and is commonly defined by how much oxygen it takes to cover a given distance at a fixed speed (Saunders et al. 2004). RE significantly correlates with performance and has been used in conjunction with other factors as a model to explain running performance. In a study on well trained runners (VO2max of ~72), Conley and Krahenbuhl showed that RE could explain 65% of the variation in race performance among the group of 12 runners (Conley et al., 1980). While the problems with the measurement of RE itself will be discussed later, it is often forgot in the research that RE is a measure of gross efficiency, meaning that it is the result of both internal and external components, so that mechanical, neural, and metabolic efficiency play a role. The RE measurement simply reflects a combination of these three efficiency components.

Biomechanical efficiency refers to the mechanical cost of running and includes such factors as energy storage and how wasteful the movement pattern is. Neural efficiency can be defined as an improvement in the communication between the nervous system and the muscles themselves. As an example, an improvement in neural efficiency could occur via more refined motor programming. Lastly, metabolic efficiency refers to factors that impact the production of energy for the muscles to use, such as fuel source or oxygen delivery. This concept of multiple types of efficiency is often overlooked, as it is possible to become more efficient in one area yet have total RE be negatively affected because of a larger decrease in another aspect of efficiency. Thus it is a balancing act to maximize total efficiency. The measure itself of gross efficiency may be flawed as it tries to integrate too many variables based on oxygen consumption, but the idea behind efficiency being important is right on. Let's look at some of the components that affect running economy.

Biomechanical Efficiency

Biomechanical efficiency refers to anything that impacts the mechanical cost of running. If you think back to the chapter on how running happens, you'll recall that active and passive dynamics combine to provide the energy needed for movement. Biomechanical efficiency is mainly concerned with the passive dynamics side, while the active side which relies on muscle contraction for movement is accounted for in metabolic and neural efficiency. There are several factors that contribute to biomechanical efficiency. If these factors are optimized then less energy is required to cover a given distance. They include elastic energy storage and return, the mechanics of the stride itself, how the foot lands, and the structure of the runner.

Stretch shortening cycle and Elastic energy return

While the most efficient way to run biomechanically is discussed in a later chapter, there are several mechanisms that improve biomechanical efficiency, one of the most important being the Stretch-shortening cycle (SSC). The SSC occurs when a muscle is actively stretched and then is immediately followed by a muscle contraction. During the pre-stretch portion, energy is stored in the series elastic components of the muscle and then the energy is released during the contraction part. Essentially it is a spring like mechanism with storage and release of energy greater than if just contraction took place. The amount of elastic energy return is dependent on several factors including the length and speed of the stretch, the stiffness of the muscle, and the time between the stretch and the subsequent contraction (Saunders et al., 2004).

In general, a stiffer muscle will store more energy than a loose muscle, although there is likely an optimal stiffness. This idea can be conceptualized by thinking of a very loose rubber band compared to a very tight rubber band. If the tight rubber band is stretched it stores a much larger amount of energy and will fly much further if let go than if a loose band is stretched and let go. In addition, the longer the delay between the stretch and the subsequent contraction the more energy dissipates. Therefore the SSC works best when a stiff muscle is rapidly stretched and contracted with little time in between. A good example of this is the calf muscle upon landing and subsequent toe off during running or the extension of the hip while sprinting. Evidence of the impact of the SSC on RE can be seen by the fact that muscle stiffness strongly correlates with RE (Dalleau et al., 1998). The fact that muscle and tendon stiffness is one factor that can improve RE, leads to the question if an increase in flexibility is such a good thing? Contrary to popular belief, a stiffer muscle is more efficient during running than a flexible one.

The stiffness of the muscle-tendon unit is dependent on several factors including active characteristics such as the muscle concentric/eccentric contraction state, and passive characteristics

such as the length and condition of the muscle, tendon, and fascia during impact. Demonstrating the impact of muscle stiffness, Kyrolaine et al found that stiffer muscles surrounding the ankle and knee created an increased SSC response, which resulted in greater force on the subsequent push off (2001). The keys for obtaining optimal stiffness and energy return are to put the body in optimal position upon impact, training the muscle and tendon to be able to absorb and utilize the forces, and training to pre-activate the muscles.

Pre-activation of the muscles before landing is a way to actively manipulate stiffness of the system, resulting in greater storage of elastic energy. Dalleau et al. demonstrated this in showing that the energy cost of running was related to the stiffness of the lower leg (1998). Pre-activation occurs as the muscles in the lower leg prepare for impact. This is done to adjust for the impact forces, essentially acting as an internal cushioning mechanism, and decrease the stress caused by muscular vibration.

In addition to the muscles themselves contributing to energy storage and release, the tendons also play a role. During impact such as the braking or contact phase of running, tendons store energy and then subsequently release that energy via recoil during the take off phase of running. The tendons and muscle complexes essentially act as a spring, storing energy upon ground strike and releasing energy as the runner nears leaving the ground.

In a study by Ker et al., it was found that the Achilles tendon stores 35% of its kinetic energy, while tendons that are in the arch of the foot store 17% (1987). It has been estimated that without the two aforementioned mechanisms of storage and release of elastic energy, the VO2 required would be 30-40% higher (Saunders et al., 2004). To properly utilize these elastic mechanisms, the body has to be in optimal position biomechanically and the tendons have to be trained to utilize the forces. Rapid movements such as sprinting or plyometrics will train the tendons to be better able to utilize the energy.

Additionally, the use of these elastic energy systems depends greatly on a person's biomechanics. Running in a certain way will elicit a greater elastic storage and return as it will put the muscles and tendons in a better position to be able to store and use energy. For example, forefoot running (or barefoot running) has been shown to improve RE when compared to heel striking with shoes (Squadrone & Gallozzi, 2009). One possible explanation is that forefoot running results in the runner landing with the ankle more plantar flexed, this may allow the subsequent stretch reflex on the calf and Achilles-calf complex to utilize more elastic energy then landing in a dorsiflexed position at heel strike where the calf complex is already in a stretched position which minimizes the SSC. Another possible explanation put forth by Dr. Daniel Lieberman of Harvard is that forefoot striking utilizes the elastic components of the arch of the foot more so then a heel strike does. Lastly, in a heel strike the ground contact time is longer as the foot has to transition from the heel touching the ground to pushing off with the ball of the foot. This lengthened ground contact means more time between energy storage and release. Thus more energy likely dissipates and is

lost. Regardless of the explanation, it appears that a non-heel strike would be beneficial to maximize efficiency. This is but one example of how the position of the body impacts energy usage.

Maximizing Elastic energy use:

1. Biomechanics- How you run plays a large role. Ideally, foot strike would be mid or forefoot underneath your knee and close to your center of gravity. An often overlooked use of elastic energy is in the muscles in the hip. Good hip extension allows for the use of this rubber band like effect that takes care of the subsequent forward recovery leg movement.

2. Reactive or Plyometric training- Short hops, jumps, and bounds with the focus on minimizing the time spent on the ground will work. Sprinting is about the most specific form of plyometric activity that can be done for runners, yet it is often underutilized. Doing 60-100m accelerations is a great way to work on using elastic energy. This type of work will train you to reach force development faster, minimizing ground contact, and will optimize the stiffness of muscles and tendons.

Racing on grass vs. mud vs. track:

Have you ever wondered why some runners are better in Cross Country than track? The use of elastic energy plays a role. When running on a softer surface, such as mud or grass, more energy is dissipated instead of being returned. Runners who rely a lot on energy return tend to do worse in races, like cross country, where the surface is soft and better on surfaces where they get good energy return, like the track or road.

Stride characteristics

There are several biomechanical characteristics that impact the efficiency of the running gait. One factor is their individual stride length. Since speed is equal to stride length x stride rate, optimizing and manipulating those variables is of importance. In a study on fatigue during a marathon, it was found that RE decreased in the later stages of the marathon partly due to a decrease in stride length (Hausswirth). This leads to the question of whether manipulating stride parameters can optimize RE. Evidence has shown that runners are most efficient at their self selected stride lengths and that when made to run with shorter or faster stride lengths, RE was worse (Cavanagh & Williams, 1982). However, it should be noted in the Cavanagh study that not all runners self selected their optimal stride length. In another sutdy they found that 20% of their subjects selected stride lengths that resulted in an elevated RE (Morgan et al., 1994). It is of interest that all of these subjects were overstriders. Similarly, in the Cavanagh study, of the 10 subjects who did not select optimal stride lengths, 7 were overstriders. It has been hypothesized that overstriding results in a reduction in the ability to use stored elastic energy. This hypothesis was backed up by the findings of Kyrolaine et al. who found that increased braking force, which would occur in overstriding, results in an increased RE (2001). In regards to optimizing stride length, it appears that it functions based on a U-shaped curve, where too short and too long of a stride length results in increased VO2, therefore hampering RE (Morgan et al., 1994).

In addition to stride parameters such as length and frequency, vertical oscillation contributes to RE. Excess vertical oscillation decreases RE and is one of the factors that helps explain why long distance runners are typically more economical at slower paces than middle distance runners (Saunders et al., 2004). In fact, several studies have found that runners are more efficient at running velocities at which they frequently train (Jones & Carter, 2000). There are two important things to remember with vertical oscillation that are sometimes missed. Remember, that it is excess vertical oscillation that decreases RE, not vertical oscillation itself. A certain amount of vertical lift will be needed to cover ground. If a person were to try to stay completely flat, their stride length would be severely limited.

Tying into this idea, it should be noted that RE is measured at slower speeds due to the need of a steady state in oxygen consumption when doing the measuring. Since runners who focus on shorter events, such as middle distance athletes, tend to have greater vertical oscillation, it is likely that this is a more efficient way to run at faster speeds. A greater vertical oscillation is needed to cover ground in the air and reach their optimal stride length. Based on these two ideas, I would advise against minimizing vertical oscillation and instead recommend finding the optimal vertical component. It is not unlike the basic physics problem we all got in high school, where when shooting a cannon the optimal angle is needed to shoot the furthest distance. Too great of an angle would result in a large vertical component but very little distance covered. On the other hand, a very small angle would result in a line drive shot that would again fall short of maximum distance. The running stride is very similar in that we need to come off the ground at an optimal angle to maximize stride length.

Foot strike

Whether a runner should use a heal strike or a midfoot/forefoot strike is often debated and can play a role in biomechanical efficiency. Williams and Cavanagh suggested that a heel strike is more economical due to decreased muscle activation required to provide cushioning, because heal strikers let the shoe do the cushioning (1982). However, it has not been established if the muscles used to alter cushioning play a role in fatigue during a race. Also, as previously mentioned a forefoot strike potentially utilizes elastic storage and return to a much higher degree than a heel strike thus negating the muscle activation consequences. Research by Ardigo et al. backed up the benefits of a forefoot strike, showing that a forefoot strike results in a shorter ground contact time and time of acceleration, both beneficial adaptations (1996). Given these facts, forefoot or midfoot strike are likely to be more economical, especially at faster speeds which require rapid force production and short ground contact times, such as middle distance events, or even distance events at the elite level.

It is likely that there will be individual differences in regards to foot strike, but forefoot striking may be more efficient mechanically, and perhaps more importantly may be needed for the

increased force production coupled with a short ground contact time that is seen in faster running (Kyrolaine et al., 2001). Additionally, forefoot striking allows for greater use of elastic energy storage and return from both the Achilles tendon and the arch of the foot. Similarly, Lieberman et al. also found that a forefoot strike allowed for more conversion of translational energy into rotational energy (2010).

Looking at the foot strike characteristics of elite athletes may provide some clues. Unfortunately there are only a limited number of research studies on the topic. My own experience in taking video of some of the best runner's in the world, and more recently using a high speed camera to do so, has been that the majority of elite runners have a mid or forefoot strike.

Hasewage et al. looked at foot strike at the 15km mark in a Japanese half marathon (2007). In the study, out of the 283 runners, 74.9% were rear-foot strikers. The rest were mid-foot and forefoot strikers. That leads many to conclude that rear-foot strike may be the way to go. However, if we look at a couple of other factors the picture gets a little cloudier. When you separate out the top 50, instead of using the entire group, those who mid-foot or forefoot strike jumps from ~25% to 38%. That significant difference showed that there was a tendency for more mid/forefoot strikers to be faster.

Secondly, if we look at ground contact time, there was a linear relationship with the faster runners having less ground contact time and GC increasing as you got slower and slower. In addition, forefoot/mid-foot strikers spent significantly less time on the ground than their heel strike counterparts (183ms vs. 199ms). These findings led the researchers to conclude:

> "The percentage of RFS increases with the decreasing of the running speed; conversely, the percentage of MFS increases as the running speed increases. A shorter contact time and a higher frequency of inversion at the foot contact might contribute to higher running economy."

However, the story is not complete. There are still two other factors that one has to take into account when looking at this study. First off, the video was taken at ~9.3mi into a 13.1mi race. In other words it takes place late in the race when fatigue has already set in. As fatigue sets in, more mid-foot and forefoot strikers become heel strikers. Thus, when you look this deep into a race, it could potentially skew the percentages.

Secondly, the study was done in a large Japanese road race. While there were several elite Kenyans and other nationalities, of the top 283, the vast majority were Japanese, which is very significant. This means that technically, the results are only generalizable to Japanese runners. Why is this significant? Because of how the Japanese historically train and how they historically run. Due to the heavy emphasis on very high mileage and moving everyone to the longer distances (half marathon and marathon) with neglect to the shorter distance races (1500,5k), the running style

of Japanese runners is much different than Americans, Europeans, and even Africans. Similarly, the traditional ideas taught by Japanese coaches at the time favored a running style that was more flat/scoot around type running. One of my coaching mentors started spending time going to Japan educating Japanese coaches on running mechanics at around the time of this study. We've spent many hours discussing what their views were, what the runners were being taught, and how their athletes were running, and his experiences confirm this idea.

Due to these factors, it's impossible to take this study and generalize it to anyone except Japanese runners. In addition, the study shows that speed is correlated with both foot strike and ground contact time.

In the only other studies to look at foot strike in faster runners, Hayes and Caplan evaluated foot strike during high level 1500m races in Britain. These studies are unpublished at the moment but were presented at the European College of Sports Science meeting in 2009. They looked at foot strike and ground contact time on each lap. This will not only give us an idea on foot strike implications but also on how that changes with fatigue. The conclusions that can be drawn based on the research about fatigue and training are very interesting.

In the 1500m, the range of times was from 3:45 to 4:22 with the average being 3:56. Once again, ground contact time-ground contact time was related to foot strike. Forefoot strikers spent 161ms on the ground compared to 169ms for mid-foot and 192ms for heel strike. The difference between heel strike and the other two are pretty remarkable. What is interesting is that ground contact increases on every lap.

Foot strike also changed based on lap. Initially on lap 1, 34.6% were forefoot striking, 46.2% mid-foot, and 19.2% heel striking. On lap 4 the picture changed slightly. More of the mid-foot strikers in particular had switched to heel striking (heel striking increased to 27 %.) What all of this means is that fatigue presumably caused a change in foot strike.

Before delving into the meaning of this, let's look at the results of the other study on 800m runners quickly.

Average time 1:55 (range: 1:47-2:01)

	Foot strike	Ground Contact
Forefoot	35%	156ms
Mid-foot	48%	161ms
Heel	17%	177ms
	Lap 1	Lap 2
Ground Contact	156ms	168ms

What does this all mean? It's hard to come to any firm conclusions, but the possible implications are very interesting. In regards to foot strike, there is a relationship between running speed and foot strike. Do all fast runners forefoot strike? No, but there is a tendency for the faster runners to forefoot/mid-foot strike more so than the slower runners.

Critiques of the fore/mid foot striking idea say that foot strike is a function of only speed. While speed certainly plays a role in where you strike to an extent, it's unlikely that it plays as much of a role as people make it out to. This means that someone is not going from a straight heel strike while running easy to a forefoot strike while running 800m pace. The previously mentioned Lieberman study provided the first evidence showing that forefoot strikers struck forefoot regardless of condition. Similarly, if we look at the data in the studies above, you can see that the percentage of foot strike types is remarkably similar despite the significant increase in average running speed between the events (from 63-64sec per lap down to 57.5sec per lap). Even though these are different people running each event, if foot strike was solely a function of speed, like many have claimed, then you would expect to see a definite trend away from heel striking as the group got faster.

Perhaps most interesting is what seems to happen during fatigue. Ground contact times increase in both studies, regardless of foot strike. If we remember back to what typically impacts ground contact time, it provides some interesting clues. Ground contact changed even when foot strike did not, so we can eliminate that possibility. The other two possibilities are that leg stiffness and use of elastic energy changed, which is entirely possible, but impossible to know or that rate of force development was diminished. There is some research showing fatigue changes leg stiffness and stretch shortening cycle fatigue. The most likely scenario is that fatigue is impacting the body's ability to produce force in as short a time period. The foot has to be on the ground longer in order to generate sufficient force to maintain the pace.

Lastly, in the 1500m study it was interesting to see a change in foot strike pattern. During the last lap, you saw an increase in heel striking. This would seem surprising as generally one of the faster laps in the race is the last as people try and kick it in. The question is why do runners switch to a heel strike under heavy fatigue? One possibility is that stride length tends to decrease with fatigue and runners are trying to compensate by lengthening their stride, but instead of doing it by pushing off and covering more distance, they simply let their lower leg reach out. Another possibility is that fatigue may impact fine control of the lower leg. As fatigue builds up, the body tends to ignore less critical functions, so the sensory feedback that tells the higher motor centers that we are heel striking are filtered out and ignored. While early on, the brain paid attention to the feedback and didn't want us to heel strike so that injury was avoided; under heavy fatigue the body is more worried about protecting critical functions so impact forces due to heel strike are seen as unimportant. Lastly, in a surprising twist, our body could make us heel strike subsconciously as a way to slow us down. While this seems contradictory, on a subconscious level it knows that if we

slow down fatigue will start to dissipate, therefore heel striking might be a sort of protective mechanism when we are venturing too far away from homeostasis. Regardless of the cause, it's obvious that lower leg biomechanics can play a role in not only performance, but also in resisting fatigue.

Structure

One last factor can impact biomechanical efficiency and that is the individual variance in the structure of a person. There are certain structural characteristics that make a person more efficient. A reduced lower leg mass in elite African runners has been hypothesized to partially explain their superiority in RE (Larsen 2003). This would be due to the fact that a reduction in mass at the extremes would reduce the energy cost of swinging the lower leg through during running. A study by Scholz et al. found that runners with shorter heels had better RE (2008). This is due to the fact that a shorter Achilles tendon moment arm results in greater elastic storage. Additional structural factors that affect running economy include body fat percentage, pelvis width, and foot size (Saunders et al. 2004).

Since an individual's structure can not, or only minimally, be changed, I won't go in to great detail on the subject. The good news is that there are a wide variety of runners with varying body shapes who have been successful on the elite level.

Optimizing biomechanics and Running Economy

Looking at how biomechanics affects RE is a difficult proposition. Most scientists ignore the fact that runner's have years of motor programming that has already developed neural pathways for their individual stride, thus are very efficient neurally at that stride. It is possible that a runner could be running less mechanically efficient, but has adapted to that stride after years of training so that their RE is relatively good. For example, in looking at Cavanagh's study, the question becomes are runners most efficient at their selected stride length because that is their optimal stride length mechanically or is it because they have adapted to that stride from years of running and thus has become their "optimal" because it has become an engrained motor program? In other words, with training could a different stride length become more efficient? Unfortunately, no long term training studies have been done to clear up this picture. As will be discussed shortly, there may be a flaw in measuring economy or efficiency that may be clouding the research results. The way economy is measured greatly effects the conclusions that can be drawn. At this point in time, I'm not confident in the measurement to be able to draw any conclusions from the research.

Comparison studies between elite and non-elite runners may help to provide clues on what optimal running mechanics consists of. In a study by Leskinen et al., world class 1500m runners (~3:35 for 1,500m) were compared to national class (~3:49) runners (2009). It was found that the elite runners demonstrated a more efficient use of the hip joint, and potentially more elastic energy

utilization. The importance of hip extension was also noted in the study by Kyrolainen et al. (2001). This should make sense, as hip extension can both generate a lot of force, and if done correctly results in a stretch reflex occurring that helps cycle the leg through the recovery phase.

The manipulation of biomechanics to improve RE in distance runners is often a subject of controversy. As mentioned previously, the complexity of efficiency makes discerning the effects on changes in biomechanics on RE a difficult one. This is due to fact that when changing biomechanics it should be expected that a short term reduction in economy would occur due to the fact that new motor programming has to take place and muscle recruitment patterns might change. Therefore, doing short term studies on changes in RE via mechanically manipulations might not be a good idea. However, evidence exists that some mechanical changes may improve RE. The study by Morgen et al. changed stride length using audio and visual feedback that resulted in an optimizing of step length and an improvement in RE (1994). This study points to the idea that mechanical manipulations can improve RE. Lastly, the practical experience of many of the world's best track coaches who focus on manipulation of technique provides evidence from the practical world.

Neuromuscular Efficiency:

In addition to the various components that impact mechanical and metabolic efficiency, neuromuscular characteristics need to be taken into consideration. This efficiency can be broken into two categories, factors which improve the neural signaling and motor programming of the running motion and those that improve the muscle force production itself. Recall how the running movement occurs. Reflexes and motor programs at the brain and spinal level combine that result in the running motion. Although it is often not considered one, running is a skill, just like hitting a baseball or swinging a golf club. Similar to those skills, practice is needed to improve efficiency at the activity. As the movement is practiced the body becomes more efficient by refining the motor program, learning exactly what muscles to recruit, what ones to inhibit, and the exact number of muscle fibers needed. It is through this refinement that the movement becomes better coordination and efficient. Research has demonstrated these claims, consistently showing that repeated practice results in improved muscle fiber recruitment and movement control (Bonacci et al., 2009). A more efficient recruitment pattern decreases RE because of the intricate linking between VO2 and RE.

Neuromuscular efficiency can also be seen through muscle activation studies comparing untrained with moderately or well trained individuals. In a study done comparing cyclists, differences in muscle recruitment were found. Novice cyclists showed greater variability in muscle recruitment between pedals, more variations of recruitment between each individual, more muscle co-activation, and longer muscle activation periods than well trained cyclists (Chapman et al., 2008). These results point to the idea that training can improve neuromuscular characteristics. The fact the

neuromuscular adaptations do occur gives an explanation for how resistance or explosive type training could improve RE, which will be discussed in detail later.

Applying Force to the Ground:

The muscle contraction itself also plays an important role in neuromuscular efficiency. The ground reaction forces (GRF) and contact time influence both neural and biomechanical efficiency. GRF is explained by its namesake, it is simply the forces that occur when the foot makes contact with the ground. These forces are typically shown in their vertical form as the force exerted upward as the foot hits the ground and then as the person applies force to the ground. GRF can also occur in two other planes, side to side and front to back. Ground contact time is self explanatory and refers to the amount of time that the foot is in contact with the ground. These two factors combine to partially determine speed. To be successful, a runner not only needs to impart a large amount of force into the ground for propulsion, but they need to do so in a short period of time, often around a tenth of a second. Thus, it is not only about force production, but about rate of force production.

A study by Weyand et al. uniquely demonstrated this by comparing ground reaction forces in running, hopping, and backwards running (2010). They demonstrated that during the other activities higher peak forces could be developed but ground contact time was longer, thus it took longer to develop such forces. During running, the forces weren't as high, but were generated in a very short amount of time due to the short ground contact period. Their study demonstrated that total force development is not the limiter in how fast someone can run, but rather it is how quickly force can be developed.

In terms of RE and long distance running, a runner would be most economical if they produced only the amount of force necessary to run at a given pace. In a study by Wright and Weyand, they found that the determining factors for VO2 were rate of ground force application and the amount of muscle mass activated (2001). Therefore these two factors combine to impact RE. Backing up this idea, Heise and Martin found that the amount of vertical GRF, when combined with the time it took to apply this force, explained 38% of the variability of RE between individuals in their study (2001). Therefore, maximizing rate of force development so that ground contact is minimized creates a more efficient runner.

Muscle fiber characteristics

Lastly, the characteristics of the muscles themselves impact efficiency both metabolically and neuromuscularly. In particular the muscle fiber type is of particular interest. There have been several studies that have found a correlation between ST fiber percentages and improved RE (Svedenhag & Sjodin, 1994; Saunders et al., 2004). Other studies have found correlations between cycling efficiency and ST fiber content (Coyle et al., 1992). This is partially due to the ST fibers being

better equipped to utilize oxygen due to its increased mitochondria, myoglobin, and Krebs cycle enzymes. With an increase in mitochondria, less oxygen will be used per mitochondria chain. This would result in a more efficient use of oxygen, resulting in a decrease in VO2. Another explanation is based on the mechanical efficiency of the different muscle fibers. For a specific muscle fiber its peak efficiency occurs at one third its maximum contraction velocity (Coyle et al., 1992). Given that endurance training and competitions rely on slower muscle contractions, ST fibers will be closer to their peak efficiency during endurance activities. Backing this up, Coyle et al. found that during endurance cycling, the contraction velocity of the vastus lateralis was close to the peak efficiency rate of ST fibers, leading to greater mechanical efficiency (1992). Thus, ST fibers are likely more efficient when an activity requires slower contractions, while FT fibers are most efficient during activities that require very rapid shortening.

In addition, the makeup of the individual muscle fibers themselves could impact RE. Titin has been shown to have elastic properties which aid in the storage and then subsequent utilization of that energy. Thus Titin could potentially spare chemical energy use (Kyrolaine et al., 2001). It is possible that each fiber type stores elastic energy to a different degree.

Metabolic Efficiency

The processes that result in energy production determine metabolic efficiency. How much oxygen is needed by the muscles, the efficiency in delivering and utilizing that oxygen, and how well substrates can be utilized all play a role. Each of these factors can be improved to a large degree with training.

Oxygen's role should be readily apparent after reading the chapter on how Oxygen is delivered and utilized. Additionally, the traditional measurement for running economy relies on VO2 at a given speed to quantify efficiency. That explains why efficiency is often looked at in terms of the percent of VO2max that a person can run at for a given race distance. The lower percent of maximum he can run at it, the more efficient the runner is thought to be. Efficiency is impacted by many of the steps from oxygen intake to transportation to utilization that were discussed in Chapter 3.

Substrate Utilization

The preferred fuel source in endurance races that last less than a couple hours is glycogen. The body has a finite supply of glycogen stored in the muscles, while it has a relatively large supply of fat. Due to the limited glycogen stores, in longer distance races, such as a marathon, glycogen depletion is a major source of fatigue (Brooks et al., 2004). However, in races under an hour, which consists of most of the distance races (3k, 5k, 10k, half marathon), glycogen depletion is generally not considered a major source of fatigue, but instead plays a large role in determining the training

load (Costill & Trappe, 2002). Given its importance and limited supply, how efficiently glycogen can be used is an important factor in training runners. Reducing the rate of glycogen use at a given speed will improve the metabolic efficiency of a runner.

For the marathon, substrate use and total glycogen stores greatly influence performance. While glycogen is never fully depleted during such a race, it seems that the body induces a slowing of the pace as glycogen levels drop, presumably to limit the reduction in glycogen stores. Well trained runners use a mixture of fuel sources of carbohydrate, fat, and a minimal amount of protein. Thus to delay fatigue from low glycogen, either an increase in total glycogen storage, or a shift in being able to use more fat as a fuel source at marathon pace is necessary. Long runs and high mileage training tends to increase the total glycogen supply of the muscles. A change in the fuel source ratio also comes from long aerobic running and moderate paced aerobic running. Additionally, research has demonstrated that dietary manipulation, such as training in a fasted state, is one way to increase the use fat as a fuel. Therefore, for a marathon runner, some runs and long runs should be done without taking supplemental fuel as low glycogen during the run seems to be the signal for a shift in substrate use.

Even though glycogen depletion is not a direct cause of fatigue in shorter events, adaptations that change substrate utilization can aid performance in middle distance events. With the high volumes of training done, glycogen depletion during training can impact training intensity and adaptation. A large percentage of training is done at easy to moderate paces (Esteve-Lanao et al., 2007). The intensity of this running allows for significant contribution of fat as an energy source, which would help alleviate the strain on glycogen stores during high volume daily distance running. One training adaptation that shifts the balance between fat and carbohydrate use towards fat usage is mitochondria density (Brooks & Mercier, 1994). An increase in mitochondria and mitochondria enzymes, which occurs with endurance training, allows for greater use of fat as a fuel.

However, it is also possible that glycogen depletion could play a role in fatigue in shorter races. While glycogen depletion is usually only thought of in terms of long duration exercise where total glycogen depletion is a problem, it can occur at a local muscular level. Glycogen breakdown occurs within each working muscle fiber, in addition to some total body glycogen breakdown stimulated by hormones. It can not be transported from fiber to fiber, except when converted to lactate, transferred, and then reconverted to glycogen. Given this, glycogen depletion can occur in individual fibers before total glycogen depletion occurs, and the selective depletion depends on the intensity of the activity and the use of that fiber (Abernathy et al., 1990). Therefore it can play a role in fatigue in much shorter events. With glycogen depletion of a muscle fiber, that fiber can no longer perform work at the same intensity, therefore with fibers getting glycogen depleted, there are less total fibers to cycle in and do work, potentially causing fatigue due to the increased demand put on the other fibers. This can be seen in animal models that show after partial glycogen depletion, anaerobic performance was impaired (Lacombe et al., 1999). Research done by Jacobs et

al. showed the specificity of glycogen depletion when they compared a group that almost exclusively depleted ST fiber glycogen with one that partially depleted both ST and FT fibers (1981). In the ST only group, maximum strength was not impaired but muscular endurance was. In the FT and ST depleted group, maximum strength was impaired, presumably because of the glycogen depletion in the FT muscle fibers.

Substrate Utilization and the actual Running Economy measurement

Due to the way RE is commonly measured, what the energy source is will effect the measurement. RE is commonly calculated by dividing the rate of oxygen uptake by running speed. VO2 is used because, when measured at a speed using almost entirely aerobic energy sources, it represents the amount of ATP used. However, the common measurement does not take into account that the energy equivalent of O2 depends on which substrate is used. Fats, carbohydrates, and proteins all provide a different amount of energy per liter of O2.

When expressed as a unit of caloric cost, RE is more sensitive to changes in substrate use. This is particularly important in distance events that are limited by fuel issues, such as glycogen depletion. Thus while substrate utilization might not be as big of a worry for a middle distance athlete in terms of efficiency, for a marathon runner it is one of the deciding factors. A marathoner who is able to utilize fats more will spare his glycogen stores, delaying fatigue. For the marathoner it is not only about total glycogen stores, but the rate at which those are used. His goal is to run as fast as he can while using glycogen at a rate that will put him depleted just after the finish line. Training to be able to utilize a greater amount of fat at marathon pace will increase performance. For our marathoner, he would be more concerned with looking at RE in terms of caloric cost, because he is less concerned with RE as measured in terms of VO2 as that is not his race limiter. That brings us to one of the problems with RE measurements. It can be expressed in several different ways that all take into account different efficiencies (metabolic, biomechanical, or neural) to a higher degree.

Problems with Running Economy

As mentioned previously, one problem with RE is that it can be measured and expressed in several different ways. The most common ways are expressing it in terms of VO2 at a given speed, and VO2/distance. There are two ways to report VO2 using running speed. One is to express it in terms of the VO2 at an absolute speed, while the other is to express it as VO2/speed. These rely on the idea that below a certain speed (around Lactate Threshold) the oxygen cost is independent of speed. However, Vickers demonstrated that this is not the case (2010). When measuring economy at an absolute speed, runners will be running at vastly different relative intensities that will change several factors. Biomechanical factors such as ground contact, ground reaction forces, stride lengths

and frequencies, and are all speed dependent. One individual at 6 minute mile pace may need a very low percentage of their max force development, while another runner might need a much larger percentage, thus changing ground contact times. That is but one example of how speed affects biomechanics. In addition, substrate use would be vastly different among groups of individuals at an absolute speed. Since races are of a given distance, expressing RE as the oxygen cost per unit of distance seems to make more sense, but it still neglects the effect substrate use has on VO2.

One last way to that RE is reported is caloric cost. In this measurement, the Respiratory Exchange Ratio (RER) is used to calculate the caloric equivalent of Vo2, which can be used to determine the amount of energy used per unit of distance. A study by Fletcher et al. compared RE expressed as VO2/distance and caloric unit cost (2009). They stated that the traditional way of measuring RE appeared to be flawed for several of the reasons discussed earlier, and for the practical reason that faster runners tended to have worst economy which makes little logical sense. In their study, Fletcher et al. found that the better runners had a lower energy cost of running (better economy) and that RE expressed as caloric unit cost was much more sensitive to changes in speed. One of the interesting findings was that RE using the VO2 measurement showed that RE was independent of speed. The caloric unit cost showed that RE got worse as speed increased, which would be the expected outcome as when speed increases reliance on carbohydrate also increases. This debate on how to measure RE demonstrates that we have a long way to go in understanding efficiency. It also makes one question the conclusions based on studies looking at what improves RE.

One other problem with measuring RE is whether to use gross or net VO2. Gross VO2 refers to the total oxygen consumption, while net refers to the total oxygen consumption minus the resting value. It has been hypothesized that net would better reflect active energy usage and correct for a speed-related bias that is present when measuring RE as VO2/speed (Vickers, 2005).

RE is used as a means of quantifying total efficiency. However, given that one is basing efficiency off of VO2, the merits of this approach may be in question. According to Noakes' CGM VO2 may be a result of a central governor regulating VO2 by muscle activation. If this is true, then RE reflects the degree of muscle activation at a given pace. This still means it is a valid measurement and it may reflect efficiency of muscle activation. However, the degree that it can be used to measure and quantify mechanical, metabolic, and neuromuscular efficiency is unknown. Separating efficiency into these three different categories might aid in clearing the quiet muddy picture of what effects RE, or in this case total efficiency. The interactions of the different types of efficiency might also explain the mixed results in changing biomechanics or why there are contrasting results on what is the most efficient way to run in the literature. In a study by Purkiss et al. they demonstrate a model of calculating internal work, which allowed for distinguishing efficient and inefficient running styles in elite runners (2003).

Also, the measurement of RE may also be suspect if we look at a recent study by Baden et al. In that study, RE changed when the participants did not know when they were stopping exercise, even though they ran for the same amount of time as during the test when they knew how long they were exercising for (20 minutes) (2004). The results of these studies demonstrate that another mechanism may regulate RE based on feedback. The fact that RE could change solely based on outside feedback such as when exercise would be completed, is interesting and may lead to reevaluation of the way we test RE. Lastly, efficiency needs to be put into context when using it as a performance measure. The fact that performance is the main concern and not just efficiency should be taken into consideration. It is possible, because of the way we measure efficiency, that a parameter that improves RE could be detrimental to performance. For example, increased vertical oscillation decreases RE, but if that vertical oscillation results in a far greater stride length, it is possible that the person has increased efficiency per unit of distance traveled. Taken by itself, vertical oscillation might be detrimental, but taken in context of the whole locomotion a slight increase could prove beneficial. An example of this can be seen in a study done in converting runners to the POSE technique of running. After 12 weeks of training, the runners reduced their vertical oscillation significantly, by 1.52cm on average, but their RE was significantly impaired (Dallam et al., 2005). This was due to a significant reduction in stride length. Thus, it is the optimization of the variable, and giving hard recommendations may not be appropriate.

In conclusion, the measurement of RE may need to be called into question as a measure of total efficiency. This would have significant repercussions in the interpretation of many studies. Looking at other methods of calculating efficiency may be more beneficial, as the two groups mentioned above have done (Fletcher et al., 2009, Purkiss et al., 2003). Further more, evidence from the world of cycling may shed some light on using a measure based on VO2. Using a measurement of VO2 economy, cyclists were most efficient at pedaling rates that were relatively low, 40-65 revolutions per minute (rpm), when professional cyclists preferred pedal rates that were much higher, over 90rpm. These higher rates consistently show a much higher sub maximal VO2. Rather than coming to the conclusion that the elite cyclists were wrong and needed to have lower pedaling rates, Belli and Hintzy instead expressed efficiency as the energy cost of cycling, in terms of joules per kilogram per minute (2002). Using this method, it was found that the energy cost was lowest at pedal rates that matched those preferred by elite cyclists. The authors concluded that energy cost represented biomechanical efficiency much better than the economy measurement. A similar scenario could easily play out in running if we looked at stride rate and length instead of pedal rate. These results may explain some of the discrepancy between what coaches suggest is correct biomechanically and what appears in the research, the heel strike versus mid-foot strike and the degree of vertical oscillation would be prime examples. The problem isn't with the importance of economy or efficiency, but with the actual measurement used to define economy.

Chapter 7: The Brain-Muscle Connection

The aforementioned VO2max, Running Economy, and Lactate Threshold make up the traditional trio of running performance, but there are several other factors that contribute to performance such as anaerobic, neuromuscular, and psychological factors. While distance running and even middle distance running is primarily an aerobic activity, anaerobic systems play a significant role. In the 1,500m estimates of anaerobic contribution have generally been found to be about 20%, while in the 3,000m event for men 16% anaerobic contribution was found (Hill, 1999). Despite this fact, for years the traditional research ignored the anaerobic contributions to performance in their models. In the following chapter, we'll delve into some of the neglected contributors to running performance work.

Understanding the entire picture is crucial for athletes and coaches looking to maximize performance. It is easy to focus on the main contributors, but when we consider that well trained runners are thankful for any improvement they can get, the smaller contributors can not be neglected. Traditional training programs focus mostly on the aerobic aspect of running, neglecting the neuromuscular and even anaerobic. Since the time of Lydiard, the anaerobic contribution has been misunderstood. It is not the evil system that it is often portrayed to be, but instead requires a little more fine tuning and balance than the aerobic system. Let's take a look at these nontraditional factors and see what they mean in regards to performance.

Neuromuscular and Anaerobic factors in performance

Since the majority of distance running performance is tied to traditional factors focused on aerobic components, most of the research has followed that route. However, recent research has shown that anaerobic and neuromuscular factors play a role in improving distance running performance. One such study was done by Paavolainen et al. which found that better performance in a 10km run was partially due to increased pre-activation and decreased ground contact time (1999). Pre-activation could potentially increase RE due to a tightening of the muscle-tendon system, which would increase elastic storage. In another study by Paavolainen, performance on a maximal anaerobic running test was significantly correlated with running performance over 5,000m (1999a). Lastly one study found that anaerobic work capacity and power contributed significantly to a model of explaining running performance in Cross-Country and over a 10km race (Sinnett et al., 2001). These studies demonstrate that anaerobic factors are crucial for distance running success.

There are many components that affect anaerobic capacity and power. Anaerobic power refers to the rate of energy turnover for the anaerobic systems, while capacity is the total energy

turnover. Factors that affect both of these include muscle fiber type, recruitment, and force production.

Muscle fiber type

Muscle fiber type will largely influence anaerobic capabilities. The differences between ST and the various FT muscle fibers (FT-a,x,c) make this clear. FT fibers have a higher Myosin ATPase activity, which means it also has a quicker shortening velocity (Fitts & Widrick, 1996). Due to this increased shortening velocity, FT fibers also have greater power outputs. FT fibers also have more muscle fibers per motor neuron, which allows for greater force production than ST fibers, which have fewer fibers per neuron. Lastly, FT fibers have a larger Creatine Phosphate store, more Glycolytic enzymes, and a larger Sarcoplasmic Reticulum. The larger SR will result in enhanced Calcium delivery (Stephenson et al., 1998). It can be seen then that, the fiber type makeup of the muscles of an athlete will help determine his anaerobic capabilities.

Conversion in muscle fiber type was previously believed to be impossible. However, recent research suggests that that view is erroneous. Changes within Fast Twitch fibers in humans have been demonstrated in numerous studies. Conversion from FT-x to FT-a is now commonly accepted. A full conversion from FT to ST is still debated in humans. If we look at research in animals, this conversion can be seen (Pette, 1999). However, it takes a large amount of chronic low-frequency stimulation to get this conversion to occur. In fact, some research demonstrates that it takes a large amount of damage for this conversion to occur (Pette, 1999).

Another piece of evidence for fiber type conversions is common sense. We would have to believe that every single distance runner pre-selected themselves into distance running, since the almost every runner tested shows a large percentage of Slow Twitch fibers. Lastly, if we look at longitudinal studies that looked at muscle fiber types in high level endurance athletes, changes can be seen. A longitudinal study by Rusko et al. found that after 8 years of training and a doubling of training volume, the percentage of ST fibers in a group of Cross-Country Skiers increased by 11% (1992).

Why do I bring up the possible change in muscle fiber type? One reason is that it could partially explain why runners tend to reach peak performance levels later than sprint or power type athletes. It could be because distance runners need large amounts of training, or damage, to complete this muscle fiber type conversion. Secondly, this fiber conversion could also partially explain the success of African distance runners. Perhaps their large volumes of unofficial training early on, such as running to school or even the active lifestyle that they live, allows for this conversion to take place at a younger age. Africans tend to reach peak performance at younger ages then their Western counterparts and this could partially explain that phenomenon.

The long term changes in muscle fiber type are important for the coach to understand. It points to the idea that training can not be seen through a segmented season to season viewpoint, but should instead be seen through a lifetime and long term view. The athlete changes from the start to the end of his competitive career and training should be adjusted to accommodate and even facilitate those changes. That is why we can not get stuck in a cookie cutter coaching plan or have one dogmatic philosophy.

Fatigue and the CNS

From a neuromuscular standpoint, fatigue can occur either centrally or peripherally. Central fatigue means that it occurs sometime between the CNS initiating and the muscle receiving the signal for contraction to occur. Peripheral fatigue occurs between the time when the muscle receives the signal and actual contraction takes place. Fatigue at both levels is generally reflected by changes in motor unit recruitment, decrease in relaxation time, and a decrease in neural signaling among other changes. A build up of metabolic by-products, depletion of fuel sources, or a decrease in central drive can all affect the neuromuscular system by decreasing force output. Attention will be given to changes that occur along the route of the CNS signaling for muscle recruitment to take place. The focus is not on what exactly causes neural fatigue, as that differs for each event, but on how it manifests itself during a race and how to combat it. There are several different factors that can be altered to prevent fatigue from a neuromuscular standpoint. Changes that can occur neurally that may enhance performance include increased muscle fiber recruitment pool, decrease in time to activate muscle, and increase in rate of force development.

The CNS decision on the specifics of muscle fiber recruitment affects both anaerobic attributes and neuromuscular characteristics. There are several potential ways that muscle fiber recruitment can delay fatigue. First, an increase in the muscle fiber pool allows for more fibers that are available to do the work, thus delaying fatigue. Secondly, during fatigue a decrease in the Action Potential signal strength and velocity occurs (Leppers et al., 2002). Training to decrease the degree of activation needed for those fibers to be recruited allows for the ability to recruit such fibers when fatigue is high and the neural recruitment signal is not as strong.

In a study by Nummella et al. a significant correlation was found between EMG levels at the 3km mark of a 5km time trial and the average velocity that the 5k was run at (2006). Essentially, the runners who could maintain a higher level of muscle activation during the middle section of the 5k had better finishing times for the entire distance. This shows that maintaining muscle recruitment during a distance race is an important factor in success. If we tie this back into the fatigue models, it's clear to see why. The runners whose recruitment is dropping are having fatigue build up earlier than the others. Depending on the fatigue model, training to increase muscle fiber recruitment during hard training may be one way to combat this problem. If we can "force" recruitment during

heavy fatigue with high force activities, it will train the maintenance of recruitment during a race. Some preliminary evidence for this can be seen with strength endurance work.

Another interesting find in this study is the increase in EMG at the end of the race when the runners demonstrated the "end spurt" or "kick" phenomenon. The ability to recruit additional motor units under heavy fatigue at the end of the race impacts ones ability to increase the speed in the closing stages of a race. This idea could give an explanation why heavy resistance training or sprinting improves distance running performance, as it increases the total muscle fiber pool that can be recruited. Some evidence for this idea is that in the Nummela study, the velocity of this final lap end spurt was not related to VO2max or RE, but to the vMART, suggesting that neuromuscular abilities are what determine the end spurt, and not the oxygen dynamics.

In distance running, the various muscle fibers work so that they are rotated so that some fibers do the work while others rest, then the others cycle in and take over. This works well at low intensities, but as intensity increases more force is recruited, so more fibers have to be activated to do the work and time that they can "cycle off" and rest is reduced. As fatigue sets in, fibers begin to fatigue, and harder to recruit fibers are called upon. When fatigue gets intense, there is a decrease in the number of muscle fibers recruited as the neural drive decreases and the number of fibers that are fatigued has reached a high level. This is where increasing the muscle fiber pool comes into play. Even during maximum voluntary contractions, a person can never fully recruit all of his fibers. There is always a reserve. Research has demonstrated that well trained athletes are able to tap into a higher percentage of that reserve then untrained. For distance runners who rarely ever do high force activities like sprinting, there total recruitable fibers will be lower than those runners who have trained to recruit fibers during high force activities. The increase in fiber pool allows for more muscle fibers to be recruited or cycled in, thus delaying fatigue.

Another neural factor is rate of force development. As has been covered already, the amount of time that a runner has to produce force while the foot is on the ground is very limited. Weyend et al. demonstrated that we do not have time to reach max force levels even during sprinting (2010). Therefore, it is more a matter of rate of force development. If distance runners can be trained to increase their rate of force development, this allows for several beneficial adaptations. At slow speeds it means they are working at a lower percentage of their max force production capacities, which should extend endurance. During race speeds, it also means that they can generate the necessary forces required to run at their pace faster. This means that ground contact times can be reduced, as not as much time is needed to generate the force. Previous research has shown that decreased ground contact times results in less time having to support the weight of the body and thus improving Running Economy. One result of training that improves rate of force development is a decrease in the time required to activate a muscle. Research using explosive strength training by Mikkola et al. demonstrated that with training endurance athletes can be trained to enhance rapid muscle activation (2007).

Further research in the area that anaerobic and neuromuscular factors play in regards to distance running is needed. The ability to close fast in the later parts of the race has been suggested as what separates the best runners, and it has been speculated that this might be due to an ability to increase motor unit activation in the later parts of the race (Noakes, 2003). The aforementioned data by Nummella et al. seems to support this notion (2006). The previously mentioned field observations in elite Kenyans being able to run at a MAXLass during a race and then dramatically increase lactate at the end might also lend credence to this idea.

The CNS is an often overlooked factor in running performance. As distance coaches we are used to recognizing fatigue that results from running lots of mileage or hard workouts. What we don't grasp is CNS fatigue. The fatigue experienced after doing 50m sprints is a totally foreign concept. Distance coaches need to recognize CNS fatigue and implement a recovery strategy for it just like we do for recovery from a long run. Additionally, coaches can plan strategies to improve the neuromuscular performance of an athlete by exploiting training practices to improve muscle recruitment and coordination.

Recognizing CNS Fatigue:

Any activity that requires a large neural demand will result in CNS fatigue. For instance, all out sprints, heavy weight lifting, or very taxing interval work all will result in various amounts of CNS fatigue. Adequate rest and recovery from such high demand CNS work is needed. It is why sprinters do not do a large volume of repeated all out sprints day after day.

Several strategies can be used to determine if CNS fatigue is present. Explosive tests such as a broad jump are useful. As are reactive tests such as a repeated hopping exercise in which you measure ground contact time. If ground contact time is increased, it's likely that there is CNS fatigue present. Lastly, how an athlete feels or looks during faster strides before a workout or run can give the coach an indication of fatigue. Often times using one of these methods prior to a workout or the day before can tell a coach if the athlete is prepared for the next workout or needs more time to recover.

Temperature Regulation

An increase in body temperature is another mechanism that can cause fatigue. When core temperature gets to a dangerously high level, people either slow down dramatically or stop exercise. An interesting fact is that people terminate exercise at very similar temperatures of around 40 degrees Celsius. This termination of exercise happens regardless of starting temperature, pre-cooling of the body, or rate of heat loss and storage (Gonzalez-Alonso et al., 1999). There is a theory that exercise in hot conditions, where fatigue is likely due to reaching high core body temperatures, is governed by an anticipatory regulator. This anticipatory regulation system monitors rate of heat storage and adjusts muscle activation to delay or prevent overheating. Evidence for this theory comes from studies that have studied EMG levels in a variety of temperatures. Lower EMG levels

have been seen in hotter temperatures, which should not be surprising, but the anticipatory regulation can be seen that EMG levels are lower early in the exercise when core temperatures were essentially the same (Tucker et al., 2004). Regardless of the exact mechanism, it can be seen that slowing the rate of heat storage can enhance endurance. In addition, other mechanisms related to heat and core temperature increases can potentially impair performance, even before the critical core temperature is reached.

At higher temperatures more of the blood is redistributed to the skin to aid in cooling. When this happens, blood flow that is normally going to the muscles is reduced, potentially hampering performance. Also, at high muscle temperatures, ATP production is decreased due to a loosening of the coupling between oxidation and phosphorylation in mitochondria (Brooks et al., 2004).

Delaying heat gain and core temperature increases depends on the bodies cooling mechanisms. Heat loss can occur via radiation, conduction, convection, and evaporations. Radiation allows for heat loss due to a difference in temperature gradient. When the skin temperature is higher than its surroundings, more heat radiates from the body. This process accounts for 60% of heat loss (Brooks et al., 2004). Conduction is the transfer of heat via direct contact and accounts for around 3% of heat loss. Convection is the transfer of heat to air or water and accounts for about 12% of heat loss at room temperature (Brooks et al., 2004). Blood flow to the skin aids the process of convection. Heat travels with the blood from the core to the skin, which then allows for heat loss to the air. In this way, heat is transferred from the core to the environment, aiding in temperature control.

Lastly, Evaporation accounts for about 25% of heat loss in normal conditions (Brooks et al., 2004). For heat loss to occur, sweat must evaporate. In humid conditions, evaporation of sweat is greatly reduced, thus decreasing heat loss through evaporation. In addition, in very hot temperatures where the outside temperature is greater than the body temperature, evaporation is the only means of heat loss since the other methods are dependent on having a temperature gradient. An acclimatized person can sweat up to 4 liters per hour, while a person not acclimatized can only sweat about 1.5 liters per hour (Brooks et al., 2004). Therefore, heat acclimatization to get this affect could potentially enhance endurance performance.

For athletes competing in hot environments recognizing that heat and temperature regulation may be the cause of fatigue is paramount. Practices can be designed to minimize the effect of temperature on fatigue. For instance, during the 2004 Athen's Olympic marathon several elite runners used ice vests pre race to decrease their core body temperature because the temperatures were extremely warm for a marathon. Their coaches recognized that heat regulation was going to play a bigger part than usual and designed a strategy to combat that. Recent research has suggested that drinking ice cold fluids helps to combat the core body temperature rise. Using strategies such as these can make a large difference in hot weather races.

The Psychology of it all

Racing is a very intense activity. Dealing with pain or mental toughness is often cited as a reason for success in the sport of distance running. The ability to deal with fatigue and pain could lead to improved performance. Despite the rigors of training and racing, running remains one of the most popular activities in the U.S. A look at the motivation of runners and how they deal with pain is essential in understanding what makes a successful distance runner.

Motivation

The motivation of runners has been analyzed with a variety of different approaches. Clough et al. (1989) took an interesting approach looking at running in terms of other leisure activities. They found that runner's motivation could be divided into six groups: Well Being, Social, Challenge, Status, Fitness/Health, and Addiction. The most important factors were challenge followed by fitness/health and well being. While the first four factors were similar to those seen in other leisure activities, the last two factors were different. This study set the stage for acknowledging that there was something extra that set running apart from other similar activities.

While studies initially compared runners as a whole group, latter studies pointed to the idea that there are different groups of runners motivated by different factors. One such study was done by Slay et al. (1998) in which they found that obligatory runner's motivation differed from non-obligatory runners. The obligatory runners were more motivated by negative or external factors such as guilt of stopping, and weight control. In another study on committed runners who ran more than 40 miles per week, it was found that the committed runners were more motivated by mastery, competition, and weight regulation (Scott and Thorton, 1995). Lastly, Ogles and Masters (2003) found that marathon runners could be divided into several groups based on their motivation profiles. These groups were: Running enthusiasts, lifestyle managers, personal goal achievers, personal accomplishers, and competitive achievers. These studies demonstrate that runners' motivation differs based on their level of running, goals, and experience.

Motivation also tends to vary based on skill level. Although not done with runners, a study by Chantel et al. (1996) looked at the differences in motivation in elite athletes. They looked at motivation in terms of Deci and Ryan's Self Determination Theory, using the Sports Motivation Scale (SMS) as a way to measure Self Determination. The Self Determination Theory states that there is a continuum of motivation that includes three different levels of internal motivation, three levels of external motivation, and one level of amotivation. Internal motivation refers to participating in the activity for its own satisfaction or pleasure. In other words, it is motivation based on the activity being an end in itself. External motivation includes doing the activity as a means to an end or for some external reason. Finally, Amotivation is a lack of motivation and is first on the continuum.

Chantel et al.(1986) found that the best athletes had higher levels of amotivation and non-self determined extrinsic motivation. Non-self determined refers to external regulation and introjected regulation. It should be noted that this study was on Bulgarian athletes who were likely influenced by communist practices. Looking at what motivates elite Kenyan runners, Onywera et al. (2006) found that they were primarily motivated by economic incentive, followed by talent and tradition. This finding would seem to match up with elites having higher extrinsic motivation.

The motivation to run is different at each level of running so coaches have to deal with a wide range of motivations. The key is finding out what motivates each individual runner and to exploit that. In an ideal world, runners would have both high intrinsic and extrinsic motivation, meaning they are motivated by internal and external factors. However, the reality is that coaches need to understand where each runner falls and exploit it. Additionally, as a coach, it is important to try and influence the motivation of an athlete so that he can enjoy the sport in the long term. For example, trying to slowly change someone who is motivated by a fear of failure to a more healthy motivation is important. The bottom line is that knowing what makes your athlete tick is valuable information that can be used to enhance their performance.

Pain Tolerance

Studies have shown that perceptions of pain differ within subjects and genders (Koltyn et al., 1998). Since the pace that a person is running during a race is self controlled, they have the ability to slow the pace down consciously before ultimate fatigue takes place. The degree of pain and how well this is dealt with mentally could affect when or if this voluntary slow down occurs. With a slowing of pace comes some relief from the pain, it makes sense then that the ability to deal with or fight this pain could impact performance.

In a study by Garcin et al. showed that perceived exertion changed based on the sport that one participates in. Even though participants from three different sports were running at their same individual percent of Maximal Aerobic Speed, the distance runners perceived the exercise as lighter. In addition, the distance runners felt that they could continue for a longer time (2003). In a similar study, Garcin et al. found that Rating of Perceived Exertion (RPE) was related to aerobic fitness level on a graded exercise test, meaning that the lower aerobically fit the athlete; the more strenuous they felt the exercise was (2004). Lastly, a study by Marcora et al. demonstrated that mental fatigue can impair performance (2009). In comparing groups who performed 90min of cognitive demanding tasks to a control group that watched documentaries, they found that the group that performed the cognitive tasks performed significantly worse on a cycle to exhaustion.

While readily apparent and intuitive to the coach, the idea that pain perception plays a role in controlling performance is a novel concept in Science. We've all seen this phenomenon as athletes and coaches, in the form or end of the race kicks, but it is nice to see that Science is catching on.

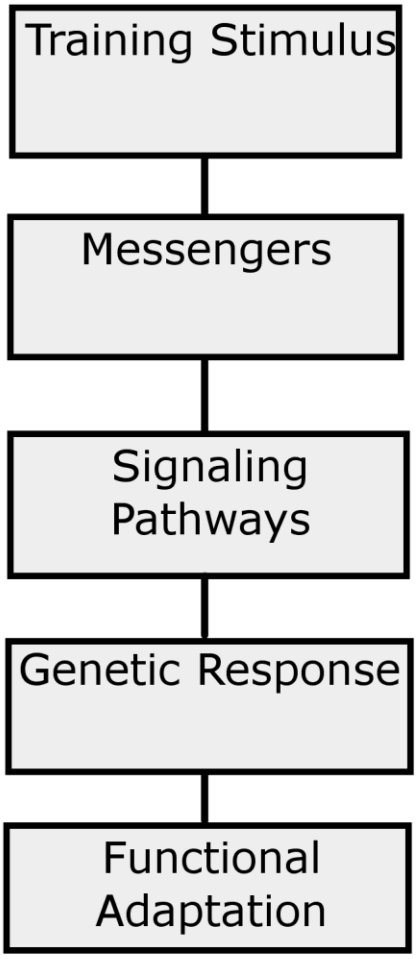
What this means is that while fatigue does occur via several physiological mechanisms, it is not catastrophic, meaning there is always a reserve leftover. Instead, fatigue is a signal that gets processed both consciously and subconsciously. How an individual deals with this perception can have a large impact on their performance. This can be seen in the aforementioned studies. There is always a larger reserve left. Marcora et al. demonstrated this in one study in which they had cyclists perform a 90min cycling test to exhaustion. Upon completion of the test, they then had the participants immediately complete a 5 second all out sprint on the bike while measuring power output. Despite riding to "complete" exhaustion during the 90min ride, participants were able to increase the power generated by 3 times during the 5sec add on. What this means from a science standpoint is that there is always a reserve left and that fatigue is an integrated phenomenon of physiological processes giving feedback which are then interpreted consciously and subconsciously. Using data from his studies, Marcora has developed a theory on fatigue that is based on motivation. His contention is that we consciously choose to slow down based on our perception of pain. Essentially, when we choose to slow down depends on our motivation levels to continue.

The Psychological and Physiological are intricately linked and we have much to learn in these areas of study. High level runners are likely to have higher degrees of both intrinsic and extrinsic motivation which allows them to delve into the deeper depths of perceived pain. Additionally, well trained runners probably have a callusing sort of effect, where they get used to certain levels of discomfort and pain during practice, so that they perceive it as being less painful then other runners. The important take away message is that they are psychologically and physiologically linked. It is a mistake to blame poor results on an athlete being "weak" as that is too simplistic of a viewpoint. Instead, recognize that pain is the result of a combination of many factors and work towards improving an athletes ability to handle it in the correct fashion.

Chapter 8: The Genetics of Training

Knowing all of the physiological components that can change with training is important, as is knowing what kinds of training improves those components, but how do we get from the workout to those functional changes? The body must go through a series of steps that takes it from the training stimulus all the way down to the genetic level and then back up to a functional change. This process is called the signaling pathway and is demonstrated in the chart below.

While the coach who reads this may get discouraged by the references to certain complex sounding names, it is important to not get caught up in the minutia but understand the concept. How a training stimulus results in a tangible adaptation such as increased endurance is paramount for making good decisions on workout planning. By knowing the adaptation steps that the body goes through, the specific aims of a workout and the recovery post workout can be set up to optimize this process. For example, if we know adaptations related to fuel use are triggered by low glycogen stores, then that gives us information that we can use to create a training stimulus that creates low glycogen stores. In this case, a long run would be the training stimulus which would seem obvious, but what might be overlooked is that knowing low glycogen stores is the signal for adaptation we might want to forgo a carbohydrate drink on that long run.



Steps of Adaptation

Training Stimulus

The first is that a training stimulus is applied; this is what occurs during the workout. In order for the workout to elicit adaptation, it has to be significant enough both in terms of intensity and volume to disturb the body's homeostasis. The body's homeostasis can be disturbed in a variety of ways ranging from stressors that will impact the mechanical, metabolic, neural, endocrine and cardiovascular

systems. In reality, all of these systems function together. Two examples of this process are a high intensity endurance workout at altitude might decrease the Oxygen saturation levels, while a lengthy endurance workout might result in a depletion of glycogen. In these cases, the endurance workout at altitude changed the natural levels of oxygen in the blood, and in the long endurance workout example the workout caused a change in glycogen levels. These changes, glycogen and oxygen levels, will act as triggers for the rest of adaptation which will be discussed shortly.

Essentially, the body must be put in a state of crisis to trigger the stages of adaptation. The workouts goal is to bring about this "crisis." Outside factors determine the degree of training stimulus needed for adaptation. We all know that a untrained person can go walk for 30 minutes and get some sort of training effect, while it is unlikely if that will produce any likely effect besides possibly enhancing recovery, for a well trained runner. Training state must matter then. This effect can be seen on the genetic level, as research by McConnel et al. found that AMPK activity, which is one of the signaling pathways discussed later, increased by 9-fold after 2 hours of cycling in untrained people, but after the subjects completed 10 training sessions, AMPK activity did not increase after the same amount of exercise (2005). This demonstrates the importance of what a person is accustomed to in training. A larger stimulus is required to disturb homeostasis. In addition to training state, the physiological makeup of the person and their nutritional state determines the effects of the training stimulus. An example of this effect is seen when runners take some sort of carbohydrate during a run versus taking nothing. The carbohydrate changes the subsequent signaling pathway response entirely.

Messengers

The primary messengers act as triggers to the various signaling pathways. It's best to think of the messenger's role as that being like a type of alarm. Whenever there is a large change in a messenger, it's as if an alarm goes off that triggers action. In this case, the action is to activate subsequent pathways. The messengers act to tell the body that homeostasis has been disturbed and that action should be taken to fix that problem. Thus, most of the messengers are either depletion or too high of a build up of certain substances, or drastic changes in the ratio of substances.

These signaling events are important for both coaches and scientists, as knowing what events need to happen to trigger adaptation allows for the manipulation of training to try to accomplish this. With that in mind, the following is a list of the messengers that have currently been discovered and the signaling pathways that they trigger:

Messenger	**Signaling Pathways initiated**	**Functional Results**
Mechanical Stretch-frequency and intensity	CaMK, MAPK and IGF pathways	Hypertrophy, fast to slow fiber type conversions,
Changes in Calcium	CaMK, MAPK, protein	Hypertrophy, slow twitch

levels in the cell	kinase C	fiber type conversion
NAD: NADH ratio		
Low muscle glycogen	AMPK and MAPK	Increased mitochondria
ATP:AMP ratio	AMPK	Increased mitochondria
Decreased blood Oxygen levels	HIF-1	Increased EPO and Red Blood cells
PI3-k and Akt	mTOR	Muscle Hypertrophy

MAPK=Mitogen activated protein kinase IGF-Insulin Like Growth Factor
PI3-k= Phosphatidylinositol 3-kinase
CaMK=Calcium/Calmodulin pathway

Signal Pathways

The messengers activate various signaling pathways. Signaling pathways are basically the steps taken to initiate adaptation. In our analogy, the pathways represent the response to that alarm, and the protocol that has to be followed to ensure that next time the same thing happens; the alarm won't go off again. The signaling pathways are like the collecting of the materials needed to fix the problem. Then, the next step of genetic adaptation is the process of using the materials collected to build our final product that will ensure the alarm doesn't go off.

The signaling pathways consist of a series of reactions and steps that ultimately result in some sort of gene expression. The exact steps differ with each pathway. It is this gene expression that ultimately leads to the functional changes and will be discussed next. Some of the better researched pathways will be discussed and what functional changes are results of these pathways.

Pathway	**Gene changes**	**Effects**
AMPK	Increase in AMPK= 1. ↑ PGC-1 2. inhibit MTOR	1. ↑PGC-1= ↑mitochondria 2. inhibits protein synthesis
HIF-1	EPO gene	Increased Red Blood cells
mTOR	1. results in either EIF4E or p70S6k	1. Muscle hypertrophy
CaMK	1. ↑ PGC-1	↑mitochondria
MAPK	1. ↑ PGC-1	↑mitochondria

MAPK- mitogen-activated protein kinase
PGC-1=peroxisome proliferators-activated receptor gamma coactivator C
CaMK=Calcium/Calmodulin pathway

Genetic adaptation

While each signaling pathway differs, most work by increasing gene expression. As mentioned earlier, think of the pathways as collecting the material, and the step of genetic adaptation as building the final project. The building process is accomplished first by transcription. Transcription is the process of copying a string of DNA into mRNA. This gives us the blueprint of what we want to build. mRNA can then go through the process of translation. Just like it sounds, translation is taking all that information that we copied and translating it, or better thought of as assembling it, in a nice usable package. Translation turns the mRNA into a protein. In regards to training, the protein formation is often the step that results in some sort of functional adaptation. For example, mitochondria biogenesis will improve endurance via several mechanisms.

It is important to note, that I only mentioned the pathways that we know a good deal about right now. It is likely that there are many other pathways for which adaptation occurs. The important take away is in learning how adaptation works. This allows for the designing of training based on the desired effect. You do not need to know the specifics to apply this concept. Look at the glycogen depletion example. Just use logic and a good training workout can be deduced. If the marathon is an event that is dependent on the optimal fuel use ratio between carbohydrates and fats, how do we signal the body to change that ratio so that there is more reliance on fat. We need to "embarrass" it and send the signal that more fat needs to be used. How do you do that? Simple, by running a workout that depletes the glycogen stores to a significant enough amount that the body adapts to make sure that it does not run low on glycogen the next time. It accomplishes this by increasing the use of fat as a fuel and/or as increasing glycogen stores. This example just took you through the process of adaptation without mentioning a single complex name. Use this technique to design solutions to how to train for the upcoming race.

Long Term Adaptation

Until now, we have focused on just one training session, but that is seldom how it works in the real world and in reality one training session is not going to be enough to elicit much adaptation. Following a single bout of training, mRNA levels increase and peak between 3-12 hours post exercise. The levels will remain elevated until around 24hrs post workout (Coffey & Hawley, 2007). With each increase in mRNA, a new level of protein synthesis is reached. So, while mRNA levels may fall back to resting levels after a day, the proteins that are created in this process remain elevated for much longer periods of time. For example, mitochondria protein turnover half-life is about 1 week. As mRNA levels return to baseline, protein levels level off. With each subsequent training stimulus that increases mRNA levels, the protein amounts are increased to a new level. Therefore, for long term training adaptation to occur, repeated bouts of exercise are needed to increase mRNA levels and continually keep progressing protein levels to the next level. If there is

too long of a gap between training bouts then protein levels can potentially fall to the preceding level, thus detraining occurs.

Training Applications

Although the study of signaling pathways is relatively knew, we are starting to understand the training that affects these pathways. As previously mentioned, low glycogen training has been validated in the research recently. The idea is to train low, compete high, or in other words do some training in a low glycogen state, while competing fully stocked with glycogen. In events like the marathon, this approach would seem to enhance performance. Once again, the research is behind the coaches and is simply explaining a phenomenon used for years in training. As far back as the 1950's, Ernst Van Aaken coached several elite marathoners who would practice low glycogen strategies by doing doubles without eating in between or doing long runs with no supplemental food. The fact that coaches figure out what works in training and the scientists come in later and explain why it works is the norm.

Where the study of signaling pathways really contributes to the practical world is in its demonstration of why the never ending debate of volume or intensity is not an either/or situation. The AMPK and Calcium-Calmodulin pathways both ultimately result in an increase in mitochondria development. What is interesting is that the AMPK pathway seems to be highly activated by short intense workouts such as 30sec sprints, while the Calcium-Calmodulin pathway is activated by prolonged endurance exercise (Laursen, 2004). What this demonstrates is that there are multiple ways to get the same adaptation. Two entirely different workouts activate two different pathways that give the same result. That explains why runners in particular need a wide variety of training stimulus. It also could explain why athletes who switch coaches from one emphasizing lots of long steady running to a high intensity lower volume coach, or vice versa, seem to have breakthroughs for the first year before stagnating. Perhaps the mitochondria increases of one pathway are near maxed out and when there is a dramatic change in training, the other pathway has more adaptation potential.

These signaling pathways do not work in isolation. They can interact and this may partially explain why strength gains are affected by endurance training. In particular the mTOR pathway seems to interact with the AMPK pathway. In looking at these two pathways, the main interaction explains why it is harder to gain strength while doing endurance training than it is to gain endurance while strength training. The AMPK pathway inhibits the mTOR pathway and thus inhibiting protein synthesis. The inhibitory effect explains the interference effect between the two types of training.

How Nutrition affects Genes:

Beyond the obvious example of low glycogen training, your nutritional habits can impact these pathways and the resulting adaptations. For strength gains, supplementing with Amino Acids with strength training works to increase strength because Amino Acids activate the mTOR pathway.

AMPK activation can be manipulated via nutrition too. AMPK activation is partially dependent on glycogen stores. AMPK generally stays active during a workout, but as soon as a workout is complete can be turned off by taking carbohydrate post workout. This quick shut off for AMPK would be a good idea for strength training athletes who want to minimize its effects, but for distance runners that might not be a good idea. While glycogen replenishment is a good idea, one has to wonder if deactivating AMPK as soon as exercise is done by chugging some Gatorade is a good idea all of the time?

The EPO example:

To demonstrate how the process of adaptation, signaling, and genetic response works in full, I'll briefly take you through the process of how altitude increases EPO production and ultimately red blood cells.

To set some hypothetical parameters, we will look at a moderately intense aerobic training stimulus (done around the lactate threshold) in a hypoxic environment. Going to a hypoxic environment, whether natural or artificial, puts a new stress on the body causing the various systems of the body to react and adapt to its new environment. Training, even at sea level, similarly causes numerous changes to occur within the body as it attempts to react to the stress of the particular training load. Therefore, when exercising at altitude, the body not only has to respond and adapt to one stimulus, it has to respond to two, a training and hypoxic one. When these two stressors combine, the body's primary objective becomes insuring adequate oxygen delivery to the working muscles. It accomplishes this through a variety of mechanisms.

First, let's look at some of the larger changes that occur when training in a hypoxic environment. Two main factors will determine the physiologic and metabolic response; the intensity of the exercise and the level of hypoxia. Both of these factors will affect the amount of oxygen delivered to the muscles through a variety of factors. The level of hypoxia will first affect the hemoglobin oxygen saturation due to the decrease in the pressure gradient which occurs because of the reduction in the partial pressure of oxygen at altitude (Rusko 2004). This greatly affects VO2max, as each 1% drop in oxygen saturation below 95%, decreases VO2max by 1-2% (Dempsey and Wagner, 1999). This drop in oxygen saturation even occurs at moderate to low intensities, as was demonstrated in well trained athletes seeing drops even when training at 60-85% VO2max (Peltonen 1999). The drop in oxygen saturation is linearly related to the drop in maximal heart rate that occurs at altitude (Rusko 2004). Due to this drop in HR max, cardiac output is reduced at altitude.

The nervous system also plays a role in controlling the response to hypoxic conditions. Due to the decrease in oxygen concentration, muscle activity is reduced in hypoxic conditions (Peltonen 1997). This reduction in muscle recruitment may be a way of the CNS governing performance. It has been suggested that the decrease in VO2max at altitude is the result of the CNS controlling exercise, instead of the decreased recruitment being a cause of the reduction in VO2max (Noakes et al. 2001). What the exact controlling mechanism of the response of the CNS to hypoxic conditions is remains unknown, but the CNS does play a role in the overall response to training in hypoxic conditions.

All of these changes that occur while training in hypoxic conditions lead to an eventual decrease in oxygen levels in the blood, and a decrease in the muscles themselves. This reduction in oxygen concentration in the blood and at the muscular level is the stimulus for the mechanisms behind our eventual desired outcome, an increase in RBC mass. A reduction in oxygen concentration of the blood activates the Hypoxia Inducible Factor-1 (HIF-1) pathway in tissues where EPO production can take place (i.e. kidney, liver, the brain) (Stockmann et al. 2006).

HIF-1 is a main oxygen homeostasis regulator in the body. Two subunits, HIF-1α and HIF-1β, make up the HIF-1 complex. Under normal conditions, HIF-1β is present, but HIF-1α is constantly being degraded by the proteasome (Dery 2005). When oxygen levels are lowered, the degradation of HIF-1α is inhibited, this stabilizes HIF-1α. The stabilization allows for HIF-1α to bind to transcriptional co activators and enter the nucleus of the cell. Here, HIF-1α binds to HIF-1β, forming an HIF-1 transcriptional complex (Marzo et al. 2008). This HIF-1 complex then binds to the Hypoxia Response Element (HRE) on the EPO gene. This in turn leads to EPO expression (Stockmann et al. 2006).

EPO then needs to be transported to and bind with EPO receptors. EPO receptors can be found on erythroid stem cells in bone marrow (Marzo et al. 2008). The binding to the receptor on the cell membrane results in a signaling cascade that results in the activation of the transcription factor STAT-5 and two enzymes, PI3K and MAPK. These enter the nucleus and induce transcription of specific genes that result in the inhibition of apoptosis, programmed cell destruction (Marzo et al. 2008, Jelkmann 2004). The end result is that this prevention of destruction of developing RBC results in an increase in RBC.

A larger RBC mass means a larger oxygen carrying capacity, which ultimately results in increased oxygen delivery to the muscles. Oxygen delivery has been shown to be a major limiter of VO2max (Bassett & Howley 2000). In studies done on blood transfusion of RBC in elite endurance athletes, increases in endurance performance and in some cases VO2max have been significant (Calbet et al. 2006). In one particular study done on elite athletes with an average VO2max of 80 ml kg−1 min−1, time to exhaustion in an endurance test and VO2max were both significantly increased (Buick et al. 1980). Therefore, increased oxygen delivery results in increased aerobic capacity and the functional change of improved endurance.

This leads to reasoning behind the selection of the training intensity. The idea behind the selection of a training intensity around lactate threshold is due to our desired outcome. In order to elicit the reduction in oxygen levels in the blood and muscles, the intensity needs to be high enough that it will do this. As was stated earlier, Peltonen et al. found that oxygen saturation was reduced at even sub maximal intensities of between 60-85% of VO2max (1999). In well trained individuals, this intensity corresponds well to that of LT. Secondly, the intensity has to be low enough to allow for a significant volume of training to take place. The duration spent training has to be long enough to allow for activation of the pathway responsible for the desired adaptations. The signaling of the HIF-1 pathway under hypoxic conditions has been shown to already show increases in HIF-1α in the first 2 minutes. However, maximum HIF-1α did not occur until 1 hour of hypoxia, with max half times occurring at between 12 and 13 minutes. In addition, the reduction of HIF-1α with reoxygenation occurred quickly, also within 2 minutes, and was back to normal within 32minutes (Jewell 2001). These results suggest that a sufficient duration is necessary to elicit maximum gains via the HIF-1 pathway. Lastly, in a study by Zoll et al. they found that training at an intensity that corresponded to the ventilatory threshold increased mRNA concentrations of the HIF-1α, giving credence to the theory discussed above (2005).

In looking at the research and the pathways involved in EPO production and RBC increase, it can be seen that in order for hypoxic training to increase RBC mass a sufficient intensity and duration is needed. The intensity must be high enough so that a drop in oxygen saturation occurs, while being low enough so that sufficient time can be spent training at that intensity for the pathway to be activated. In addition, enough repeated bouts of this type of training are needed in order to ensure that EPO synthesis achieves continued increase to new levels. Without repeated bouts, it is likely that mRNA will return to baseline levels, and EPO will quickly follow, thus not allowing for a sustained increase in RBC.

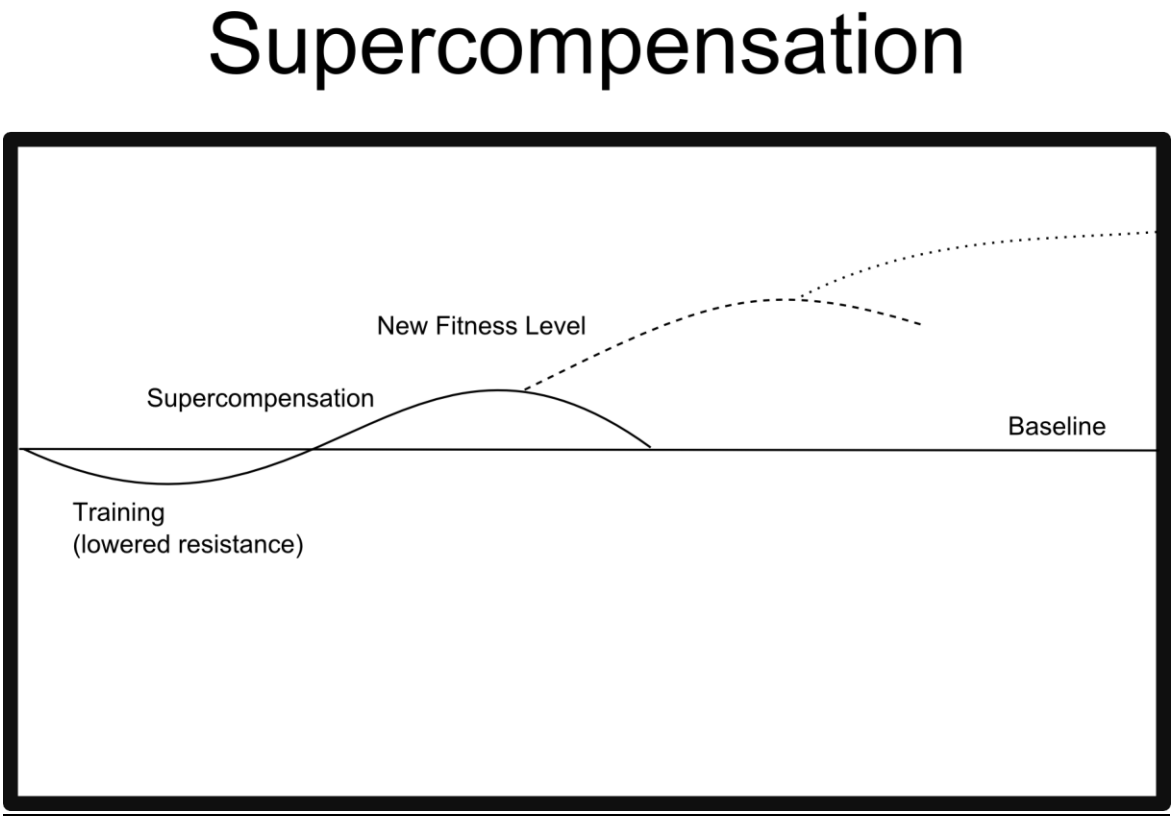
Chapter 9: Theories of Training Adaptation

The training of endurance athletes is a complex process that is not entirely understood. Coaches of endurance athletes have generally taken an approach on focusing on improving various physiological parameters, a system based on tradition and trial and error, or a system based on race pace being the central focus with various percentages of race pace used to elicit different training adaptations. The wide variance in distance running training is the result of a reliance on training methods passed on both coaches that have been successful, and skepticism of the science behind training. Before delving into what the current models of training are, looking at the volume of training done, the intensity of training, and the distribution of the training load and how all of those effect physiological parameters will give insight into what it takes to be a successful distance runner.

General Adaptation Syndrome and Dose-Response

Before analyzing the research done on the various ways to manipulate training for increased performance, it is essential that the overall process behind how the body responds to a stimulus and adapts is discussed. Hans Selye's General Adaptation Syndrome (G.A.S.) provides the basis for the process that the body goes through during training. In Selye's original work, he found that the response of animals to a variety of stressors followed a similar pattern, which became the G.A.S. (Selye, 1978). It consists of three distinct stages.

The first stage is the Alarm stage. This refers to the initial response of the body to a stress where a lowering of resistance is seen, and is often classified as our "fight or flight" response. The next stage is the Resistance stage, which consists of the body adapting to the stressor and an increase of resistance to the stress. The last stage occurs if the stress is continually applied and is the exhaustion stage. In this stage, resistance to the stress is lower considerably, and the body cannot respond effectively to the stress (Selye, 1978).



Selye's principle of adaptation can been applied to exercise training. In popular exercise literature, the application of the G.A.S. is often referred to as the principle of supercompensation (Bompa & Haff, 2008). The principle of supercompensation provides a theoretical guideline for the response and adaptation to an exercise stimulus. Similarly to Selye's theory, when some sort of training stimulus is applied, there is an initial alarm phase, where fatigue occurs and the performance level is decreased. Following this alarm stage, with recovery, there is an adaptation phase where fatigue subsides and adaptation takes place so that there is a supercompensation where performance increases to a level above that which it was before the training stimulus was applied. A new training stimulus can then be applied to go through the process again. If too little recovery is allowed, then the body never fully recovers or adapts and can enter Selye's exhaustion phase, or as it is commonly referred to, overtraining (Bomba & Haff, 2008).

This model on stimulus and adaptation allows for the planning of the work and recovery during the training process. Obviously, recovery is needed to allow for adaptation during the resistance or adaptation phase, without which the person would enter an overtraining phase. In addition, a strong enough stimulus is needed to elicit adaptations, and on the other hand the stimulus can not be too intense, thereby lowering resistance to a level which requires too much recovery time and does not result in an adequate supercompensation. Thus assigning the correct intensity and volume of a workout to elicit an optimal response is essential in proper training

The dose response relationship is another model that helps explain the optimal load of a workout. The dose response relationship originates from medicine and explains the interaction

between the dose, the total lode of a stimulus (i.e. intensity, volume, frequency), and the response, or resulting training effect (Iwasaki et al., 2003). In looking at a typical dose-response curve, it can be seen how the training dose affects the resulting outcome. Too low of a load given will result in little or no adaptations, and contrasting to that, too high of a load will give diminishing adaptations. Busson created a model that showed that an inverted U-shape best explained the optimal dose-response relationship between training and performance (2003). The top of the inverted U demonstrated an optimal dose of training, while either too little or too much training dose decreased the effect on performance.

It is also important to consider what effects this relationship. Just as a child needs less Advil than an adult for the same effect, a particular individual will need more or less of a training stimulus depending on a variety of factors. In doing research on large groups of participants it is easy to sometimes forget this relationship. An individual will react differently to a stimulus based on his physiological make up and his training history, among other factors. This individual response is an often neglected parameter when discussing training dose and can shift the dose-response curve. Manzi et al. demonstrated this by showing that Autonomic Nervous System parameters were dose related when accounting for the individual training dose, instead of taking a group only approach which had previously shown no relationship (2009). This can be seen in the constant search for an optimal intensity or volume which can be seen in both the popular literature, such as Arthur Lydiard's proclamation of 100 mile weeks being optimal, or in research, such as the search for an optimal training intensity, which seems to have varied between VO2max and Lactate Threshold (Midgeley et al., 2006; Lydiard, 1998). The recognition that each individual will respond differently to a stimulus, instead of giving recommendations that appear to be a one size fits all recommendation, would be one step towards closing the gap between research and coaching in the sport of distance running.

Individuality of Adaptation

In a viewpoint article, Kainulainen points out the individuality of the training response and the relative neglect of this phenomenon (2009). In this article, the author points to a study by Vollaard et al. that showed that while on average improvements were seen in a variety of endurance parameters after six weeks of endurance training, the individuality of the response was widespread with some showing even negative responses to the training, even though the training was the same for all individuals and done as a percent of VO2max to provide some measure of individuality to the training program (Vollaard et al., 2009; Kainulainen, 2009). The study showed that there was a wide range of adaptation in maximal and sub maximal tests including VO2 parameters, muscle enzyme activity, and metabolite levels. Another interesting finding in the study is that low responders for an increased VO2max were not low responders in other parameters. The

change in VO2max did not correlate with the change in performance on a time trial, which is a significant finding demonstrating that perhaps more attention should be paid to changing in performance instead of manipulating physiological parameters such as VO2max. One has to question the training recommendations based on training designed at improving parameters such as VO2max, with the assumption being that performance will improve because of it, when studies show that change in VO2max are often not linked with a change in performance. This phenomenon of varied response is not new and can be seen in a wide array of training situations, such as altitude training for example (Chapman et al., 1998).

The individual response goes beyond short term adaptation. In a study by Gaskill et al. they tested the yearly training model for 14 Cross Country Skiers (1999). During the first year all skiers performed the same kind of training that consisted of high volume training with only 16% of the training being performed at or above LT. At the end of the year, subjects were split between responders, those who showed the most improvements, and non-responders. The responders trained the same way the following year, while the non-responders slightly reduced their training volume and increase the total intensity. Following the 2nd year, the non-responders showed significant improvements in race times, VO2max, and LT. The responders also showed similar continued improvements in race times. These results point to the individual nature of training adaptation and show that some subjects will thrive off of different training stimulus. The mechanism behind this different response has not been found yet. It is speculation but a difference in the fiber type distribution and strength of aerobic and anaerobic capacities might explain the different reactions to training. Perhaps an individual with a tendency towards more FT fibers responds better to higher intensity training that recruits those muscle fibers. Similarly, an athlete with a predominance of ST fibers might respond better to more volume.

The neglect of the importance of the individual dose-response relationship could be of consequence when trying to determine the effects of various training modalities. Furthermore, as pointed out by Kainulainen, perhaps the problem is the standardization of training paces as percents of VO2max. The aerobic and anaerobic contribution at a fixed percent of VO2max can vary widely from individual to individual, as can be seen in the fact that the Lactate Threshold can occur at a range of percent V02max (Brooks et al., 2004). Similarly, in the Vollaard et al. study there was a 40-fold range for increase in lactate levels at 70% VO2max, showing the wide individual range of response to the exercise stimulus (2009). They acknowledged this point and suggested standardization on parameters that more directly effect power output. In practical terms for trained distance runners, it probably makes more sense to standardize paces in relation to their recent race performances or based on percentages of goal race pace in well trained runners.

Chapter 10: Volume and Intensity of Training

In the following chapter a review of what the research literature says on volume and intensity of training will be done. Coaches and the research community appear to have two opposing views on optimal training. To attempt to find out why this may be, a review of the research is needed.. In this chapter, we'll explore the two different viewpoints on volume and intensity of training and see if we can make some practical sense out of the two contrasting views.

Volume of Training

The volume of training is generally quantified as miles per week ran. How much is needed to optimize performance has been a source of great debate in popular distance running training for many years. Following World War II, the contemporary training was based on a lower mileage higher intensity training program which can be seen in the training of Gerschler and Franz Stampfl. It was not until the late 1950's, early 1960's that the volume of training became a focal point. With the success of Arthur Lydiard's athletes, typical training volumes that eclipsed 100 miles per week (mpw) quickly became the norm (Lydiard, 1998). Up until the early 1980's, high mileage was predominant in the elite distance runners of the world. In the 80's and early 90's, a new line of thinking focused on moderate mileage (~50-70mpw) with more high intensity work gained popularity. This was led by the British coaches, Peter Coe, with his success in coaching his son Seb Coe, and Frank Horwill (Coe & Martin, 1997). The rise of sports science during this period of time also seems to have led to the emphasis on higher intensity work with lower volume. While Coe and others had success off this approach, the dearth of American distance running success on the world level during the 90's is sometimes blamed on this change in emphasis.

With the rise of Kenyan and Ethiopian distance runners, ideas on training volume shifted again towards higher mileage totals. The East African runners demonstrated that relatively high volumes of training might be necessary for reaching the levels that they were. Some author's have speculated that East African runners are actually in the moderate volume, high intensity line of training, but this neglects the popular literature and more importantly the scholarly literature on the subject (Berg, 2003). Billat et al. found that East African runners in their classified Low Speed Training group averaged 172km per week, while high speed runners averaged 158km per week (2003). Both of these values are substantially higher than the values done by lower mileage programs such as Coe's. In addition, these values were recorded while the subjects were in racing season, which is not the period of highest volume. Therefore, the idea that East African runners are of the lower volume model is not substantiated. Lastly, training programs by authors who have coached elite Kenyan runners also demonstrate that they take a relatively high volume approach (Canova & Arceli, 1999). One important note is that their intensity is also fairly high, and in this

author's opinion, much higher than the higher volume programs of the 1960's and 70's, perhaps this is due to their large volume of running and activity during their childhood. This could potentially allow for them to handle higher intensities while still maintaining higher volumes. While examining the historical volumes of training of elite runners gives us an idea of what might be best, the question is what does research say about volume of training?

There have been a few studies that have tried to determine the impact of training volume on performance and physiological factors. Daniels et al. found no change in VO2max while increasing training from 20-30km/wk to 50-70km/wk (Daniels, 1978). Similarly, a study by Costill comparing changes in VO2max as two marathon runners coming back to training after a 6-month layoff, VO2max increased progressively as mileage increased up until ~100mpw, while further increases in mileage, all the way up to 200 mpw, resulted in no further changes in VO2max (Costill & Trappe, 2002). The fact that VO2max does not seem to change after a certain amount of mpw is sometimes cited as a reason that higher mileage is not needed. However, it is important to remember that VO2max is not the be all end all and does not accurately reflect performance. Two studies have shown that with increases in mileage, performance was improved, even with no change in VO2max (Costill, 1967; Daniels, 1978).

Two different studies have also looked at the relationship between amount of training done and RE. Theoretically, one adaptation that could occur with a large volume of running is an increase in RE due to increased efficiency, both mechanical and metabolic, because of the repetitive nature of running. In a comparison of a spectrum of runners spanning from untrained to elite, the findings suggests that high volumes of running could increase RE (Costill, 1967). Similarly, Mayhew et al. found that the amount of years running significantly correlated with RE (1977). Contradicting this data, Pate et al. did not find a correlation between RE and training volume (1992). The connection between RE and volume of training remain inconclusive.

Lastly, it's important to recognize the limitations of research studies on the impact that volume of training has. Many of the adaptations that may occur due to increases in volume may take longer than the couple of weeks that most studies last. For instance, a longitudinal study by Rusko et al. found that after 8 years of training and a doubling of training volume, the percentage of ST fibers in a group of Cross-Country Skiers increased by 11% (1992). While obviously not solely due to the volume increase, these changes show the long term adaptations that take place in endurance athletes. Another study by Fiskerstrand and Seiler found that the performance changes (+10%) in world class rowers from 1970 to 2001, occurred not with of an increase in intensity of training but with an increase in volume of easy training (2004). The amount of low intensity training per week increased from 30hrs to 50hrs per week, while the amount of supra-maximal training decreased from 30hrs to 7hrs per week. The lack of longitudinal studies prevents making any research based recommendations on what the ideal volume of training is for distance runners.

Intensity of Training

While volume of training is often the first factor most think of, intensity of training plays just as important of a role. Distance runners typically perform several higher intensity workouts per week which are separated with easier runs. The intensity of both the harder and easier runs affects the adaptations that take place and lead to improved performance. The intensities used range from easy jogging all the way up to sprinting. There has been much research aimed at finding the effects of different training intensities. Most of these studies separate training intensities into sub maximal, LT, VO2max, and supramaximal training. While the efficacy of separating training into zones is debatable, we will look at the effects of training at these various intensities for the sole reason that this is how most of the research splits the training up.

In general, the knowledge of what intensity to train it is in the dark ages as far as research goes. This is definitely one area of interest where coaches are light years ahead of the scientists. Given that fact, I will lightly touch on some of the pertinent research done on training at various intensities, then give a brief overview of a practical approach to assigning training intensities.

Lactate threshold

For distance runners, training at or around LT has risen in importance in recent years (Billat et al., 2003). This is partly due to the observation that VO2max does not seem to change in elite athletes, yet performance and LT continues to improve (Legaz Arrese et al., 2005). For this reason, training at LT has become en vogue.

For recreationally trained runners, LT seems to impact LT, RE, and VO2max. Four different studies with recreational athletes found that adding between 1 and 6 LT type sessions per week increased VO2max by between 2.5-8.1% after 6-8 weeks of training. In addition, two studies measured changes in LT and found that it increased by 3.3% and 10.7% respectively. It is worth noting that the 10.7% increase was seen when 6 sessions of LT training were done per week, while the other study added two such sessions per week. Lastly, improvements in RE were measured and seen in two studies with improvements of both 1.8% and 3.1% (Billat et al. 2004; Franch et al., 1998; Hoffman et al., 1999; Yoshida et al., 1990).

While the improvements in all three variables are impressive, what happens in well trained runners? Acevedo and Goldfarb investigated adding training sessions at an intensity near LT in well trained runners (average VO2max=65.3 ml/kg/min). The added training sessions resulted in a nonsignificant improvement in VO2max of 0.7% but a significant increase in LT of 5.1%. In addition, performance in a 10k race was improved by 3.1% (1989). In an interesting study Jemma et al. found that after 3 weeks of interval training at an intensity that usually corresponds with LT (~85%VO2max), performance over a 40km time trial in well trained cyclists was improved (2005).

They measured EMG changes and found that the performance improvements could have been due to changes in muscle recruitment.

This improvement in LT and performance without an increase in VO2max is consistent with the idea that VO2max seldom changes in well-trained athletes and performance improvements come from other factors. The changes in EMG found during the Jemma study are interesting and point to the notion that factors outside the traditional endurance performance model need to be considered when conducting training studies.

VO2max

Many studies have focused on training at around the velocity of VO2max. The belief is that spending time at VO2max may help increase VO2max or stress the aerobic capacity maximally, thus serving as the most effective stimulus for adaptations. There have been numerous research studies on the effects of training on VO2max. In untrained subjects, training at or near VO2max increases VO2max (Berg, 2003). In two separate studies, adding intervals at or around vVO2max in recreational trained athletes increased VO2max by 5.0% and 6.0%, and RE by 2% and 3% (Franch et al., 1998; Smith et al., 2003).

In well trained athletes, as previously mentioned, changes in VO2max are rare, as can be seen by a study by Legaz Arrese et al. which found improvements in performance without changes in VO2max in elite runners (2005). Several specific studies have been done on a variety of interval training programs at the velocity at VO2max (vVO2) in well trained runners. Out of four studies that tracked changes in VO2max, only one study found any significant changes in VO2max. Three of these studies showed improvements in performance, with one showing no change. Only one study measured changes in RE, and it showed an improvement. Lastly, two showed an improvement in vVO2, while two showed no change (Acevedo & Goldfarb, 1989; Smith et al., 1999; Billat et al., 2002; Billat et al., 1999). The data on training at vVO2 remains inconclusive on a number of levels. Performance seems to improve the majority of the time, while the exact mechanism to this performance improvement is unknown. Since only one study measured changes in RE and those were positive, it is possible that this could be the main benefit. The Jemma et al. study also points to the idea that neuromuscular changes such as altered recruitment pattern could play a role in alteration in performance during high intensity training (2005). Even though the training is generally termed VO2max training since it corresponds with that parameter, it should be remembered that training at that intensity will result in adaptations other than those related to VO2max.

Supramaximal Training

Training at speed faster than vVO2max has been shown to improve endurance performance. It is possible that this occurs through several mechanisms. Training at speeds faster

than VO2max could potentially alter many of the previously discussed anaerobic and neural factors that impact endurance performance. In addition, faster training could have an impact on the traditional factors in performance; VO2max, LT, and RE.

There have been a limited number of studies with supramaximal training involving runners only. The limited number of studies involved recreationally or moderately trained individuals. Two different studies found contrasting results. In a study by Franch et al. had participants do intervals at 132%vVO2 and found a significant increase in VO2max of 3.6% and a nonsignificant increase in RE of .9 (1998). On the other hand Bickham et al. found that doing sprint intervals at 90-100% perceived maximum effort increased time to exhaustion at 110%VO2max and a decreased VO2max by a nonsignificant 2.2%, (2006). Interestingly, this study showed an increase in MCT1, which as mentioned earlier may play a role in assisting with the oxidation of lactate. These results point to the need for more studies on supramaximal training in distance runners, and the measurement of more variables that may impact performance.

It is possible that supramaximal training could induce changes that could negatively affect distance running performance. High intensity interval training could elicit Glycolytic adaptations that could impair aerobic processes. In several studies using very intense interval training, increases in lactate production, and the enzymes PFK and LDH have been found (Billat, 2001). As mentioned in the LT section, an increase in lactate production, and an increase in LDH, which could represent more pyruvate being converted to lactate, could alter the balance between lactate production and clearance. This in turn would negatively impact LT. The degree to which this theoretical argument holds up in well-trained runners needs to be investigated, because it is likely that in untrained subjects any sort of training stimulus that had some aerobic component would improve LT.

Cycling studies provide us some clue of the general effects of such training on endurance performance. Several studies have looked at the impact of supramaximal intervals of 30-60sec in length. Laursen et al. found an improvement in power output by 5.3% over a 40km time trial after 4 weeks of interval training, an increase in power following an incremental test (+4.0%) and an increase in VO2max by 2.2% (2002b). Stepto et al. found an increase in power over the 40km distance by 3.4% without a significant change in VO2max (-0.6%) (1999). Laursen et al. found an increase in power by 5.7% at Ventilatory Threshold (VT), which closely relates to LT, an increase in maximum power attained in the step test (+4.7%) and an increase in VO2max by 3.5% following four weeks of training (2002a). In a study by Creer et al., EMG measurements were taken before and after 4 weeks of interval training to assess neural factors (2004). They saw significant increases in peak power and VO2max. As well as an increase in EMG readings, showing increased motor unit activation.

The cycling studies all point towards improved performance following the addition of supramaximal training. The exact mechanisms for these improvements vary with each study. The

measured increases in maximum power in the Laursen studies, combined with the Creer study measuring increased motor unit activation, could lead credence to the idea that neural adaptations may be the main reason that supramaximal training improves performance in endurance athletes. In addition, if Noakes CGM is taken into consideration the increase in VO2max seen in relatively untrained individuals after supramaximal training could be a result of increase muscle activation.

Training at speeds faster than VO2max is a regular part of most distance runners training, yet there is a lack of information regarding its effects on well trained runners. This is probably due to the heavy emphasis on training at VO2max and LT. That emphasis is likely due to the fact that those quantities are easily measured and provide points of reference. Cycling studies help show that other factors besides these may be the true mechanisms to why supramaximal training improves performance. More running specific research is needed on this topic to discern the exact mechanisms.

Interaction of Volume and Intensity of Training

In the research literature there has been a trend towards the recommendation of high intensity versus high volume training (Berg, 2003). This is likely a result of the short term nature of research studies. Given the changes that occur with high intensity versus lower intensity and longer duration training, it is not surprising that high intensity training shows greater benefits over only a couple of weeks. This has led to the conclusion that high intensity training is preferential over higher volume to reach the same adaptations. However, it is questionable whether this preference has merit. Seiler and Tønnessen found that elite endurance training consists of about 85% of the training being performed at low to moderate intensities in several different groups of elite endurance athletes (2009). Furthermore, they point out that elite training is more likely to represent optimum training than what is demonstrated in the lab setting because of a likely evolution process of training by elites.

Contrasting the research theory, Esteve-Lanao et al. demonstrated that it was the amount of easy running that impacted running performance over a 10km race, not the amount of high intensity training (2007). The experience of elite endurance athletes also contradicts the research findings that high intensity, low volume training is optimal. As pointed out already it seems that the best runners combine a higher volume of training with some higher intensity training. Due to the high volume of training, the large percentage is low intensity, but the absolute volume of relatively high intensity training still reaches levels of 20mi or more per week (Billat et al., 2003). In a paper by Laursen, he points out two key findings, first, without a background of high volume training, high intensity training can maintain performance, but seldom improve performance (2009). Secondly, while the adaptations of high intensity and high volume training may be similar, they may occur via two different pathways. Most of the training studies manipulating intensity of training add high intensity work to athlete's training schedule that have been doing a large volume

of low intensity training. This interaction that high intensity training must be preceded by high volume training is fundamental in popular literature on distance running, but neglected in the research.

Addressing the first point made by Laursen, high intensity training is more beneficial when preceded by what is termed a base in popular literature. In a study by Quinn et al., two different groups performed either continuous exercise or interval training for 12 weeks (2002). At the end of the 12 weeks, the groups switched, so that the continuous group now did 12 weeks of interval training. It was found that the group that did continuous and then interval training improved significantly more in terms of time to exhaustion (15% to 5.3%) and VO2max (7.4% to 3.6%) than the group that did interval and then continuous training. These finding supports the idea that a base of moderate work needs to be established before intense training is done for maximum benefits. This could also explain why in most studies which simply replace a portion of the low to moderate training in their subjects with high intensity training, performance is improved by so much. This interaction of prior training's impact on training during a study should be taken into consideration when evaluating study results.

According to Laursen, high volume training may signal for adaptations through the calcium-calmodulin pathway, because of the increase in intramuscular Calcium seen in long duration, high volume training (2004) In contrast to this, high intensity training may signal adaptations through the adenosine monophosphate kinase (AMPK) pathway because of the increase in AMP seen following high intensity training. Both of these pathways eventually lead to PGC-1α, which is a transcriptional cofactor that will result in the typical adaptations seen in endurance athletes such as mitochondria biogenesis (Laursen, 2004). Laursen thus concludes that there are two very different ways to achieve similar adaptations. This points to the idea that to maximize these adaptations both stimulus are likely needed, as one pathway may be more difficult to activate than the other depending on the individual and their training status.

In conclusion, it is likely the time course of changes and the length of the research studies which is creating this discrepancy between real world recommendations and findings and the findings in the lab. High intensity training is likely to show quick adaptations, explaining its greater impact on performance within a short period of time. The long term effect of repeated high intensity training is not something that has been researched in the literature. The experience of coaches points to a long block of high intensity training leading to decreased performance, mostly likely due to overtraining (Lydiard, 1998). Lastly, some research points to the idea that high intensity training may need to be preceded by higher volume training to see maximal benefits (Quinn, 2002). This possibility needs to be further researched and considered when evaluating training research.

Training in the Real World

While studies on training are interesting, they should be taken with a grain of salt. In research we have to isolate one variable to see its effect. In the real world, variables are never done in isolation. This creates a situation where training in the lab is completely different than training in the real world. The training effect is the result of a combination of all of the training stimuli applied to the athlete.

Research on concurrent endurance and strength training has demonstrated this effect by showing that the genetic adaptation that occurs is partially dependent on whether the strength or endurance training was done last during the day's training. This is but one example of the limits of research based endurance training. Another problem is that the research looks at averages and everyone as a group, when a coach has to look at people as individuals. This creates a situation where researchers are looking for the magic workout or training intensity that improves performance on average. It does not exist.

In the real world, we know that training can not be classified into easily manageable zones. Yes, certain workouts will have a greater effect aerobically or anaerobically, but as discussed previously an individual's reaction to training will vary drastically. This might partially explain the drastic difference in opinion of optimal training between most Scientists and most coaches. The scientist viewpoint has been explained in detail, so a brief counterpoint of the coaches' viewpoint on intensity of training is in order.

From a coaching standpoint, there should be no zones of training. All paces are connected in some way and elicit slightly different adaptations. The central premise of deciding training intensity should depend primarily on two factors, the race distance the individual is aiming for, and the individual makeup of the runner. The race distance and goal pace should give the coach a pace that can be defined as specific. From this pace, as we get faster or slower, the intensity becomes more general, all the way out to the extremes of jogging or full out sprinting.

Generally, coaches start with using very general training, on both sides of faster/slower spectrum, and work towards specificity. Further details on this are discussed below in the periodization section. The key difference is that all the intensities build upon each other, or in other words are supported by each other. The term base is often used and applies to this idea. A base of easy aerobic running needs to be built before slightly more intense aerobic running needs to be done. Similarly, a base of pure speed and muscle fiber recruitment provides the foundation for slightly slower but longer work. The goal of training is to use both sides of the spectrum to increase the speed that you can handle over a specific distance, or in other words to extend the length of time you can run at your goal pace until it reaches the full race distance.

The workouts themselves are not done in the way that most research based workouts are in that you run a single set of repeats at a given pace. Instead there is mixing of intensities for a

desired goal. Workouts also have built in progression to take the athlete to the next adaptation level by manipulating pace, intensity, rest interval, etc.

This is a very brief and general overview of the training process from a coach's standpoint, and just one of many theories. The key differences between research designs and real world training can be summed up with the idea that research is looking for the magic workout and magic training intensity that enhances performance, while coaches use a large mixture of workouts at a variety of intensities to reach peak performance. In addition, research is focused on the group average, while coaches recognize the individuality of training.

Supplementary Training

Training that is not running specific also has been shown to improve distance running performance (Jung, 2003). Resistance training and plyometrics are the two types of training that are most studied and beneficial. Despite the evidence pointing towards performance benefits, there has been a hesitation by many runners to include these types of training. Part of this hesitation is due to the idea that resistance training increases size and weight, while the belief that only VO2max, RE, and LT impact running performance has also held back the acceptance of supplemental training. However, as discussed already, it is now known that anaerobic and neural factors contribute to success in distance running.

There are many different types of resistance training which complicates quantifying its effect on endurance performance. The types of resistance training that will likely aid runners are those which do not result in a significant increase in body weight and muscle size. An increase in body weight would decrease relative VO2max and RE. A significant increase in muscle size would result in a decrease of the capillary and mitochondria density (Brooks et al., 2004). For these reasons, training that focused on improving the neural and anaerobic factors that improve distance running performance without a subsequent increase in weight or size would be desirable. Following this idea, plyometrics and heavy training are likely to be the best for endurance athletes.

The majority of studies using traditional weight lifting have shown no change in VO2max in well trained athletes (Jung, 2003). The studies that have found a change in VO2max were on untrained subjects. For example, on well trained subjects, Hickson et al. found no change in VO2max when resistance training was added to cyclists training (1988). In trained individuals it is unlikely that VO2 will be high enough to elicit a change in VO2max. However, if one applies the theory of Noakes' CGM, it is possible that changes in VO2max following resistance training could be attributed to changes in muscle activation which transfer to the task of running. Of the traditional factors, RE and to a lesser extent LT appear to be the most affected by resistance training.

RE seems to be the primary factor that is improved with this training. Three studies using trained runners and replacing part of their training with plyometric or explosive type training have

shown improved RE, while one study using heavy lifting demonstrated improved RE. Spurrs et al. found an improvement in 3k time along with an improvement in RE following plyometric training (2003). Similarly, Saunders et al. found that following plyometric training improved RE by 4.1%. In addition, neuromuscular factors were improved, such as decrease time to reach maximal dynamic strength, and increased average power during a plyometric test (2006). Lastly, in well trained runners, Paavolainen et al. found that following 9 weeks of plyometric type training, the runners improved their 5k races by 3.1% and RE by 8.1% (1999). They also improved their vMART, and that improvement significantly correlated with their improvements in performance.

One study has been done on the effects of heavy weight lifting on endurance performance. Millet et al. studied the effects of 14 weeks of heavy weight training on various parameters in triathletes. They found that RE, maximal strength, and power during a hopping test were all significantly increased (2002). Another study by Esteve-Lanao et al. found that a periodized resistance training program resulted in the maintenance of stride length compared to a control group and a non-periodized resistance program (2008). This study shows another potential benefit of resistance training.

These findings on the effects of plyometric and resistance training lead to the idea that neural and anaerobic factors from strength training could be responsible for performance improvements. As discussed in the neuromuscular section, it is likely that changes in motor unit recruitement, changes in anaerobic capacity, or alterations in ground contact time or muscle-tendon stiffness could explain improvements in performance following resistance training.

Training Frequency

How often runners train and the distribution of training intensities across training are two other factors that are debated. In distance running, it is accepted that well trained runners train every day or at least six days out of the week. This is needed to reach the higher volumes of training that are done. One debate surrounds whether training once or twice per day is most beneficial. There have even been reports of elite Kenyan athletes running up to three times per day (Tanser, 2008). Billat et al. found that elite Kenyan runners trained between 10-14 times per week (2002).

To the author's knowledge only one training study has been done comparing training once versus twice per day in moderately trained runners. In a study by Mostardi and Campbell they added a 7mi run to one groups training on 4 days per week, while the other group ran only once per day, continuing there daily regimen. After 4 weeks, no differences in improvements in VO2max were found. However, VO2max was tested on a bicycle ergometer, which may impact results since the participants were track athletes and ran during their training. Both groups saw significant improvements in mile time, but no difference between the groups (1981). More research needs to be done on this topic as obviously the research results conflict with the real world training suggestions.

Several theoretical arguments can be made for twice versus once per day training, especially during in-season training. During in-season training, a large bulk of the training of distance runners is used as recovery or general aerobic running (Daniels, 2005). In well trained runners, at this point of the season, it is unlikely that a low training stimulus would elicit further adaptations. It is possible that splitting these recovery runs into doubles would further promote recovery while maintaining volume.

In theory, glycogen depletion could be minimized to a greater degree than with one single run, as glycogen repletion would be easier achieved following two separate 5 mile runs as compared to one 10mi run. Two separate runs could lead to greater glycogen repletion due to a greater window of insulin sensitivity following each run. Secondly, the hormonal response of such hormones as Growth Hormone, which plays a role in recovery and adaptation, could potentially be greater in two separate runs, as compared to one longer run. Significant GH release in endurance runs at low and moderate intensities occur in relatively short runs (Gilbert et al., 2008). In addition, some researchers have suggested that training in a fatigued state may enhance subsequent adaptations. It has been found that training in a glycogen depleted state enhances gene transcription of several markers of training adaptation (Yeo et al., 2008; Hansen et al., 2005). In two studies on training every day versus training twice every other day, increases in enzyme activity have been more significantly increased in the twice every other day group. In the study by Hansen et al., they used knee extensor exercises with one leg being trained every day and the other twice every other day (2005). The twice every other day leg should significant better time till exhaustion at the end of the training, along with the increased enzyme activity. In the study by Yeo et al., they compared two different groups using cycling as the means of training (2008). In their study enzyme activity was higher in the twice every other day group, but performance was equally increased in both groups. More research needs to be done using study designs that more accurately reflect the training done in the real world, meaning that training once every day versus twice per day, even with equal volume, should be researched instead of twice every other day.

In addition to the frequency of training done, the frequency of the more intense workouts also plays a role. Recommendations in the popular literature suggest that 2-3 intense workouts should be done per week (Daniels, 2005). However, little research has been done on this subject with runners. As will be discussed later, recovery plays a central role in the adaptation process. Without adequate recovery, athletes could be pushed into a state of overtraining or overreaching (Billat et al., 1999). Therefore, finding the optimal distribution between low intensity and high intensity workouts should be considered. In a study by Billat et al., they put this to the test having subjects perform 3 low intensity workouts, 1 LT workout, and one high intensity training (HIT) workout per week (1999). They then increased the intensity of the training by having the subjects perform 2 continuous, 3 HIT, and one LT workout per week. The subjects did not show any further

improvement with the increased density of hard workouts and showed markers for overtraining. This study gives an idea on the density of hard workouts that subjects can handle.

Chapter 11: Periodization of Training

How we plan and implemented our training is of the utmost importance for insuring success. It is important to remember that periodization does not refer to some exact method of training planning, but instead simply refers to how one goes about planning the training. Training volume, intensity, and frequency are manipulated throughout a season to bring about peak performance. There are numerous theories of how to plan and manipulate training to bring runners to maximum performance, yet there is no research testing any of these periodization models. Periodization is well studied in strength training (ACSM 2009).

Periodization is the process of dividing the training into smaller periods of training where the emphasis, or the target, of the training is altered during each period. Generally a manipulation of the volume, intensity, and frequency of workouts is done to bring a person to peak performance. Traditionally, periodized training has been broken down into several phases. The largest of these is a several year cycle that was originally based on the 4 year cycle of Olympic athletes. Below this is the Macrocycle which consists of a period of training that lasts from months to up to a year. The next level is that of the mesocycle which consists of several weeks of training. A microcycle is the next phase that consists of several days to a week or more of training. The last level is that of the individual workout (Issurin, 2008).

Training can be periodized on several different levels. On a microcycle level, the variation of volume and intensity of workouts on a day to day basis is often done. In endurance training this often means the alternation of harder workouts, those performed at intensities near or above LT, with easier runs performed at relatively light intensities. In popular coaching literature, there are many different theories on how to manipulate training. It is generally accepted that harder days are followed by one or two easier days before another hard day is repeated (Daniels, 2005; Hudson, 2008). There has been no research studies that the author is aware of that have investigated this manipulation of training on the microcycle level. It is not known whether this alternation of hard and easy training is optimal or whether some other manipulation is more effective. This is one of the many research limitations in endurance sports. One study on strength training clearly shows that altering what is done each day is more effective than altering only every 4 weeks on the development of strength (Rhea et al., 2003). Whether this can be applied to endurance is unknown.

On a larger scale of mesocycle and macrocycle level, debate surrounds how to manipulate volume and intensity over the course of a season. In endurance training, only one study has been done comparing periodized training and non-periodized training. The study by Soungatoulin et al. found that after 16 weeks of training, the periodized cyclists performed significantly better in a time trial (2003).

The intricacies of periodization can be debated on many different levels. Looking at a macrocycle level of training, the manipulation of overall volume and intensity is often debated. Volume and intensity are inversely related (Bompa & Haff 2008). Several contrasting views are seen in the manipulation of these variables. The traditional approach, emphasized by Arthur Lydiard, consisted of very high volumes of work early in the season that consistently lowers as the season progresses, with a proportionate increase in intensity (Lydiard, 1998). In contrast to this approach, Simmons has suggested an ever increasing total volume of training, with a corresponding steady increase in intensity (Simmons & Freeman, 2006). There is no research to suggest how the volume and intensity of training should progress on a macrocyle level.

Volume and intensity can be manipulated on a meso and microcyle level too. Several different approaches to manipulating these variables are seen in the popular literature and even in theoretical literature. Some coaches subscribe to a consistent approach to volume in that within a period, volume is kept relatively level from microcycle to microcycle (Lydiard, 1998). In contrast to this approach, volume is altered every week with a high and then a low volume week of training. Another approach is an escalating approach to volume where volume is increased every week during a mesocycle. Some of these programs also institute the inclusion of a down or recovery week where volume and/or intensity is decreased to allow for adaptation (Bompa & Haff, 2008). This approach makes sense in theory in regards to Hans Selye's General Adaptation Syndrome which explains the mechanisms behind stress and adaptation. In theory, the down week would allow for recovery, adaptation, and taken to a new level of resistance. How many weeks of higher volume or escalating volume can be done before a down week is not known. There has been no research done to investigate any of these theories or ideas. This leaves the manipulation of volume on a mesocycle level largely based on experience and trial and error.

How one should periodize the training to target specific adaptations is also a source of debate (Issurin, 2008). The debate surrounds whether a linear/block or non-linear/mixed periodization schemes is more beneficial. In a linear/block periodization scheme, periods of training that are almost solely devoted to developing a select few biomotor abilities, while work on other abilities are kept to a minimum. The next period or block of training would then shift its emphasis to a different biomotor ability to be its focus. For example, during the first period of training, a several week period may be spent doing exclusively endurance training. This may be followed by a period of training done to develop high end aerobic endurance, and then by a period designed to develop speed endurance (Bompa & Haff 2008).

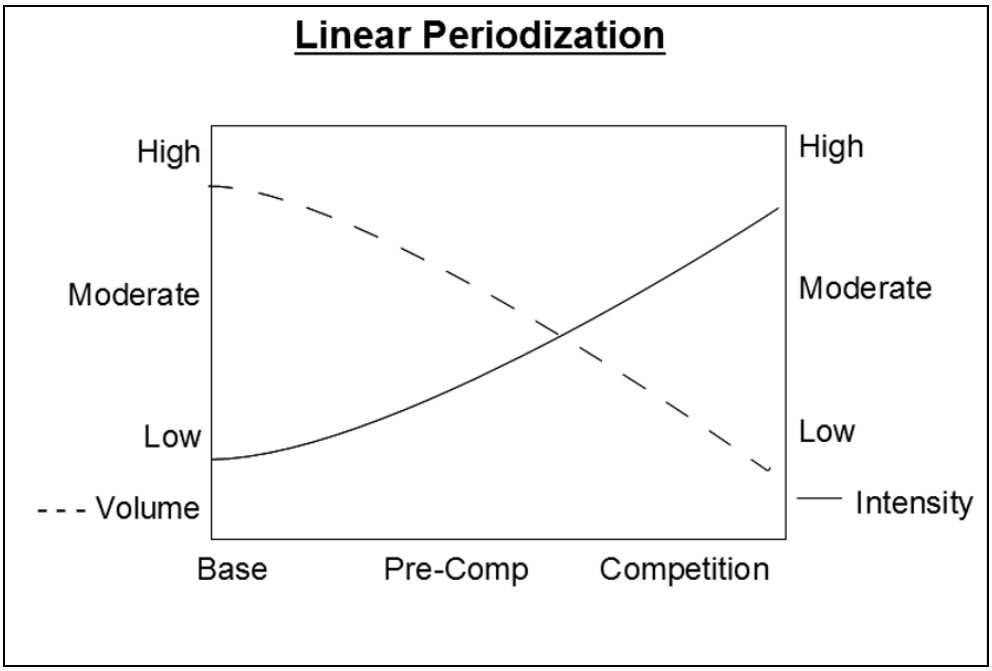
In a non-linear/mixed periodization scheme, the emphasis during each period would change, but nothing would be almost entirely neglected (Hudson, 2008). Essentially, in each period there would be an element of endurance training, speed, speed endurance, etc. just the emphasis on each one would change. For example, during the first period of training, endurance may be emphasized with lots of aerobic running, but a weekly sprint session may be included. In a later

period, speed training might be more emphasized and thus a higher volume or frequency of that work is done, but training focused on endurance would also be done, just not to the degree it was done during the period when it was emphasized.

In regards to endurance training, the benefits are largely theoretical and anecdotal. In resistance training one study has been done comparing linear and nonlinear approaches. In a recent study by Monteiro et al. nonlinear periodization showed higher increases in strength and more consistent increases in strength across a 12 week training period when compared to a linear and a no periodization group (2009). In this study, three different workout types, consisting of 3-4 sets of either 4-5 reps, 8-10 reps, or 12-15 reps, that were aimed at different adaptations were used. The periods of training were split into 3 blocks of 4 weeks each. The linear group did all 3 sets of 12-15 reps the first period, then all 8-10 rep sets the second, and finally all 4-5 rep sets the last. The non-linear group did two of the three rep types each period, with it altering each period. The non periodization group did solely 3 sets of 8-12 reps. While the design of this study was for strength increases, it shows that a non-linear approach was more consistent in strength increases after each period and produced greater total increases, which both are criticisms of a non-linear approach (Issurin, 2008). No similar studies have been done using endurance training so whether the results would be similar in endurance training is unknown, but it does provide theoretical support for a non-linear approach of training. This calls into question the merits of having a period of training focused solely on one adaptation such as is seen in many endurance sports where a large base period focuses only on steady aerobic running.

Periodization is a complex process that occurs on many different levels. In looking at the coaching literature, the general periodization for distance runners can be divided into down into two theoretical models. The two models include a linear model and a funnel to specificity model.

Periodization in endurance sport



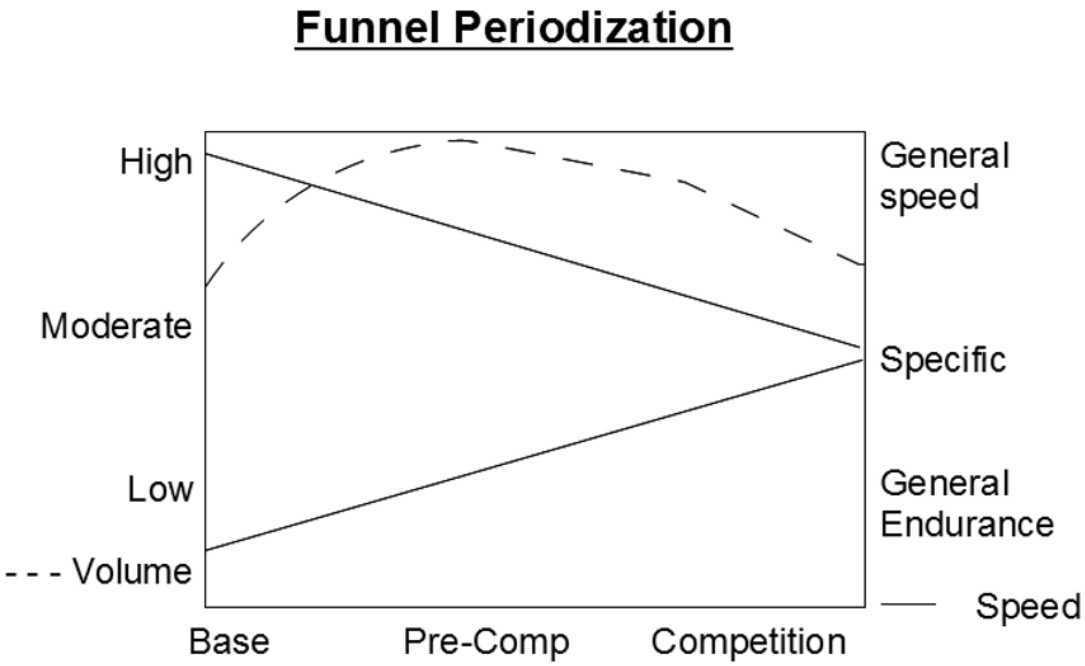
Linear periodization

The linear model is considered the classic approach to distance training. It gained popularity and influence following the success of Arthur Lydiard's athletes in New Zealand. The Lydiard approach, which is the basis for the linear approach, started with a several month period called Marathon Conditioning, where high volumes (~100mpw) of easy to moderate running were done. In addition long tempo or fartlek sessions were done during the end of this period. Following this phase, mileage was decreased and the intensity was increased with a one month period called Hill Resistance which focused on strength endurance by using a variety of hill repeats and hill circuits. The next phase consisted of track training, starting with longer intervals on the track (800's, miles) and progressively working towards shorter intervals (200's and 400's). This was followed by a sharpening and peaking phase that consisted of shorter faster work, such as 50m in and out sprints or an all out 200m, and a further drop in volume (Wilt, 1964; Lydiard, 1998).

This method is still used today by many coaches. Greg McMillan, coach of an elite training group in the U.S., characterizes his training as going from Base to Stamina to Speed (Greg McMillan, personal conversations). Similarly, Esteve-Lanau et al. described the periodization used by well trained Spanish runners in their study as a type of linear periodization. The athletes progressed from a high volume, lower intensity base period to a progressive increase in intensity and decrease in volume towards their peak race (2005).

This linear periodization model can be characterized as a system that starts with a high volume of training and progressively decreases this volume, while at the same time intensity of the workouts start low and is progressively increased. Before the peak race of the season, intensity is high while volume is relatively low. In this case, volume and intensity are inversely related. Looking at the hard workouts themselves, it can be seen that the pace of these is generally increased

from the start of the training until the end. Training might start with the harder workouts being done at LT or similar tempo intensities, and progress to 10k-5k speeds, then to vVO2 speeds, and finally speeds that are at mile pace or faster.



Funnel/Mixed periodization

The funnel model of periodization is a newer training theory that is taking hold in distance running. Its modern origins appear to be from the Italian system of training, and have been popularized by coaches Renato Canova and Luciano Giggliotti. This program focuses on the specificity of the harder workouts. In Canova's terms the harder workouts progress from General to Special to Specific (Canova, 1999; Hudson, 2008). Specific is defined as the goal pace for the peak race the athlete is training for. The further away from this pace, in both faster and slower directions, the more general the workout becomes. The workouts come from two different extremes and funnel towards specificity. On one extreme, workouts start out with highly aerobic work such as longer steady runs and then progresses towards the specific race pace with workouts getting progressively faster and shorter. On the other extreme, workouts would start very short and very fast, such as 60m sprints, and then progress upwards to longer and slower workouts toward the specific workouts.

For example, if an athlete is training for a 5k race, his goal 5k time would be the specific pace. From the aerobic side, the workouts might progress from steady runs at marathon pace to LT runs to 10k paced intervals and finally to a 5k paced interval workout. On the other extreme, workouts might progress from 60m sprints, to 200m at 800m pace, to 400's at 1500m pace, to 800s at

3,000m pace, and finally to specific work at 5k pace. Essentially, the workouts start at the extremes and work towards each other, like a funnel. Volume is manipulated in a different way then the classic linear model too. Overall volume of training is progressively increased during the first period of training, and then peaks and stays relatively constant until the peaking phase where there is a drop but only a slight one (Hudson, 2008).

Chapter 12: Training Models- Integrating Theory and Practice

There is a reason that the majority of exercise scientists don't make the best coaches, and that is the neglect of the human and practical element of training. Usually, the most successful coaches are those who come at the science from a coaching perspective. To accomplish this goal, integrating the theory into practice is an essential skill. One way to accomplish this is in coming up with models of fatigue.

Models of Fatigue:

While knowing the complexities of the science of running is good, it is easy to lose the forest from the trees, as it is hard to wrap your head around it all so that it can be practically applied. To circumvent this problem, we take those complex ideas and make them into simple models. In our case, we take the scientific knowledge and come up with simple models of fatigue which gives a road map on how to train to avoid fatigue.

The great New Zealand coach Arthur Lydiard once commented that almost anyone could run a 60 second quarter, yet very few could run four of those in a row to string together a 4:00 mile. His point was that most of the time endurance was the key to performance. In a similar vein, Renato Canova stated that every event is an event of extension. The goal of training is to be able extend your ability to last at a given pace. If your goal is to run a 2:00 800m and in your current shape you can run that pace for 650m, then your training goal is to extend your ability to run at that pace until it matches your desired goal of 800m. Once that goal is reached, you work at extending your ability to last 800m at a faster pace. This occurs via delaying fatigue. If fatigue can be delayed at race pace, then a runner can last longer at that pace before fatigue slows them down. On the other hand, if one can handle greater amounts of fatigue, performance will increase too. It is therefore the rate and maximum tolerance of fatigue that combine to determine performance. Several models of fatigue can be used to practically apply this, but first what is fatigue?

Fatigue is a catch all term that basically means we slow down. The exact mechanisms of fatigue are unknown and vary widely for each event. First, I'll briefly go over the concept of fatigue, and then look at some practical fatigue models. Fatigue has traditionally been thought of as an issue of depletion or accumulation. Either products build up that cause fatigue, or we run out of products that are necessary for function and fatigue. Initially it was thought that these products were direct causes of fatigue, meaning if Hydrogen ions (H+) got high, it directly caused fatigue.

However, more recent research has led to different beliefs. Some of the products believed to play a role in fatigue are listed below

- Increase in H+ (decrease in pH), core temperature, ammonia, Pi, and Magnesium
- Decrease in blood glucose, glycogen, BCAA, and Calcium release and uptake in the Sarcoplasmic Reticulum.

While these products can contribute to fatigue, the more recent understanding is that none of these products directly cause fatigue, and instead they are (or cause changes in substances that are) simply signals to the brain that homeostasis has been changed. An increase or decrease in these products do not directly cause the fatigue, but instead signal to the brain that if we don't change what's going on, we'll be in danger. So the brain then forces the person to slow down both consciously and subconsciously. Subconsciously it may regulate intensity via recruitment of muscles fibers. So that when the brain is receiving signals that fatigue is building up, it decreases muscle fiber recruitment to protect the body. Another proposal says that fatigue is regulated consciously, in that as these products build up or decrease, the body creates the sensation of pain, so that then the runner slows down consciously because of the high pain levels. Certain runners are able to delve deeper into the realms of pain than others, but all of us eventually give in to the ever increasing pain levels. Exactly what occurs is unknown but it is likely some combination of conscious and subconscious interaction.

All of this is nice and good, but in training it does not matter whether the products directly cause fatigue or if they indirectly cause fatigue via feedback. The bottom line is that if we delay the buildup of fatigue products, or delay the reduction in neural recruitment, fatigue is delayed. If the buildup in the products themselves causes fatigue then we are directly delaying fatigue by decreasing their buildup with training. If the build up of products indirectly cause fatigue through pain or subconscious control, then delaying the product build up still delays the signaling to the brain to cause fatigue or pain. Either way performance is improved. Given this information, we can take the complex idea of fatigue and boil it down into three useful simple fatigue models.

The Lactate or By-product theory works on the idea that fatigue is caused by the build up of some fatiguing by-product. We use lactate because it is easily measurable and corresponds with fatigue, not because it causes fatigue itself. The theory is that there is a maximum amount of these products that can accumulate during a race, so if we can decrease the rate of accumulation or increase the maximum amount that can be accumulated, fatigue will be delayed and performance will improve. Based on this theory, the training goals are as follows:

Lactate Theory training goals:

1. Decrease Lactate (fatigue by-products) overall
2. Decrease Lactate accumulation at Race Pace
3. Increase maximum Lactate that can be accumulated over race

The second fatigue theory is based on muscle fiber fatigue. In this model, fatigue is looked at as happening because of both neural recruitment of fibers and the endurance of the fibers themselves (or how long a fiber can stay recruited without fatigue). As fatigue sets in, muscle fibers begin to "fail" and other fibers are recruited to take up the slack. Eventually, there is a point where the majority of the muscle fibers are fatigued and can not maintain the necessary force production, thus fatigue occurs and the runner slows. Based on this theory, fatigue can be delayed via increasing the recruitable fiber pool, training to be able to use those fibers during a race, and lastly extending the endurance of those fibers thus delaying fatigue. This theory is an integration of both peripheral (the muscles themselves) and central (the Central Nervous System) fatigue. As with the lactate fatigue theory, It does not matter whether there is a central regulator or if fatigue happens peripherally in the muscles themselves, as the training goals are still the same. The training paradigm is as follows:

Muscle Fiber Theory:

1. Increase Maximum Fibers you can recruit
2. Increase Percent of fibers you use over race
3. Increase length those fibers can last (endurance)
4. Tying it together: Increase/maintain recruitment of fibers under heavy fatigue

Lastly, we have the overall fatigue model, which ties together local and central fatigue in a useable form. This theory integrates the brain and the muscles in a sort of sensory feedback loop. The model and its implications for training are as follows:

Simple Integrated Fatigue Model:

1. Products of fatigue build up in the muscle.
2. The brain monitors this increase in fatigue products and regulates muscle recruitment to prevent muscle from reaching dangerous level of fatigue products.
    a. Fatigue products that may signal to the brain may include lactate, H+, Ca+, heat/temperature, glycogen levels, and numerous other triggers.
3. Can train to:
    a. Delay buildup of these products thus delaying brains signal to shut down
    b. Delay signal from brain to "shut down and protect" by proving you can safely handle slightly more of that product.
    c. Increase maximum amount of these products that can be accumulated or lost before the brain starts shutting things down.

These models of fatigue allow for practical integration of science and training. The models provide an easy to understand way of looking at what causes fatigue, and in turn how to train to delay fatigue. As a quick example of how to apply these models, let's look at the muscle fiber theory. To delay fatigue we need to increase the muscle fiber pool and increase the endurance of those fibers. To accomplish these goals, we need to do workouts that cause a large increase in muscle fiber recruitment such as sprints, hill sprints, or heavy weight lifting. Following this, we need to extend the endurance of the newly recruited fibers, so the next set of workouts adds an endurance component. An example might be doing hill sprints in between sets of 800m repeats at race pace. The hill sprints would serve to recruit fibers, and then those fibers would be trained during the 800m repeats. Another option is to do long uphill repeats as more muscle fibers are recruited during uphill running then flat running.

Chapter 13: Where do we go from here? Differences between Scientists and Coaches

To conclude, I'd like to briefly outline some of the unanswered issues we have in the field in regards to running, and why coaches and scientists seem to be at opposite ends of the spectrum with their beliefs.

Unanswered questions

There are many unanswered questions in our knowledge of the limits and training of distance runners. Considerable debate exists on the exact mechanisms which limit or cause fatigue during running. From a scientific standpoint, it is hard to see how we can recommend ways to delay fatigue if the exact mechanisms that cause fatigue are still debated. Research such as the work done on Noakes' Central Governor Model and Samuel Marcora's model based on motivation are steps in the right direction in working towards this goal. Similarly in their study Vollaard et al. came to the conclusion that "Moreover, we demonstrate that VO2max and aerobic performance associate with distinct and separate physiological and biochemical endpoints, suggesting that proposed models for the determinants of endurance performance may need to be revisited (2009, pg. 1483)". Perhaps this shift in thinking will get us away from the propensity for relying almost solely on VO2max as a measure of endurance, when it is only one factor that has exploded due to its ease of measurement, and does not even change in elite runners. Which begs the question, why is so much training focused literature done on trying to impact a parameter that does not change much, if at all, in well trained runners? In addition, the validity of giving training recommendations based solely on improving parameters such as VO2max, LT, and RE, as is seen in books such as, should be called into question when changing those parameters do not result in changes in performance (Vollaard et al., 2009; Daniels, 2005). On the same subject of VO2, using percentages of VO2max as a way to quantify training should be called into question given the wide range of stimulus it might provide. It is tough to give advice on training to improve performance, when the exact mechanisms that limit performance are not known. Given this, for research that is geared towards being able to give practical guidelines to coaches on how to train runners, the focus should be on changes in performance, and not some parameter.

Secondly, for runners, I'd argue that researchers should shift away from trying to find the magic workout that improves performance and instead try and explain why workouts that successful coaches use work. In looking at the innovations in track and field training or in techniques, almost all come from the athletes or coaches first and then the science comes in and explains why it works. One prominent example is in the High Jump technique the Fosbury Flop.

There is a large gap between the research studies ideas about training and the real world coach's approach to training. For instance, many researchers and studies point to a lower volume, higher intensity approach, yet practically none of the best distance runners follow this approach. It is easy as a researcher to say that maybe the athletes would be better if they followed the opposite approach, yet as pointed out by Seiler and Tønnessen, it is more likely that elite athletes have developed more efficient training means than those developed in the lab (2009). There are several reasons for this difference, such as the length of training intervention studies. These limitations of studies are not easily overcome, so rethinking the approach or even purpose of training studies needs to be done.

This disconnect also happens in regards to the impact of training intensities. Most research studies classify training based on physiological variables such as LT, VO2max, etc. Yet, most coaches classify training based on different race paces or percentages of race paces. This means that coaches seem to be more specific during workouts with their training paces than training studies. A coach sees differences between running 800m intervals at 2:15 as compared to 2:12, yet in most training studies for someone running around 8:50 for 2miles, these paces would all be considered VO2max training intensities. The classification of training in certain zones does not take into account the differences between various paces. Similarly, the classification of training based on %VO2max should be called into question based on recent studies. This could potentially explain the wide range in results of training intervention studies and is another factor that needs to be taken into consideration.

This leads to the logical conclusion that research should shift in design to actually mimic what is done in the real world. It is very seldom seen that an athlete does the same exact interval session 2-3 times a week for 6 weeks, yet that is what is done in training studies to discern the effects of a particular workout. Perhaps, the effects of that workout would be different when surrounded by a variety of other stimulus, which is what is found in the real world.

Getting away from training studies for a minute, there is little to no research in regards to the planning and periodizing of training on the microcycle, mesocycle, and macrocycle level. Although it is a common practice, there are no research studies that show that alternating hard and easy running days is more beneficial than some other manipulation. In addition, there are no studies on the different periodization models used in training. In fact, there are no running studies showing that periodization is better than doing the same thing every day. Several studies have pointed towards the need for periodization, but none have evaluated its effectiveness. For instance, in Laursen's review, he points out endurance adaptations can occur via two different pathways that require two completely different stimuli (2004). This finding may help clear up the debate on volume versus intensity and show that it is not whether one is better than the other, but when each is used during training that is important. In other words, the periodization of the training is the key.

Lastly, given studies showing the individual response to training and the contrasting views on volume and intensity seen in both research and in popular training literature, one has to wonder whether not enough attention is given to the individual differences in people. As researchers we tend to look at things as averages of a group and ignore the individuals. The wide variety of training programs that are successful in increasing performance is perplexing. One possible answer is that adaptation to a training stress is highly individual. Attributes such as fiber type distribution may play a role in determining whether a person responds best to a high intensity or high volume type of training. This possibility needs to be investigated and more attention to the individual response to training needs to be given.

Despite the large volume of research that has been done on endurance training, the fact remains that training runners based on research only is not a successful way to train. Breaking away from laboratory type studies and looking at some of the training methods used by coaches of runners in the real world would be a step in bridging the gap between research and practical application. As of now, little credence is given to research studies by coaches or athletes, most likely because two factors. First, the studies on training seldom are applicable to real world training. Secondly, on many issues, such as volume of training, due to the measurements used and the length of studies, the research does not match up with real world experiences by world class athletes. This creates a situation where even those coaches who have a scientific background rely on theory, rather than actual research. Perhaps by mimicking that which is done by coaches, such as the periodization schemes used, research could gain credibility in terms of practical application by coaches and athletes.

The problem with applying research to the real world:

When a scientist looks at distance training they break it down and try and isolate a variable. The goal is to find out how a single workout affects a variable. It is what I call the isolation approach. This is how we get the information on what kind of training seems to increase the lactate threshold or improve buffering capacity for example. Let's look at a quick hypothetical situation of what typically occurs when athletes or coaches read such studies.

The next step in the process once a coach or runner hears of the research is that they look at the data and come to the seemingly logical conclusion that if X training improves buffering capacity for example, then runners should do X training or workout to get an improvement in buffering capacity and because buffering capacity is a limiter to certain races, that race time will improve. So the coach who does this goes out and implements this training with his runners and smugly throws around scientific explanations for why his athletes are performing the intervals they are to anybody who will listen, but something goes wrong and his runners do not improve their race times. The coach then becomes perplexed and either reasons that the science was wrong and becomes an anti-

physiology coach or they reason that the athlete did not work hard enough, thus it did not work. If the coach took his athletes to a lab, he most likely would see an increase in buffering capacity. Then he becomes perplexed and thinks the athletes are "weak" runners and not tough enough because the workout obviously accomplished what it was meant to.

The real reason for failure, however, is in the logic, not in the athletes or the coach or the science. Buffering Capacity did increase as the science said it would, but you have to look at the global effects of a workout, not just a singular effect. The singular effect was an increase in buffering capacity, but there are many other singular effects that the science did not look into or explain and all of these singular effects make up the global effect. It is this global effect that is most important. In our example, the intervals had one singular effect of increasing buffering capacity. If you did research, another effect would probably be a decrease in aerobic capabilities due to the heavy acidosis that the athlete endured to increase buffering capacity. Thus these two effects combined (and any others for the workout) would make the global effect an increase in buffering capacity with a decrease in aerobic capabilities.

This example illustrates the difference in approach of a scientists and a coach. Scientists must think in isolation as they are trying to discern the exact effects of one variable. Good coaches on the other hand are concerned with the end result, or in other terms the global effects of a workout. We have to stop thinking of the singular effects a workout might have and start thinking about the global effects it will have on the athlete. Not only do the global effects of one particular workout need to be considered, but also how the workout interacts within a training program. One example that has already been researched is the interference effect with concurrent strength and endurance training.

The takeaway message is that to be useful in the real world of training athletes, the isolation method can not be used. Training does not occur in isolation. Instead, taking a global and integrated approach to training should be done. It's my belief that this difference in perspective or approach between the researchers and the successful coaches is why for the most part the training of the best runners reflects practices established in the real world and not the laboratory. .

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